

NOVA IMS

Introduction to Computational Thinking and Data Science

Solar Energy Investment

Client / Scenario: SunVolt Energy/ Solar Energy Investment

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Submission Date: 21.05.2026

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Executive Summary

As a part of SunVolt ENERGY's expansion plan, it intends to proceed to the installation of a new solar panel facility in Portugal. However, they face a strategic decision regarding the ideal location for the project: Beja or Braga. In order to solve our client's problem, throughout this document will be presented an evaluation of climatic conditions using historical weather data, with the aim to determine which location better satisfies predefined environmental and operational criteria. The dataset in study includes several weather-related variables, such as solar radiation, climate conditions and precipitation levels. These factors directly influence solar panel efficiency and long-term energy production. By analysing and comparing these variables of the two locations in question, our objective is to conclude which location presents the most favourable conditions for solar energy generation. This report hands over the analytical approach, key findings and insights derived from the meteorological data. The results aim to support the client in selecting the optional location for the installation of the solar facility, ensuring a strategic investment decision.

Chapter 1

Problem Framing and Context

1.1 Client Context and Objective

SunVolt ENERGY is a private European investment group specialized in renewable energy infrastructure, particularly in the development and expansion of large-scale solar projects in the Portuguese market. Over the past decade, the company has developed and financed numerous wind and photovoltaic projects across Europe, focusing on sustainability, long-term environmental responsibility, and stable energy returns.

As part of its growth strategy, SunVolt ENERGY is evaluating the development of a new solar energy park in Portugal. After an initial geographic and logistical assessment, two potential locations were identified: Beja, in southern Portugal, and Braga, in the northern region. Both locations offer suitable land availability and grid access, but the final investment decision requires a deeper evaluation of environmental and meteorological conditions.

The main objective of this project is to determine the most suitable location for a new solar park by comparing Beja and Braga. Since the profitability of solar energy projects depends on factors such as solar radiation, cloud coverage, and temperature patterns, this study uses historical meteorological data from Open Meteo to evaluate both locations.

Through the analysis of this environmental data, we aim to identify which region offers greater solar potential, more stable climate conditions, and more predictable energy production.

1.2 Decision Alternatives

1.2.1 Understanding Solar Energy Generation

Solar energy is the light (photons) emitted by the sun, generated when over 500 million tons of hydrogen atoms are converted into helium every second. Solar panels harness this energy through the photovoltaic effect.

Panels utilize photovoltaic cells (typically made of semi-conducting materials like silicon) that excite electrons when exposed to sunlight, creating a flow of direct current (DC) electricity. Crucially, these panels capture *light*, not heat. In fact, excessive heat creates electrical resistance within the semiconductor, actively reducing the panel's efficiency.

1.2.2 The Competing Alternatives: Beja vs. Braga

This project evaluates two distinct geographic alternatives for a new solar installation: **Beja** (in the south of Portugal) and **Braga** (in the north). These cities represent contrasting environmental profiles that heavily influence photovoltaic generation. Beja generally experiences higher overall solar radiation and longer sunshine duration, but suffers from significantly hotter peak temperatures. Braga features cooler temperatures (which helps panel efficiency), but experiences higher humidity, much more frequent rain, and denser cloud cover.

1.2.3 Defining "Better" and Key Variables

To determine which location is "better," we cannot simply look at which city receives more sun. "Better" in this context means finding the optimal balance of environmental conditions to maximize total, consistent energy yield over time. While our dataset contains numerous environmental metrics, our evaluation will primarily hinge on a few critical factors. Such as Shortwave Radiation and Cloud Cover. We must evaluate if Braga's frequent cloud cover penalizes production too heavily compared to Beja's clear skies. However, we must also account for Ambient Temperature and Wind Speed. Because solar panels are semiconductors, Beja's intense summer heat will cause noticeable voltage drops, typically modeled at $\approx -0.3\%/^{\circ}\text{C}$ above standard testing conditions. Ultimately, our modeling will determine if Beja's superior raw radiation outweighs its efficiency-draining heat, or if Braga's cooler, windier climate can mathematically compensate for its lack of direct sunlight.

1.3 Key Questions and Success Criteria

The following analytical questions will guide the evaluation of solar potential and climate suitability for each location:

- Across all years of recorded data, which city receives the highest amount of total solar energy?
- Which city experiences more cloudy days that might block the sun and reduce power generation?
- Which location has more frequent extreme heat days that could cause solar panels to overheat and perform less efficiently?
- Which city has a steadier climate, avoiding big swings in energy production from one day to the next?
- Are there frequent high-speed wind gusts in the cities that would require more expensive, reinforced mounting systems?
- Is the air significantly more humid in one city, potentially leading to faster wear and tear on the electrical components?

The recommendation will be based on specific criteria, such as:

- **Total Energy Yield:** The primary winner will be the city that generates the most cumulative energy across the period from 2018 to 2026.
- **Reliability Factor:** The chosen city must demonstrate a higher frequency of clear-sky days and fewer "zero-production" days.
- **Temperature Balance:** Preference will be given to the location that offers high solar radiation while maintaining manageable peak temperatures to minimize photovoltaic efficiency loss.
- **Infrastructure Safety:** The recommended city should have lower average and peak wind speeds to ensure long-term structural integrity and reduce the cost of reinforced mounting systems.
- **Seasonal Sustainability:** The city must produce a stable and sufficient power supply during the winter months to ensure year-round project viability.

Chapter 2

Data Understanding

2.1 Data Sources

For a more reliable analysis, this study uses daily meteorological data collected for two Portuguese cities, Beja and Braga. Two separate datasets were used, one for each city, enabling a comparative analysis of climate variables and of their behaviour over time across different regions. Each dataset contains daily meteorological observations and follows the same variable structure, allowing a direct comparison between the two locations.

The datasets were obtained from the Open-Meteo weather data platform, an open-source weather API that offers free access for non-commercial use and utilizes open-data weather forecasts provided by national weather services.

The period covered is from January 2018 to early 2026, with observations recorded on a daily basis. Each record represents the weather conditions for a specific day in each location, resulting in approximately 2,981 observations per city.

To enhance the rigor of this study, multiple meteorological variables are included, each measured in different units, as detailed in Table 2.1 below.

Meteorological Variable	Unit
Time	iso8601
Air Temperature	°C
Precipitation	mm
Wind Speed	km/h
Wind Gusts	km/h
Shortwave radiation	W/m ²
Relative Humidity	%
Cloud cover	%
Sunshine duration	s
Sunset and Sunrise Time	iso8601
Latitude	°
Longitude	°

Table 2.1: Meteorological variables and their respective units

2.2 Dataset Structure and Variables

The variables used in this study are presented in this section, highlighting their importance in terms of solar energy analysis, as well as their characteristics. The results showed that after analysis, there were no missing values (0%).

Time – temporal variable representing the specific date associated with each observation in the dataset, essential for identifying seasonal patterns and long-term trends, ensuring that the investment model accounts for annual variability and energy supply reliability.

Temperature_2m_max – continuous numerical variable capturing the highest air temperature observed during the day at a height of 2 meters (°C), describing daily heat intensity and potentially impacting solar panel performance, as elevated temperatures can reduce efficiency.

Temperature_2m_min – continuous numerical variable representing the lowest air temperature reached during the day at 2 meters above the surface (°C), providing insight into cooler conditions that contribute to daily temperature variation and may influence overall solar system behaviour.

Temperature_2m_mean – continuous numerical variable expressing the average air temperature throughout the day at a height of 2 meters (°C), providing an overall view of thermal conditions that can influence the efficiency and performance of solar energy systems.

Precipitation_sum – continuous numerical variable denoting the daily accumulated

precipitation (mm), influencing solar energy production by temporarily reducing solar radiation due to cloud cover, while also contributing to natural cleaning of solar panels by removing dust and debris, which may improve efficiency over time.

Wind_speed_10m_max – continuous numerical variable showing the maximum sustained wind speed measured during the day at 10 meters above ground level (km/h), affecting panel temperature through natural cooling and influencing system stability.

Wind_gusts_10m_max – continuous numerical variable indicating the maximum wind gust recorded during the day at a height of 10 meters (km/h), associated with extreme weather conditions and important for system safety, as high wind speeds may require protective measures that reduce energy production.

Shortwave_radiation_sum – continuous numerical variable representing the total amount of solar shortwave radiation received at the Earth’s surface during the day (W/m^2), serving as the main energy input for solar systems and determining potential energy production, as well as enabling comparison between locations such as Beja and Braga.

Sunshine_duration – continuous numerical variable indicating the total duration of sunshine during the day (s), contributing to more stable energy output, as longer sunshine periods generally lead to fewer interruptions.

Relative_humidity_2m_mean – numerical variable measuring the average relative humidity at 2 meters altitude (%), related to solar radiation availability, as higher humidity levels can increase cloudiness and haze, reducing solar energy efficiency.

Cloudcover_mean – continuous numerical variable representing the average percentage of sky covered by clouds throughout the day (%), directly affecting solar radiation and energy production levels, particularly in cloudier regions.

Sunrise – temporal variable indicating the exact time of sunrise (HH/MM), marking the beginning of the daylight period and determining when solar energy production can start.

Sunset – temporal variable indicating the time of sunset (HH/MM), marking the end of the daylight period and influencing the total duration of daily energy production.

Zone – categorical variable identifying the geographical region where the meteorological data is collected, in this case Braga and Beja.

Latitude – continuous numerical variable defining the north–south geographic coordinate

of the location ($^{\circ}$), influencing solar radiation angle and contributing to optimal panel tilt and energy production performance.

Longitude – continuous numerical variable representing the east–west geographic coordinate of the location ($^{\circ}$), affecting the sun’s position during the day and the timing of solar energy production.

Variable	Mean	Mode	Median	Dispersion	Min	Max	Missing
precipitation_sum (mm)	4.7469	0.00	0.00	2.0671	0.00	90.700	0%
cloudcover_mean (%)	53.83	0	58	0.62	0	100	0%
shortwave_radiation_sum (W/m ²)	15.0799	8.690	14.390	0.5645	0.510	30.770	0%
sunshine_duration (s)	31432.32	0.00	34063.8	0.4647	0.00	50805.2	0%
temperature_2m_min ($^{\circ}$ C)	10.6377	12.900	10.800	0.4543	-3.800	24.500	0%
temperature_2m_mean ($^{\circ}$ C)	15.0455	11.600	14.500	0.3530	1.300	31.400	0%
temperature_2m_max ($^{\circ}$ C)	20.0208	14.900	19.100	0.3152	6.200	39.300	0%
wind_speed_10m_max (km/h)	16.8386	14.800	15.800	0.3536	5.800	47.300	0%
wind_gusts_10m_max (km/h)	36.7285	32.800	34.600	0.3476	11.500	103.700	0%
relative_humidity_2m_mean (%)	76.69	85	79	0.16	25	99	0%

Table 2.2: Summary statistics of the meteorological variables (Braga dataset)

Table 2.2 provides a summary of the meteorological conditions for the Braga dataset, helping us understand how each variable behaves and how much it varies. Looking at the precipitation and humidity, we can see that most days in Braga are dry, as the median is 0.00 mm, although extreme rainfall events do occur with peaks reaching 90.70 mm. Even with many dry days, the air remains generally humid, with an average relative humidity of 76.69%. Regarding cloud cover and radiation, the sky is about 53.83% covered by clouds on average. Since this variable ranges from 0 to 100%, it directly impacts the solar radiation levels, which show significant variability between a minimum of 0.51 W/m² and a maximum of 30.77 W/m² depending on the weather.

The sunshine duration also shows high values, with an average of 31432.32 s, indicating that the region receives a good amount of sun exposure. In terms of temperature, the values are moderate with an average around 15.04 $^{\circ}$ C, but there is a clear seasonal fluctuation since we recorded a range between 1.30 $^{\circ}$ C and 31.40 $^{\circ}$ C. Finally, the wind speed is usually moderate, averaging 16.84 km/h, but the data for wind gusts shows that strong events can happen, reaching over 100 km/h. Overall, while some factors like sunshine are relatively stable, the high variability in precipitation and radiation is a key point for our study on solar energy potential.

Variable	Mean	Mode	Median	Dispersion	Min	Max	Missing
precipitation_sum (mm)	1.54	0.00	0.00	2.93	0.00	57.60	0%
cloudcover_mean (%)	42.66	0	42	0.75	0	100	0%
shortwave_radiation_sum (W/m ²)	17.26	7.91	16.93	0.46	1.07	31.02	0%
sunshine_duration (s)	34776.6	46800.0	36523.9	0.33	0.00	48521.3	0%
temperature_2m_min (°C)	12.07	10.40	12.00	0.41	-2.20	26.10	0%
temperature_2m_mean (°C)	17.45	12.90	16.60	0.35	3.20	34.90	0%
temperature_2m_max (°C)	23.7116	17.000	22.200	0.32321	6.500	43.300	0%
wind_speed_10m_max (km/h)	16.84	14.80	15.80	0.35	5.80	47.30	0%
wind_gusts_10m_max (km/h)	37.01	34.60	36.40	0.31	13.30	100.80	0%
relative_humidity_2m_mean (%)	67.43	75	69	0.24	18	97	0%

Table 2.3: Summary statistics of the meteorological variables (Beja dataset)

The meteorological profile of Beja, as summarized in Table 2.3, highlights a region characterized by its dry and sunny climate. Precipitation levels are notably low, with an average of 1.54 mm and a median of 0.00 mm, which indicates that rainy days are rare, although a few peaks reach 57.60 mm. This trend is supported by the cloud cover average of 42.66%, which suggests that clear skies occur frequently in this dataset. As a result, solar radiation levels average 17.26 W/m², though they still vary between 1.07 and 31.02 W/m² depending on atmospheric changes.

Regarding temperature mean, the average is 17.45 °C, with recorded values between 3.20 °C and 34.90 °C, showing a clear seasonal range. The sunshine duration remains a stable factor, recording 34776.6 s. Even though wind gusts can occasionally exceed 100 km/h, the general wind speed stays moderate at around 16.84 km/h. Ultimately, these statistics show that Beja has consistent light levels and lower cloud interference, providing the necessary data for the predictive models and the analysis of solar energy potential developed in this study.

2.3 Data Quality and Initial Limitations

In this section, we conduct an exploratory analysis of the meteorological variables for Beja and Braga to understand their distributions and their direct impact on solar energy potential. Rather than merely observing historical averages, the features were evaluated based on their technical influence on photovoltaic system performance. The analysis is structured around three main dimensions: core energy availability, thermal efficiency constraints, and meteorological variability.

2.3.1 Energy Availability: Solar Radiation and Sunshine Duration

The fundamental drivers for energy generation in our dataset are shortwave-radiation-sum and sunshine-duration. While assessing the total annual radiation is essential to estimate maximum yields, understanding the seasonal distribution of these variables is equally critical. We must analyze how radiation fluctuates across different months to determine the baseline energy production capabilities of each region and ensure there is sufficient consistency during the winter season to maintain a stable energy output.

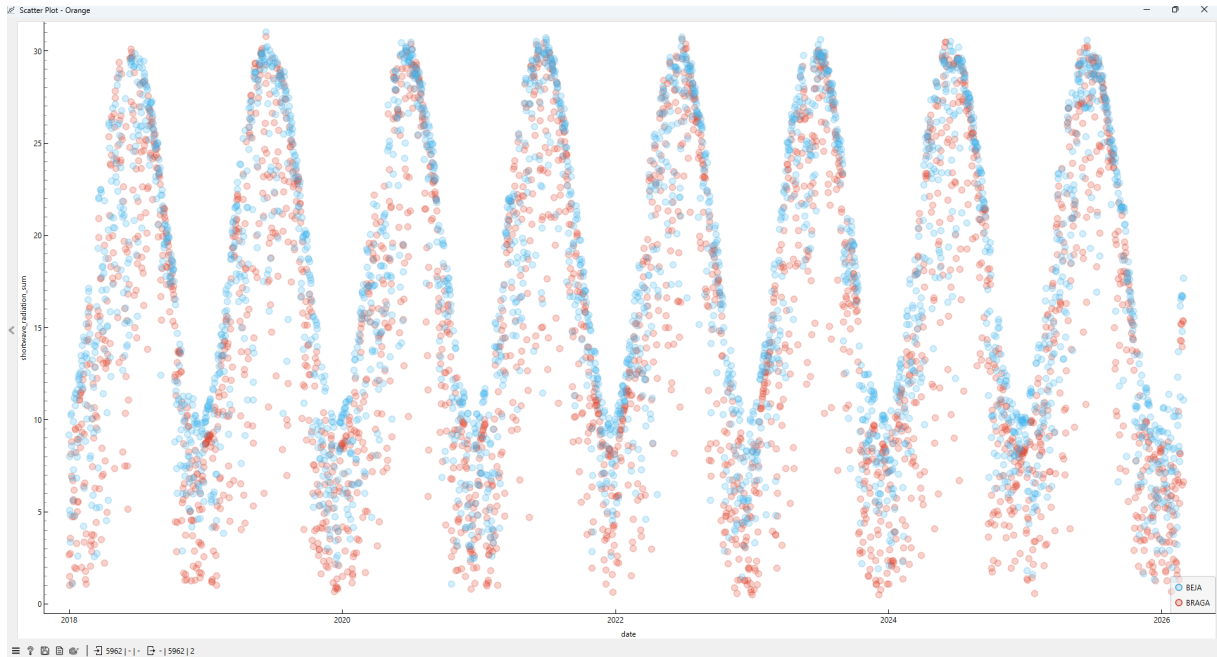


Figure 2.1: Scatter Plot - radiation seasonality

The scatter plot illustrates the distribution of daily shortwave radiation from 2018 to 2026 for both Beja and Braga. By utilizing reduced point opacity, the visualization effectively highlights data density, allowing a clear seasonal "bell curve" to emerge for each annual cycle.

Both locations follow a synchronized seasonal pattern, with solar radiation peaking during the summer months (June-August) and reaching its lowest points during the winter (December-January). However, a comparative analysis reveals that Beja maintains a higher density of points in the upper radiation brackets throughout the year. While the peak summer performance is relatively comparable between the two cities, the divergence becomes most apparent during the shoulder seasons and winter months. Beja's "cloud" of data points remains consistently higher on the y-axis, indicating a superior baseline of energy availability.

From an investment perspective, this seasonal distribution is a primary driver of project viability. The higher radiation density observed in Beja translates directly into a higher Annual Energy Production. Furthermore, the less severe troughs in Beja’s winter data suggest a more stable year-round performance, reducing the risk of extreme seasonal revenue fluctuations. While Braga demonstrates viable solar potential, its overall lower density and deeper winter declines indicate a less efficient use of the same installed capacity compared to the southern region.

2.3.2 Thermal Constraints and Panel Efficiency

A critical factor in PV performance is the ambient temperature. Solar panels are typically rated under Standard Test Conditions, assuming an operating temperature of 25 °C. As ambient temperatures rise above this threshold, the panels experience a voltage drop, leading to a measurable decrease in overall energy conversion efficiency. Therefore, analyzing temperature-2m-max and temperature-2m-mean is vital. A region with high solar radiation but frequent summer days exceeding 25 °C might suffer from significant efficiency penalties. Evaluating the frequency and intensity of these high-temperature events is a key metric for accurately comparing the true generation potential of both locations.

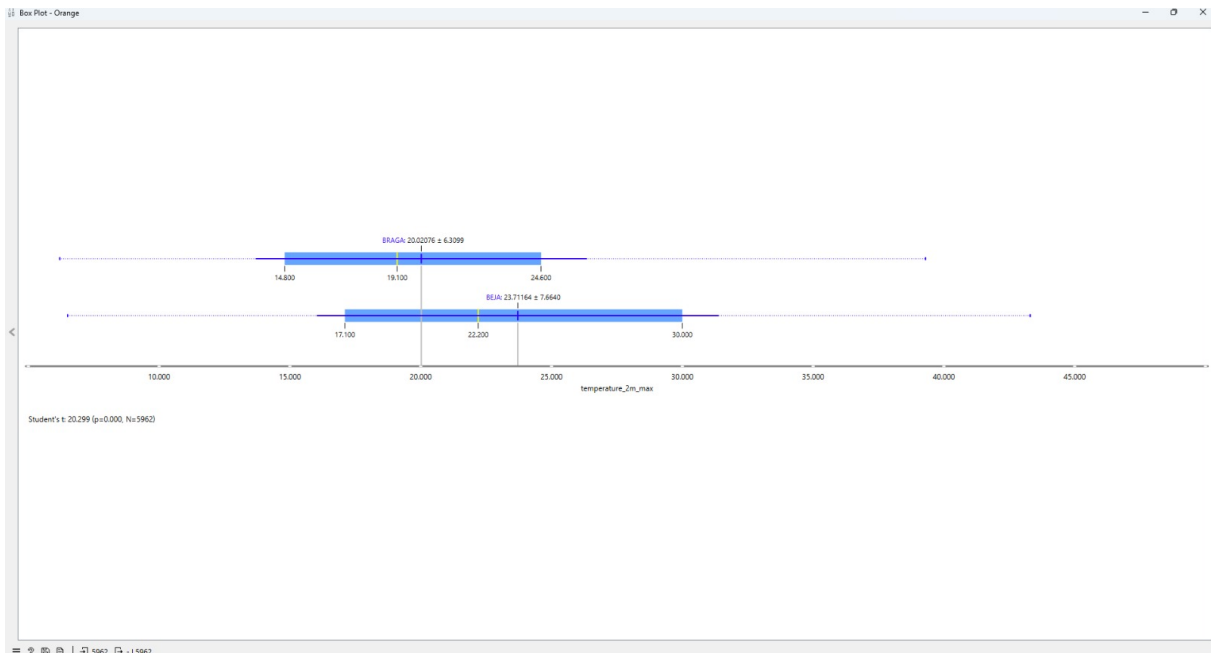


Figure 2.2: Box Plot- Thermal Constraints and Panel Efficiency

This boxplot illustrates the distribution of maximum daily temperatures (temperature-

2m-max) for both Beja and Braga. In the context of photovoltaic performance, 25 °C represents the Standard Test Conditions (STC) threshold for optimal panel efficiency. Observing the y-axis, it is clear that Beja not only has a higher median temperature but also presents a massive concentration of extreme positive outliers (heatwaves) reaching up to 45 °C. These extreme thermal events highlight a critical risk factor: despite having a higher overall solar radiation potential, a solar park in Beja will likely suffer from more frequent and severe efficiency drops due to panel overheating when compared to Braga.

2.3.3 Meteorological Variability: Cloud Cover and Precipitation

Variables such as cloudcover-mean and precipitation-sum act as direct limiters to solar irradiance. High cloud cover introduces significant volatility into daily energy production, making output forecasting and grid stability much more complex. By comparing the frequency of heavily overcast or rainy days between Beja and Braga, we can quantify the operational risks and anticipate the expected production dips, ultimately assessing which location offers a more reliable and predictable energy supply.

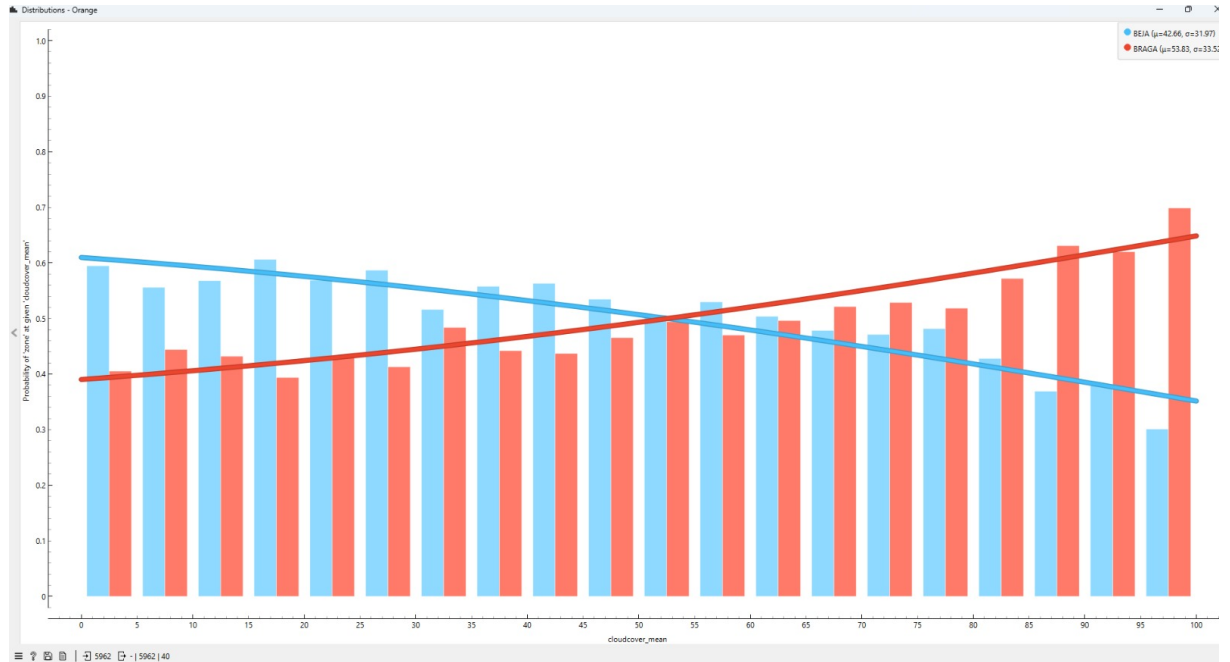


Figure 2.3: Histogram - Cloudcover

The histogram analysis of daily mean cloud cover reveals a significant disparity in meteorological stability between the two regions. Beja's profile is defined by a massive, dominant peak near 0 percent, indicating that the location enjoys clear skies for the vast majority of the year, which ensures a steady and predictable flow of solar irradiance. In contrast,

Braga's distribution is far more dispersed, with a substantial frequency of days characterized by high cloud cover (reaching 80 percent to 100 percent). From an operational perspective, this demonstrates that while Beja offers a highly reliable environment for energy generation, Braga's solar potential is frequently hampered by overcast conditions. This atmospheric variability in the north introduces higher forecasting uncertainty and more frequent dips in daily energy production compared to the stable, clear-sky conditions found in Beja.

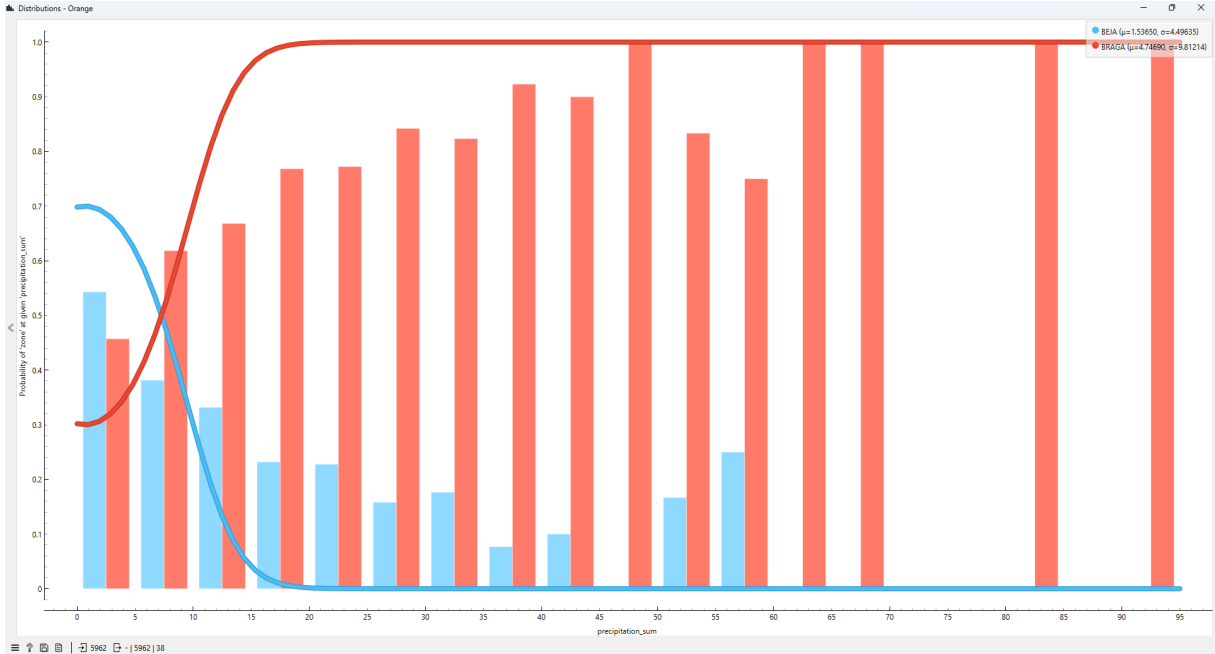


Figure 2.4: Histogram - precipitation

The histogram of daily precipitation sum highlights a sharp contrast in climatic stability between the two regions. Beja is characterized by a high concentration of dry days and a near-absence of heavy rainfall events, ensuring a stable operational environment. Conversely, Braga exhibits a much higher frequency of intense precipitation, with a long right tail showing days with rainfall exceeding 60 mm. This suggests that Braga is not only more prone to production interruptions due to the associated cloud cover but also faces higher infrastructure risks from extreme weather. Consequently, Beja emerges as the more favorable location for solar investment due to its significantly lower precipitation-related volatility.

Chapter 3

Methodology

3.1 Preprocessing and Data Preparation

The preprocessing phase is a critical step in ensuring the overall quality and predictive power of the dataset. Rather than relying solely on raw meteorological and temporal observations, a comprehensive data cleaning and feature engineering strategy was implemented. The primary objective of this stage was to reduce noise, enhance interpretability, and extract meaningful, domain specific patterns that more accurately reflect the underlying environmental dynamics.

To handle the continuous variables representing sunlight exposure and cloud cover, a discretization approach was applied. The raw continuous measurements for both metrics were transformed into three distinct, ordinal bins: Low, Moderate, and High. This categorical transformation serves a dual purpose: it mitigates the impact of minor fluctuations and potential outliers in the daily weather data, and it allows subsequent models to focus on broader, more significant environmental shifts rather than being overly sensitive to exact numerical variations.

Similarly, the temporal granularity of the dataset was adjusted to better capture cyclical patterns. Instead of maintaining highly granular data tied to specific individual days, which often introduces unnecessary noise, the timeline was aggregated and mapped into the four standard seasons. This seasonal grouping prevents the analysis from overfitting to daily anomalies while successfully preserving the overarching annual trends that dictate weather patterns and related behaviors.

Furthermore, to provide deeper contextual insights, particularly regarding energy demand and climate impact, new advanced indicators were engineered from the baseline data. The daylight-hours feature was created to measure the total duration of potential sunlight

exposure per day, serving as a crucial proxy for daily activity cycles and solar potential. Additionally, standard meteorological metrics were introduced to quantify temperature driven demands: Heating Degree Days (HDD18) and Cooling Degree Days (CDD22). Calculated using baseline temperatures of 18°C and 22°C respectively, these specialized variables effectively capture the nonlinear relationship between outdoor temperatures and the energy required for indoor heating during colder periods or cooling during warmer periods.

In addition to these transformations, further high-level explanatory variables were systematically incorporated to enrich the dataset’s descriptive and predictive capacity. A temporal indexing variable, year, was explicitly introduced to preserve interannual variability and enable the detection of longitudinal trends, structural shifts, or potential regime changes across different periods. Moreover, a derived metric termed radiation efficiency was engineered to quantify the effective utilization of incoming solar radiation, capturing the relationship between theoretical solar input and actual observed conditions, thus providing a more nuanced representation of atmospheric transmissivity and energy conversion potential.

Complementarily, a binary precipitation indicator, rain or no rain, was constructed to encapsulate the occurrence of rainfall events in a simplified yet analytically powerful format. This dichotomous feature enhances model interpretability while retaining essential information regarding weather state transitions, particularly in distinguishing between dry and wet conditions that may significantly influence environmental behavior, surface processes, and energy demand patterns.

Additionally, a composite classification variable named day-quality was introduced to encode overall daily environmental conditions into a structured categorical outcome. This variable was defined using a rule-based decision function combining key meteorological indicators, as follows:

A day is classified as “Perfect” if Radiation-Efficiency > 25 , precipitation-sum < 0.05 , and daylight-hours are equal to “high sunlight”. It is classified as “High Risk” if precipitation-sum ≥ 4.45 or Radiation-Efficiency < 10 . All remaining cases are classified as “Standard”. This formulation allows the data set to capture nonlinear interactions between solar availability, atmospheric conditions, and precipitation intensity, while translating complex environmental dynamics into an interpretable categorical signal suitable for downstream modeling and analysis.

Ultimately, this rigorous feature engineering and cleaning process transforms the raw data into a highly structured, robust, and meaningful foundation. By creating these categorical aggregations and specialized energy indicators, alongside the integration of temporally and physically informed variables, the dataset is optimally primed for deeper exploratory data

analysis and advanced predictive modeling.

3.2 Exploratory Analysis Strategy

The exploratory analysis strategy focuses on translating raw meteorological data into actionable insights for photovoltaic viability. Shortwave Radiation Sum was selected as the primary variable because it represents the fundamental energy input available for solar conversion, as we explained previously in 1.2, other variables are used in the analysis, but they are used as complementary variables, to analyze aspects that are not as important as the production of energy.

To ensure the analysis reflects real-world performance, we introduced Radiation Efficiency, which mathematically accounts for the reduction of efficiency due to higher temperatures, by applying a 0.3%/C penalty when temperatures exceed 25C, This penalty is applied to the shortwave radiation variable, making it more accurate, for production of energy.

Finally, the Day Quality index was developed as a composite metric to categorize operational success into Perfect, Standard, or High Risk days. This index synthesizes radiation efficiency, precipitation levels, and daylight duration to provide the client with a clear assessment of long-term reliability and investment risk.

3.3 Statistical Methods and/or Modeling Approach

3.3.1 Correlation Analysis: Precipitation and Cloud Cover vs. Solar Potential

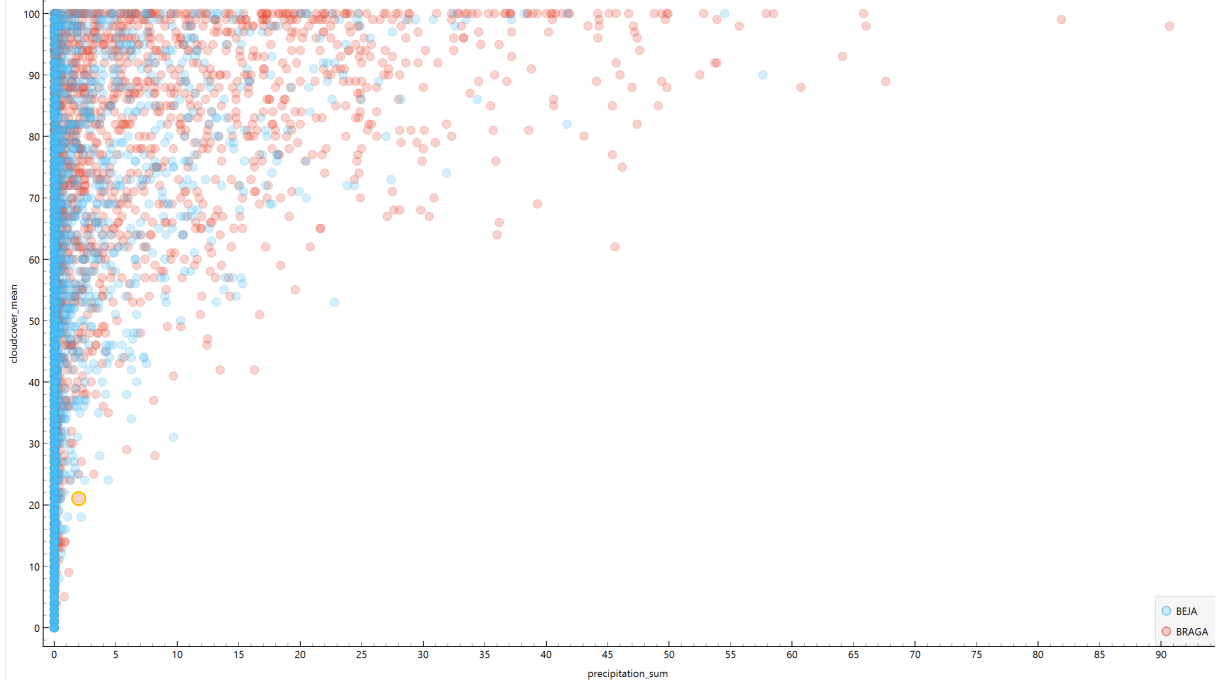


Figure 3.1: Scatter plot showing the relationship between daily precipitation sum and mean cloud cover for Braga and Beja. Points are color-coded by city (Zone) to highlight regional climatic differences.

To evaluate the feasibility of solar panel installation, it is essential to understand how local weather patterns interfere with light capture. The following scatter plot illustrates the correlation between precipitation and cloud cover. By visualizing these variables simultaneously for both cities, we aim to identify which location offers the most consistent 'clear sky' conditions, which are fundamental for maximizing photovoltaic output and reducing energy intermittency.

3.3.2 Regional Predictive Modeling: Classification and Risk Assessment

We implemented a multivariate approach using Logistic Regression, SVM, and kNN to classify the data points into their respective zones, Beja or Braga. We chose Logistic Regression to understand the specific weight of each weather variable on production, SVM to find the most robust mathematical boundary between the two climates, and

kNN to identify patterns based on the similarity of historical weather days.

To ensure the models are grounded in reality, we engineered a conjugated variable, *Radiation_Efficiency*, to mathematically account for the 0.3% efficiency loss panels suffer for every degree above 25°C, ensuring our models evaluate actual usable energy yield rather than theoretical sunlight.

Model	AUC	CA	$\hat{F1}$	Prec	Recall	MCC
SVM	0.472	0.465	0.280	0.428	0.208	-0.081
kNN	0.634	0.598	0.601	0.597	0.606	0.196
Logistic Regression	0.674	0.612	0.614	0.611	0.617	0.225
Constant	0.500	0.500	0.643	0.500	0.900	-0.001

(a) Braga

Model	AUC	CA	$\hat{F1}$	Prec	Recall	MCC
Constant	0.500	0.500	0.167	0.499	0.100	-0.001
SVM	0.472	0.465	0.575	0.477	0.723	-0.081
kNN	0.634	0.598	0.595	0.600	0.590	0.196
Logistic Regression	0.674	0.612	0.610	0.613	0.608	0.225

(b) Beja

Figure 3.2: Evaluation Metrics

On figure 3.2 (a) and (b) we have a table that presents the evaluation metrics (AUC, CA, F1, Precision and Recall) obtained through 10-fold Cross-Validation. It serves as the primary scorecard to compare the predictive power of different algorithms in identifying regional climatic fingerprints.

		Predicted		Σ
		BEJA	BRAGA	
Actual	BEJA	1811	1170	2981
	BRAGA	1141	1840	2981
Σ		2952	3010	5962

Figure 3.3: Confusion Matrix of actual versus predicted regional climatic labels.

On figure 3.3 the confusion matrix visualizes the model's accuracy by cross-referencing predicted labels against actual data. It specifically identifies True Positives and False Positives, allowing us to quantify the "Operational Risk" where one city's weather mimics the other.

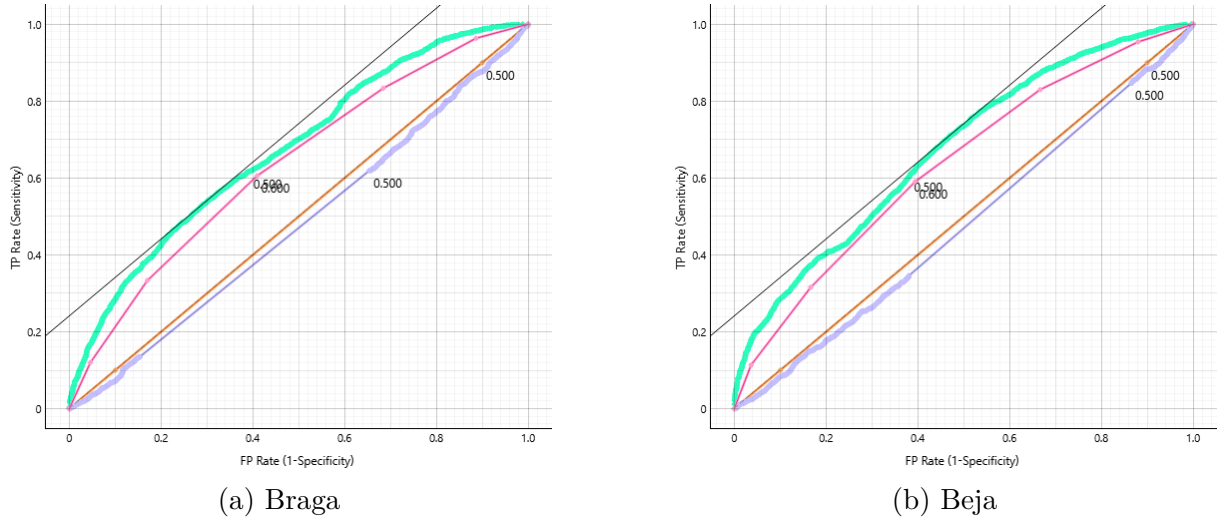


Figure 3.4: ROC curves

On Figure 3.4, the ROC curve plots the True Positive Rate against the False Positive Rate to provide a threshold-independent metric (AUC) of the model's ability to distinguish between Beja and Braga. We used this analysis to validate the discriminatory power of our models, ensuring that our regional classification is statistically significant and not based on random chance.

3.3.3 Multidimensional Analysis of Solar Potential: The Interplay of Wind, Temperature, Radiation and Location

To better understand the relationship between meteorological variables and solar energy potential, a combination of exploratory, categorical, and modeling approaches was applied to assess key determinants and support the comparison between Beja and Braga, while balancing interpretability and predictive performance.

The analysis focused on `shortwave_radiation_sum` as the target variable, given its central role as the primary driver of solar energy generation. Meanwhile, `temperature_2m_mean` and `wind_gusts_10m_max` were selected as key explanatory variables due to their significant influence on photovoltaic performance, with temperature affecting thermal efficiency and wind influencing atmospheric conditions and passive cooling, as well as playing a critical role in system stability, structural requirements, and long-term operational reliability.

As an initial step, continuous variables were discretized into categorical intervals. This transformation enabled the application of a mosaic plot to explore the interaction between temperature, wind gusts, and shortwave radiation. The choice of this method is justified by its ability to represent conditional distributions in a multidimensional context, enabling the identification of interaction effects that are not evident in simple bivariate analysis.

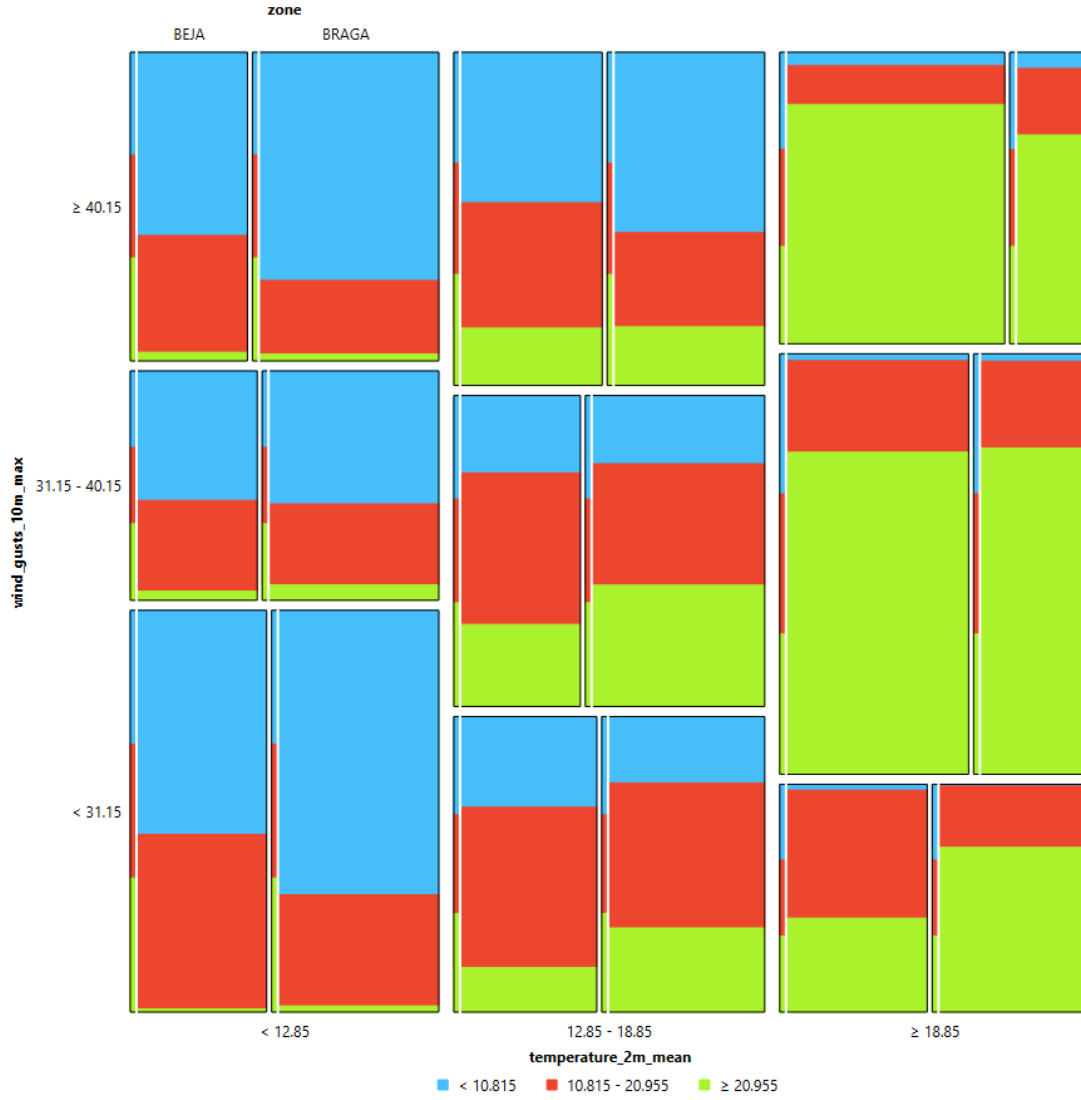


Figure 3.5: Mosaic Plot of Shortwave Radiation Levels Across Zone, Temperature, and Wind Gust Categories

The mosaic plot of Figure 3.5 illustrates how radiation levels vary across different combinations of temperature and wind categories, with rectangle size representing frequency and colour indicating the proportion of radiation levels within each segment. In particular, this visualisation supports the identification of which city is more frequently associated with lower wind conditions and higher radiation levels, which are favourable for photovoltaic energy generation.

3.3.4 Rationale for Visualization Selection: The Violin Plot Approach

The Violin Plot was selected as one of the visualization tools for this analysis due to its unique ability to combine the statistical rigor of a Box Plot with the descriptive power of a Kernel Density Plot. While a standard Bar chart only shows averages and a Box Plot focuses on quartiles, the Violin Plot reveals the "shape" of the data, allowing for a much deeper understanding of the climatic differences between Braga and Beja.

One of the main advantages of this visualization is that it illustrates data density. By observing the width of the "violin" at different temperature levels, we can immediately identify the most frequent weather patterns in each region. This is crucial for a solar energy project, as it allows us to see not just the average temperature, but exactly how many days fall within the optimal thermal range for energy production.

Furthermore, the Violin Plot excels at highlighting multimodality and asymmetry. As seen in the analysis, it clearly distinguishes Beja's bimodal distribution from Braga's downward-skewed concentration. This level of detail is essential for risk assessment, as it helps identify the frequency of extreme cold or heat events that could impact infrastructure performance, providing a level of transparency that simpler charts would fail to capture.

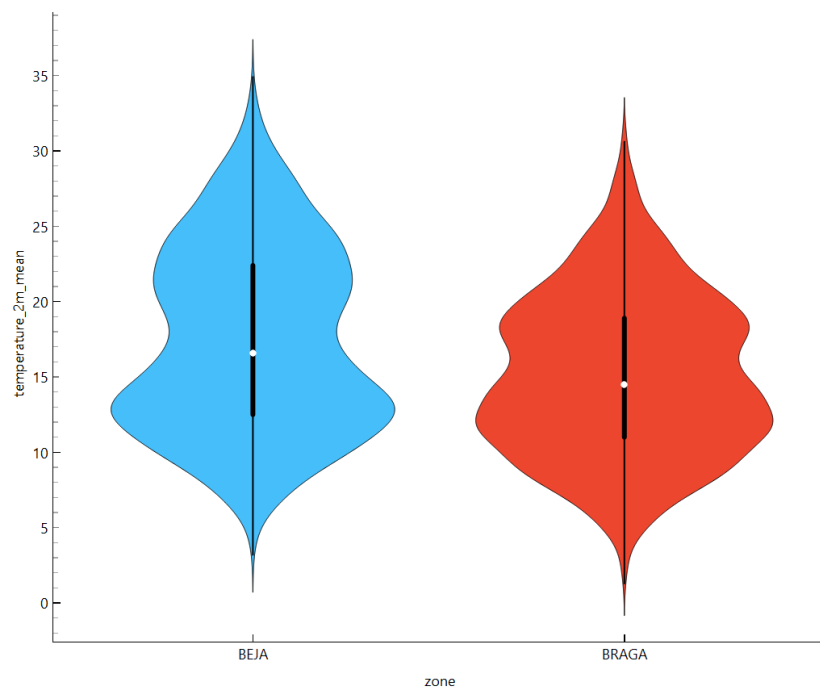


Figure 3.6: Violin Plot of Mean Temperature

3.3.5 Classification Models and Evaluation

A classification model was developed to predict rainfall conditions (rain class), distinguishing between rainy and non-rainy days using meteorological variables such as cloud cover, temperature, relative humidity, and wind speed.

Three classification algorithms were tested: Logistic Regression, k-Nearest Neighbors (kNN), and Support Vector Machine (SVM), allowing comparison between different modeling approaches.

The models were evaluated using 10-fold cross-validation, which ensures that all observations are used for both training and testing, providing more stable and reliable results than a simple train-test split.

Model performance was assessed using metrics such as AUC, accuracy, F1-score, precision, recall, and MCC, providing a comprehensive evaluation of classification quality.

Model	AUC	Accuracy	F1	Precision	Recall	MCC
Logistic Regression	0.927	0.849	0.870	0.859	0.881	0.689
kNN	0.830	0.774	0.812	0.777	0.851	0.532
SVM	0.597	0.566	0.678	0.591	0.795	0.058

Table 3.1: Performance of classification models using 10-fold cross-validation

		Predicted		Σ
		No Rain	Rain	
Actual	No Rain	3026	407	3433
	Rain	496	2033	2529
Σ		3522	2440	5962

Figure 3.7: Confusion matrix for the Logistic Regression model using 10-fold cross-validation

A confusion matrix was generated to evaluate the performance of the classification model in rainfall prediction. This analysis focuses on comparing the model's predictions with the actual rainfall conditions (Rain and No Rain), allowing the identification of how observations are assigned to each outcome.

The results are organized into true positives, true negatives, false positives, and false negatives, providing a clear overview of the different prediction outcomes.

The confusion matrix offers a structured representation of these results, showing how observations are distributed across the different outcomes and supporting a detailed assessment of the model's performance in rainfall prediction.

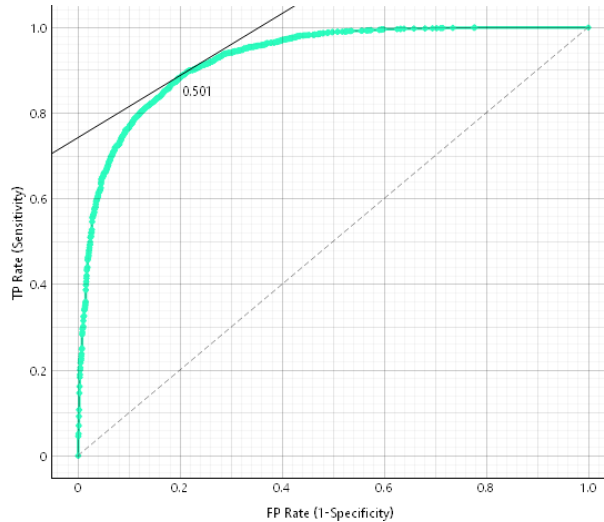


Figure 3.8: ROC curve for the Logistic Regression model using 10-fold cross-validation

To further assess the behaviour of the classification model, a Receiver Operating Characteristic (ROC) curve was used.

Rather than focusing on a single prediction threshold, this approach evaluates how the model performs across a range of possible thresholds. By progressively adjusting this threshold, the curve captures the relationship between the model's ability to correctly identify rainfall events (true positive rate) and the occurrence of false alarms (false positive rate).

This graphical representation highlights how changes in the decision threshold affect the balance between correctly detecting rainfall and incorrectly predicting it when no rain occurs.

The ROC curve is presented for the Logistic Regression model, as it achieved the best overall performance among the tested models, making it the most suitable candidate for a more detailed analysis. Through this perspective, the ROC curve provides a more complete view of the model's behaviour under different classification settings.

Graphical Analysis of Solar Radiation

In addition to classification models, graphical analysis was performed to explore the relationship between meteorological variables and solar radiation, as well as differences across locations and time.

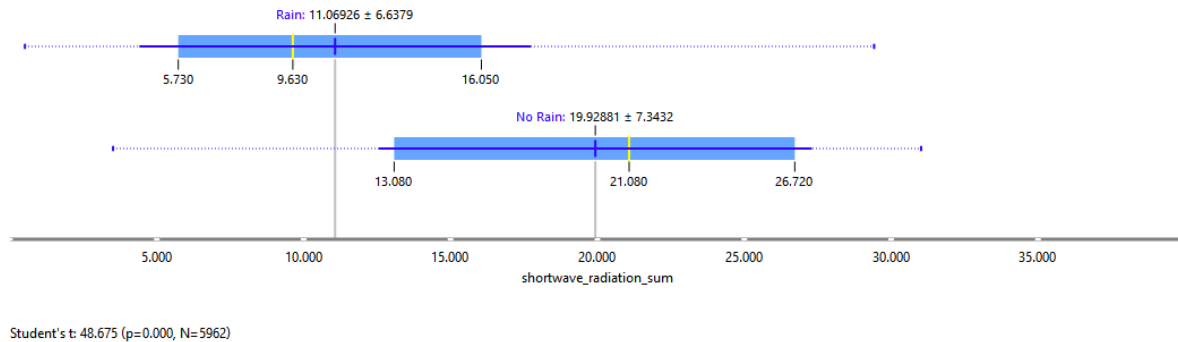


Figure 3.9: Box plot - Solar radiation distribution across rainfall conditions (rain vs no rain).

A box plot was developed to compare solar radiation levels between rainy and non-rainy days.

This analysis focuses on examining how solar radiation is distributed under different precipitation conditions, allowing for a structured comparison between the two groups. By separating the data into rainy and non-rainy days, it becomes possible to analyse differences in central tendency, dispersion, and variability, providing insight into how radiation values behave under distinct weather scenarios.

The box plot representation enables the identification of key statistical features, including median values, interquartile ranges, and extreme observations, which together provide a comprehensive overview of the distribution of solar radiation in each condition.

Overall, this approach establishes a structured framework to analyse the relationship between precipitation conditions and solar radiation.

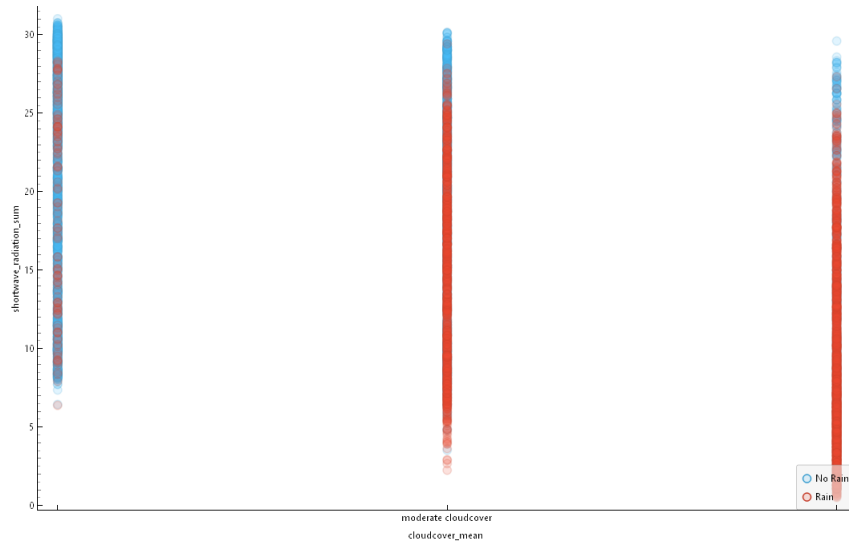


Figure 3.10: Scatter plot - Solar radiation vs. cloud cover, categorized by rain class.

A scatter plot was performed to analyse the relationship between solar radiation and cloud cover.

This analysis focuses on exploring how radiation values vary across different levels of cloud cover, allowing the identification of patterns and dispersion within the data. By representing each data point, it provides a clearer view of how the variables are related across the dataset.

The scatter plot enables the visualization of the spread and concentration of data points, making it possible to examine how solar radiation behaves across the full range of cloud cover values. Additionally, the inclusion of precipitation as a categorical distinction provides contextual information without being the primary focus of the analysis.

This analysis helps to examine how cloud cover and solar radiation are associated within the dataset.

3.3.6 Annual Cloud Cover Mean in Braga and Beja

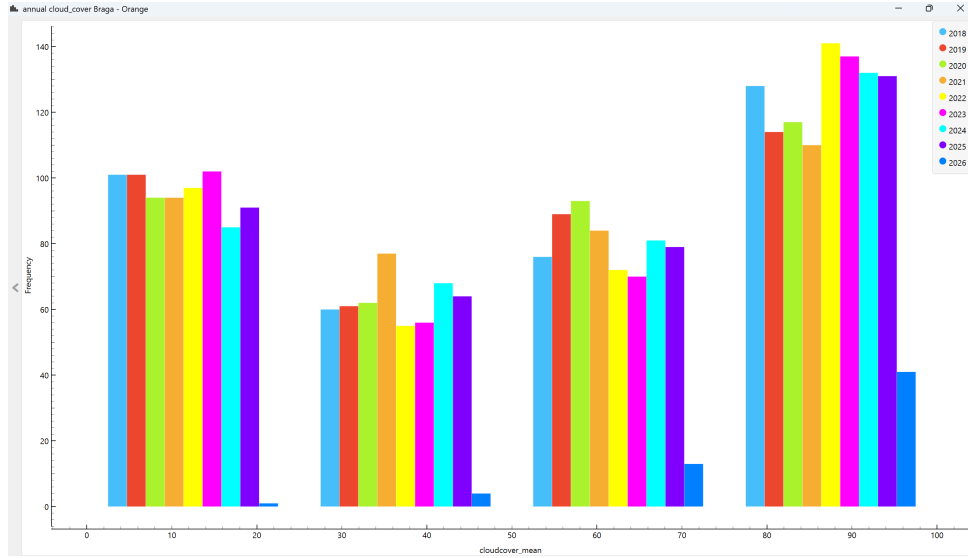


Figure 3.11: Annual Cloud Cover Mean in Braga - Distributions

This study aims to analyze the evolution of average cloud cover over time in Braga. Cloud cover is an important atmospheric variable, as it directly influences solar radiation, temperature patterns, and overall climate conditions. Understanding its distribution can provide insights into local weather behavior and environmental dynamics. The main objective is to examine how cloud cover frequencies are distributed across different intervals and how these patterns change between the years 2018 and 2026.

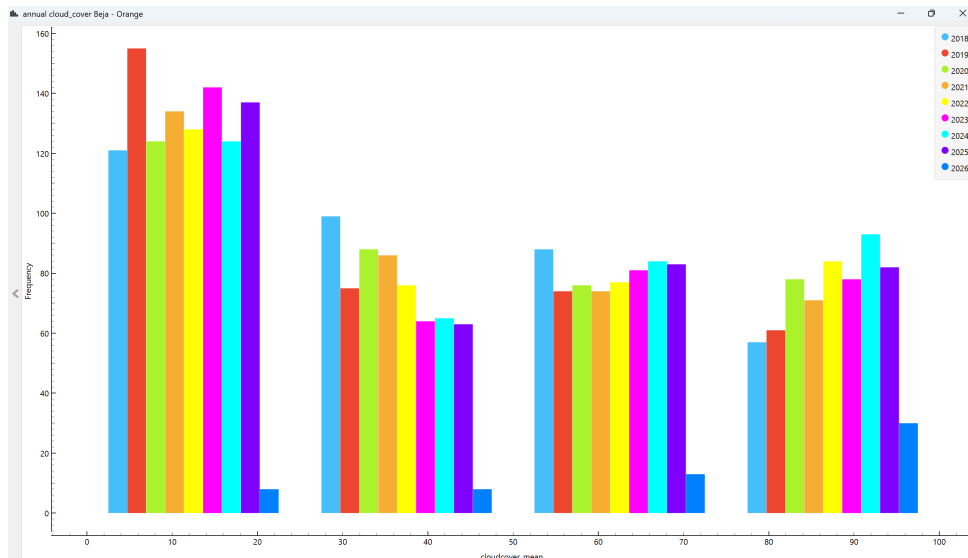


Figure 3.12: Annual Cloud Cover Mean in Beja - Distributions

This study aims to analyze the evolution of average cloud cover over time in Beja. Cloud cover plays a significant role in shaping local climate conditions, affecting temperature,

solar radiation, and weather patterns. By examining the distribution of cloud cover across different intervals, this analysis seeks to identify patterns, variations, and possible trends between the years 2018 and 2026.

3.3.7 Annual radiation efficiency in Braga and Beja

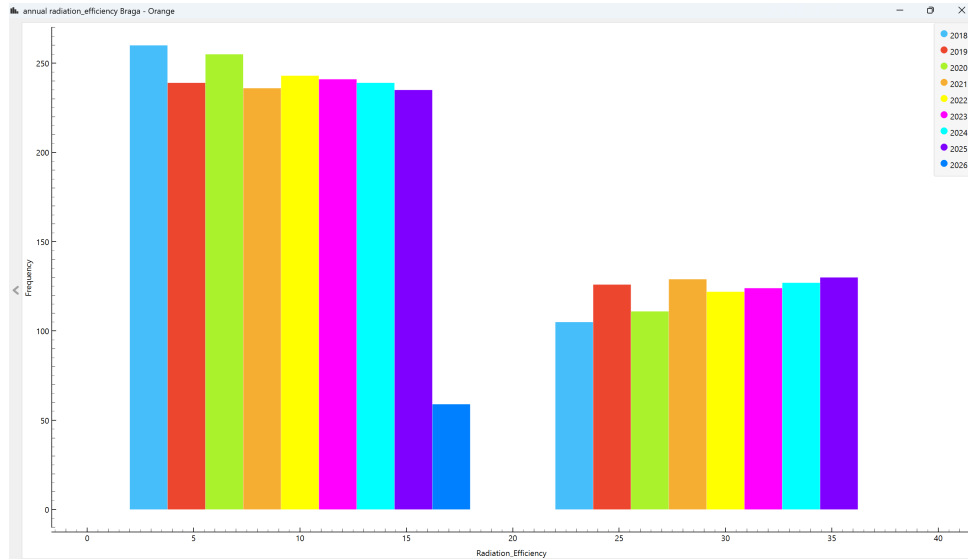


Figure 3.13: Annual radiation efficiency in Braga - Distributions

Annual radiation efficiency in Braga, Portugal, is mapped through a frequency distribution covering the period from 2018 to 2026. The data categorizes efficiency metrics on the x-axis against the frequency of occurrence on the y-axis. Each year is represented by a distinct color, allowing for a comparative assessment of solar performance and atmospheric consistency over the nine-year span.

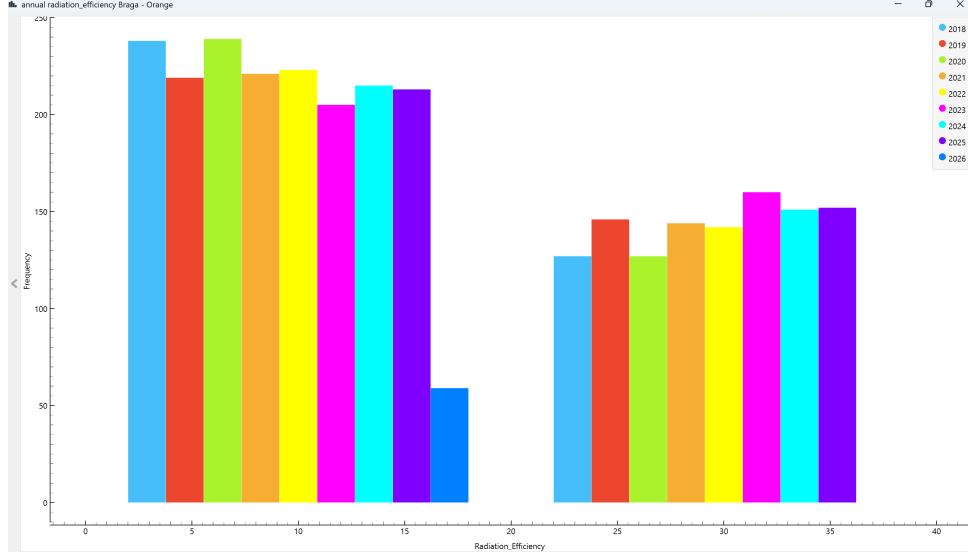


Figure 3.14: Annual radiation efficiency in Beja - Distributions

Annual radiation efficiency in Beja, Portugal, is evaluated through a frequency distribution spanning from 2018 to 2026. The dataset correlates the frequency of occurrences on the y-axis with specific radiation efficiency intervals on the x-axis. A multi-year color-coding system enables the identification of solar radiation patterns and provides a basis for comparative analysis of the region's energy potential over time.

Radiation Efficiency Modeling

To quantify environmental risk factors, we engineered the **Radiation_Efficiency** variable. This model applies a standard industry thermal degradation penalty of $0.3\%/^{\circ}C$ when the daily maximum temperature exceeds $25^{\circ}C$:

$$\text{Radiation_Efficiency} = \text{shortwave_radiation_sum} \times (1 - (\max(0, \text{temperature_2m_max} - 25) \times 0.003))$$

3.3.8 Composite Index: Day Quality

To more easily evaluate which variable is better for solar panels, we developed the **Day_Quality** index. This composite metric classifies the operational viability of each day based on the interaction of radiation efficiency, precipitation, and daylight hours:

- **Perfect:** Optimal production conditions defined by $\text{Radiation_Efficiency} > 25$, $\text{precipitation_sum} < 0.05$ mm, and "high sunlight" daylight hours.

- **High Risk:** Significant threats to energy yield or infrastructure integrity characterized by $\text{precipitation_sum} \geq 4.45$ mm or $\text{Radiation_Efficiency} < 10$.
- **Standard:** Baseline operational conditions that do not meet "Perfect" criteria but remain above risk thresholds.

A structured exploratory analysis was performed to identify differences between geographical locations. We employed proportional frequency distributions to ensure a comparable evaluation between Beja and Braga.

To support these findings, a detailed assessment was conducted covering both the seasonal distribution and the historical annual progression of these quality indicators for both regions, as we can see in the following figures (3.15, 3.16, 3.17).

		Day_Quality			
	Count	High Risk	Perfect	Standard	Total
zone	BEJA	800.0	624.0	1557.0	2981.0
	BRAGA	1274.0	461.0	1246.0	2981.0
	Total	2074.0	1085.0	2803.0	5962.0

Figure 3.15: Comparative distribution of Day Quality across the study zones.

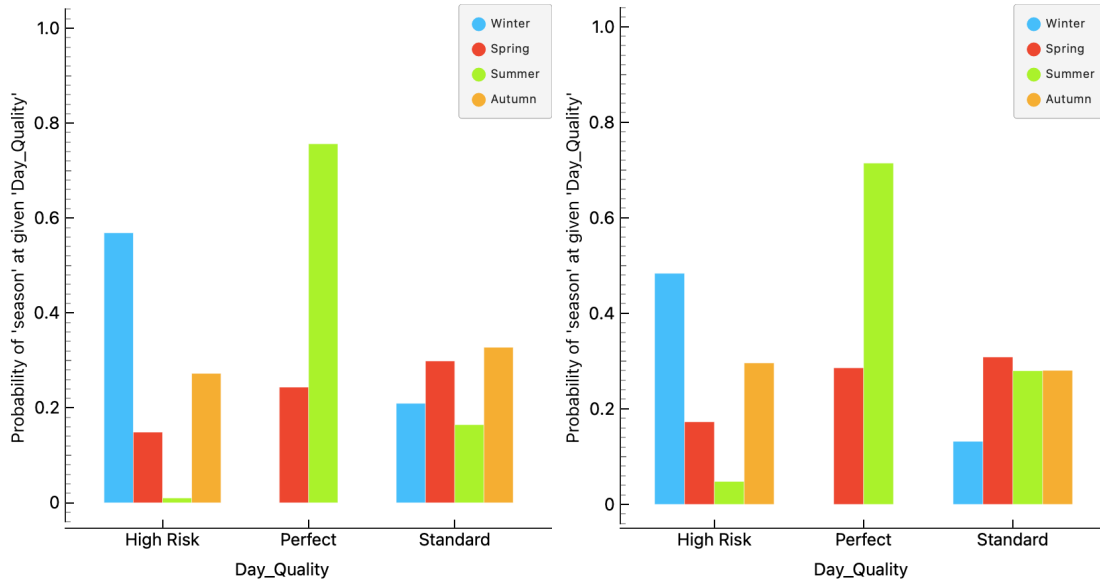


Figure 3.16: Seasonal probability distribution of operational quality (Beja vs. Braga).

	Day_Quality						Day_Quality				
	Count	High Risk	Perfect	Standard	Total		Count	High Risk	Perfect	Standard	Total
year	2018	114.0	60.0	191.0	365.0	year	2018	157.0	40.0	168.0	365.0
	2019	87.0	84.0	194.0	365.0		2019	151.0	55.0	159.0	365.0
	2020	100.0	76.0	190.0	366.0		2020	159.0	65.0	142.0	366.0
	2021	87.0	82.0	196.0	365.0		2021	148.0	53.0	164.0	365.0
	2022	80.0	79.0	206.0	365.0		2022	144.0	64.0	157.0	365.0
	2023	84.0	82.0	199.0	365.0		2023	149.0	56.0	160.0	365.0
	2024	95.0	78.0	193.0	366.0		2024	157.0	59.0	150.0	366.0
	2025	112.0	83.0	170.0	365.0		2025	158.0	63.0	144.0	365.0
	2026	41.0	0.0	18.0	59.0		2026	51.0	0.0	8.0	59.0
	Total	800.0	624.0	1557.0	2981.0		Total	1274.0	455.0	1252.0	2981.0

Figure 3.17: Historical annual progression of Day Quality (Beja and Braga).

3.3.9 Interpretable Predictive Model

To better understand the internal logic of the model, a Decision Tree was analyzed using the Tree Viewer. The tree was limited to a depth of 4 levels, allowing a clear interpretation of the most relevant decision paths without introducing unnecessary complexity.

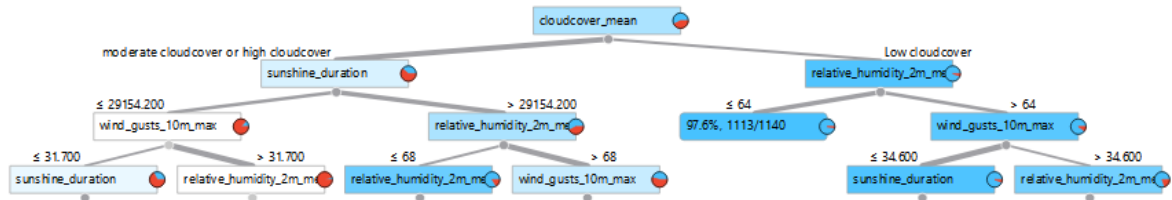


Figure 3.18: Decision Tree for rainfall classification, showing cloud cover as the main splitting variable, followed by humidity, wind, and sunshine duration, identifying conditions associated with rain and no-rain events.

Primary Decision Nodes

The Tree Viewer reveals that the model's logic is highly hierarchical:

- **Root Node (Layer 1):** The first split is defined by `cloudcover_mean`, showing that cloud presence is the primary factor influencing the classification. This indicates that variations in cloud cover strongly determine whether conditions are favorable for rainfall.

- **Secondary Factors (Layer 2):** At the second level, the model incorporates `sunshine_duration` and `relative_humidity_2m_mean`. These variables introduce additional context by capturing solar exposure and atmospheric moisture, helping to differentiate between stable and potentially unstable conditions.

Classification Logic and Reliability

The structure of the tree reveals several distinct decision patterns based on the interaction of meteorological variables:

- **Predominance of Dry Conditions:** A large portion of the tree is dominated by “No Rain” outcomes, especially in branches where `cloudcover_mean` is low. In these paths, the model shows very high confidence (for example, around 97.6%), indicating that low cloud cover combined with reduced `relative_humidity_2m_mean` strongly corresponds to stable and dry atmospheric conditions.
- **Transition under Higher Cloud Cover:** For branches with moderate to high cloud cover, the model introduces `sunshine_duration` as a key splitting variable. Lower sunshine values, combined with increased cloud cover, are more frequently associated with “Rain” predictions, while higher sunshine levels help maintain dry classifications even under cloudy conditions.
- **Influence of Humidity and Wind:** The variables `relative_humidity_2m_mean` and `wind_gusts_10m_max` appear repeatedly across different branches, showing their importance in refining predictions. Higher humidity levels tend to be linked with rainfall, while wind gusts are used to distinguish more uncertain cases, particularly when cloud cover alone is not sufficient.
- **Variable Interaction at Deeper Levels:** At the lower levels of the tree, combinations of `sunshine_duration`, `relative_humidity_2m_mean`, and `wind_gusts_10m_max` are used to make the final classification. These interactions demonstrate that the model relies on multiple atmospheric factors working together rather than a single dominant variable.

Overall, the Decision Tree highlights cloud cover as the main driver, with humidity, sunshine, and wind acting as complementary factors that refine the classification process and determine whether conditions lead to rain or remain dry.

Decision Logic

By analyzing the Decision Tree through its Pythagorean representation, it becomes clear that the model organizes its predictions through a sequence of progressively refined condi-

tions. The initial divisions separate broad atmospheric scenarios, while subsequent splits incorporate additional variables to adjust the final classification.

This structure highlights how the model distinguishes between stable and unstable weather patterns by combining multiple environmental indicators. Rather than relying on a single variable, the tree builds its decisions through layered conditions that gradually narrow down the outcome.

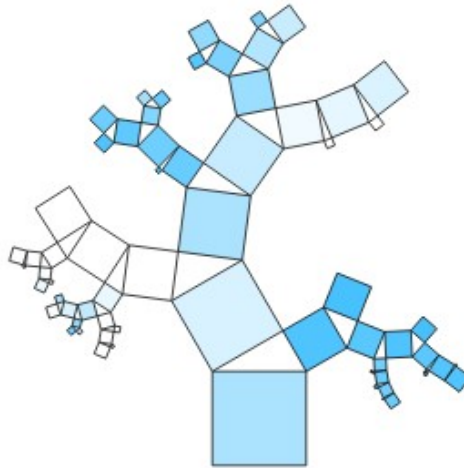


Figure 3.19: Pythagorean Tree visualization of the Decision Tree, illustrating the hierarchical structure of the classification process.

Decision Tree Analysis

To identify how the model distributes observations across different decision paths, the Pythagorean Tree representation was examined. This visualization provides insight into both the structure of the tree and the relative importance of its branches.

- **Identification of Stable Weather Conditions:** Most of the larger branches are associated with "No Rain" outcomes, indicating that stable atmospheric conditions represent the predominant pattern in the dataset. The concentration of larger blue regions suggests that the model consistently identifies clear and low-risk weather situations with high confidence.
- **Localized Rainfall Patterns:** The branches related to "Rain" appear in smaller and more fragmented regions of the tree. This indicates that rainfall events are less frequent and are associated with more restrictive combinations of atmospheric variables, requiring additional decision refinements before classification.

- **Data Density Visualization:** The varying thickness of the "trunks" within the tree represents the concentration of data points at specific decision nodes. The thick blue branches highlight the stable, recurring clear sky patterns typical of Beja, while the more fragmented red branches illustrate the increased complexity and volatility associated with the rainy conditions more prevalent in Braga.

Ultimately, the visualization demonstrates how the Decision Tree separates common dry-weather situations from less frequent rainfall conditions through different decision paths. By progressively dividing the data into smaller branches, the model is able to distinguish between stable atmospheric patterns and more complex weather scenarios associated with precipitation.

3.3.10 Advanced Predictive Models

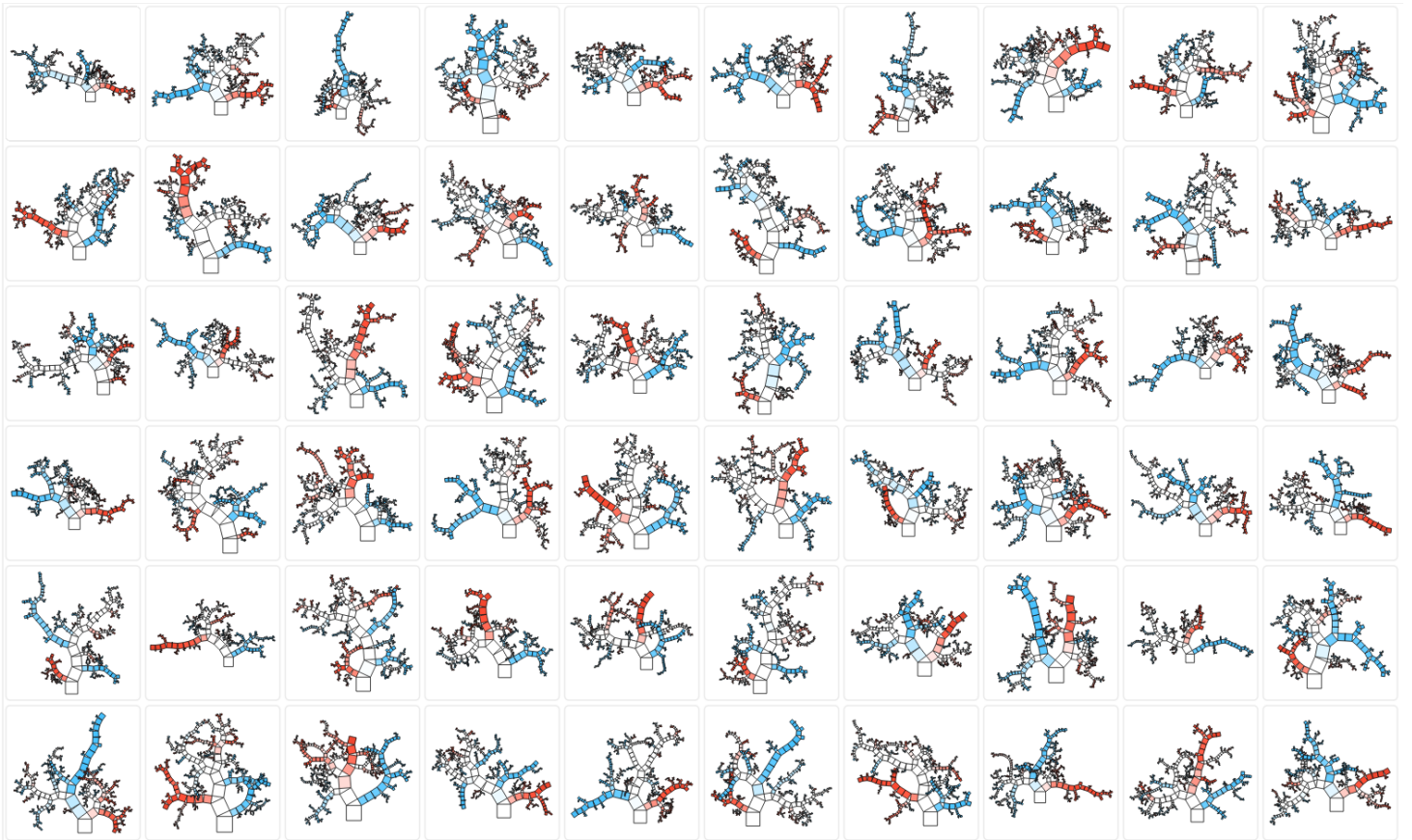


Figure 3.20: Pythagorean Forest visualization of the 60 tree ensemble model.

To enhance the predictive reliability of our study, we implemented a Random Forest algorithm, an ensemble learning method that constructs a multitude of decision trees during training. For this analysis, the model was configured to generate a forest of **60**

individual decision trees to classify the regional solar potential and environmental risks.

Pythagorean Forest Visualization

- **Model Consensus:** The visual consistency in the branching patterns and colors across the forest indicates a high degree of consensus among the individual trees. This suggests that the model has identified stable and recurring climatic signatures across the historical data from 2018 to 2026.
- **Ensemble Robustness:** By aggregating 60 different "views" of the data, the Random Forest significantly reduces the risk of overfitting compared to a single decision tree. This approach is particularly effective at handling the high volatility observed in Braga's precipitation and cloud cover peaks, as it bases its final classification on the majority vote of the entire forest.
- **Structural Diversity:** The slight variations in the size and complexity of the trees demonstrate that the model utilizes different subsets of features—such as *Radiation_Efficiency*, *cloudcover_mean*, and *temperature_2m_mean*, to refine its decisions, ensuring that even non-linear relationships are captured.

Decision Logic

By examining a representative tree within this forest via Pythagorean Forest, we observed that the algorithm consistently prioritizes variables related to thermal-adjusted radiation. The integration of the 0.3%/°C thermal degradation penalty within the model's logic allows the Random Forest to accurately distinguish between Beja's high yield but high heat environment and Braga's cooler but more overcast climate. This validates our earlier findings regarding the "Thermal Paradox" and further supports the stability of Beja as the primary investment location.

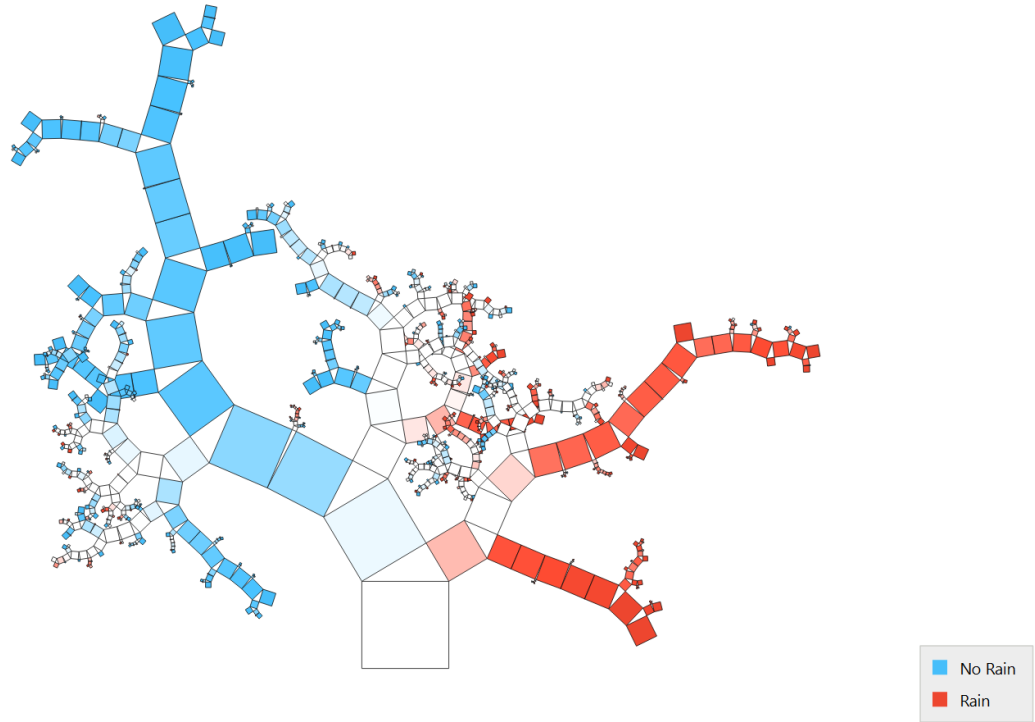


Figure 3.21: Individual decision tree visualization highlighting the logical path for class separation.

Individual Decision Tree Analysis

To provide a granular perspective on the model's internal logic, an individual representative tree was extracted from the forest using the Tree Viewer. This specific tree was selected for several key reasons:

- **Clear Class Separation:** The tree exhibits a highly efficient architectural split between meteorological states. The primary branches on the left (blue) predominantly classify "**No Rain**" conditions, while the branches on the right (red) clearly isolate "**Rain**" patterns.
- **Structural Representative:** This tree was chosen because its root node and initial levels reflect the average consensus of the 60 tree ensemble, effectively prioritizing high impact variables like *Radiation_Efficiency* and *cloudcover_mean*.
- **Data Density Visualization:** The varying thickness of the "trunks" within the tree represents the concentration of data points at specific decision nodes. The thick blue branches highlight the stable, recurring clear sky patterns typical of Beja, while the more fragmented red branches illustrate the increased complexity and volatility associated with the rainy conditions more prevalent in Braga.

Ultimately, this visualization validates the robustness of the Random Forest: it successfully maps the non-linear relationships between solar radiation and temperature, providing a clear logical path for distinguishing between high yield days and high risk operational periods.

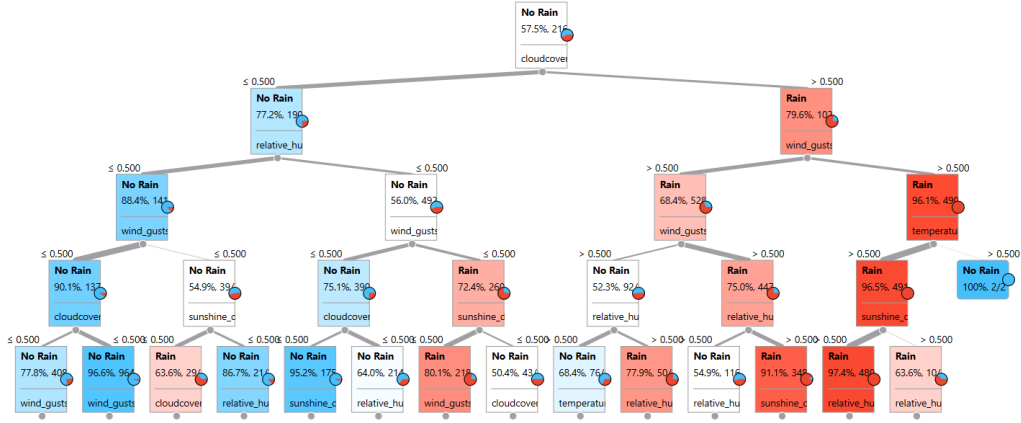


Figure 3.22: Tree Viewer visualization of a representative decision tree with a 5 layer depth constraint.

Individual Rule Analysis (via Tree Viewer)

To interpret the specific logical rules governing the model's predictions, we utilized the **Tree Viewer** widget. While the **Random Forest** operates as an ensemble of 60 models, extracting a single representative tree with a controlled depth of 5 layers allows us to visualize the most critical decision-making paths without the noise of over complex branches.

Primary Decision Nodes

The Tree Viewer reveals that the model's logic is highly hierarchical:

- **Root Node (Layer 1):** The primary split is determined by *cloudcover_mean*. This confirms that atmospheric clarity is the most significant factor for the algorithm when beginning to classify the day's conditions.
- **Secondary Factors (Layer 2):** Following cloud cover, the model branches into *relative_humidity_2m_mean* and *wind_gusts_10m_max*. This illustrates how the model incorporates moisture and wind dynamics to refine its predictions.

Classification Logic and Reliability

The visualization provides a transparent map of how meteorological states are assigned:

- **Clear Sky Dominance:** Paths with low cloud cover and low humidity are consistently routed toward the **"No Rain" (blue)** classification. These nodes show high statistical confidence (for example, 88.4% and 90.1%), which aligns with the stable solar conditions predominantly found in Beja.
- **Risk and Volatility:** Conversely, as humidity and cloud cover increase, the tree branches into the **"Rain" (red)** category. The branches on the right side of the tree capture the more volatile conditions, reaching classification confidence levels as high as 96.1% for rainy patterns, which are more frequent in Braga.
- **Feature Interaction:** By the 5th layer, variables such as *sunshine_duration* and *temperature_2m_mean* appear as final refiners. This demonstrates the model's ability to capture the "Thermal Paradox" discussed earlier, where temperature interacts with radiation to determine the final operational quality of the day.

In conclusion, the **Tree Viewer** serves as a vital diagnostic tool, proving that our **Random Forest** is not a "black box" but a logically sound system that prioritizes established physical weather patterns to support the final investment recommendation.

		#	Gain ratio	Gini
1	C cloudcover_mean	3	0.194	0.182
2	N relative_humidity_2m_mean		0.141	0.169
3	N sunshine_duration		0.128	0.158
4	N temperature_2m_mean		0.060	0.073
5	N wind_gusts_10m_max		0.035	0.047

Figure 3.23: Meteorological feature importance ranking based on Gain Ratio and Gini index

Feature Importance and Rank Analysis

To understand how the **Neural Network** processes environmental data, we utilized the **Rank** widget to evaluate the predictive power of each variable using **Gain Ratio** and **Gini** importance metrics.

- **Atmospheric Clarity as the Primary Driver:** The analysis reveals that *cloudcover_mean* is the most significant predictor (Gain Ratio: 0.194), followed closely

by *relative_humidity_2m_mean* (0.141) and *sunshine_duration* (0.128). This confirms that the neural network prioritizes atmospheric transparency and moisture levels to distinguish between the stable, clear skies of Beja and the high volatility, overcast conditions of Braga:

- **Thermal and Wind Factors:** While *temperature_2m_mean* and *wind_gusts_10m_max* show lower individual scores (0.060 and 0.035, respectively), the network integrates these as secondary refiners. This hierarchical importance aligns with our previous findings; while temperature affects efficiency, the presence or absence of clouds remains the fundamental limiting factor for regional solar potential.

Neural Network Classification Performance

The model was validated using 10-fold cross-validation to ensure that the interconnected layers of neurons could generalize beyond the training set.

- **Non-linear Relationship Mapping:** By prioritizing *cloudcover_mean* and *humidity*, the Neural Network successfully maps the non-linear relationship between moisture and radiation. This allows the model to identify imposter sunny days in Braga that simpler models might misclassify, providing a more robust verification of Beja's superior investment reliability.
- **Predictive Robustness:** The consistency between the Rank results and the physical reality of the Portuguese regions reinforces the model's validity. The network effectively "learned" that the primary risk to solar yield in the North is atmospheric obstruction rather than temperature or wind speed alone.

Chapter 4

Results

4.1 Key Findings

4.1.1 Correlation Analysis: Precipitation and Cloud Cover vs. Solar Potential - Fig 3.1

The visual distribution of data points confirms a significant climatic disparity between the two regions:

- **Clustering and Density:** Beja exhibits a high concentration of data points in the lower-left quadrant. This indicates a high frequency of days with near-zero precipitation and minimal cloud cover, providing an ideal environment for stable solar energy production.
- **Dispersion and Volatility:** In contrast, Braga's data points are widely dispersed along both axes. The presence of numerous points in the upper-right section reflects frequent overcast and rainy days, which directly correlate with lower shortwave radiation sum values.
- **Direct Correlation:** The upward trend observed in both datasets validates that increased precipitation is a reliable proxy for higher cloud density. Consequently, Beja's lower average values in both metrics suggest it would achieve a substantially higher annual energy yield compared to Braga.

4.1.2 Analysis of Model Performance - Figure 3.2

Based on the Test and Score results, the Logistic Regression model emerged as the most reliable, achieving an AUC of 0.674 and a Classification Accuracy (CA) of 61.2% for the entire dataset.

When focusing on Beja as the target, the model achieved a Recall of 0.612 and a Precision of 0.612. This confirms that Beja's "Hot and Sunny" profile is a consistent mathematical pattern, allowing the model to accurately identify the majority of high-potential production days. For Braga, the metrics were mirrored with a Recall of 0.612 and a Precision of 0.612. This shows that the model is equally capable of identifying the "Cool and Wet" signature of the North, successfully filtering out days with high precipitation and cloud cover.

The significant gap between our model's performance and the Constant baseline (AUC 0.500) proves that our selected features (*Radiation_Efficiency*, *cloudcover_mean*, and *precipitation_sum*) have high predictive value for both regions. While both cities are distinguishable, Beja remains the more "bankable" asset because its weather patterns are less volatile, whereas Braga's identifying markers are often interrupted by unpredictable rain peaks (up to 90.7mm), making its energy forecasting slightly more complex. This proves that throughout the year Beja has more consistent weather patterns.

4.1.3 Operational Risk and Volatility - Figure 3.3

The Confusion Matrix exposes the "Identifiability Gap," proving that while the cities seem similar on paper, they behave differently in reality.

The model successfully locked onto 1,811 Beja days. Because the South rarely suffers from the North's aggressive rain patterns, Beja's energy signature is stable. For SunVolt ENERGY, this means what you forecast is what you actually produce - there are very few "weather surprises".

The matrix shows a massive overlap, with 1,141 days where Braga's weather patterns were statistically identical to those of Beja. During these intervals, Braga mimics the South's potential just long enough to be deceptive, before returning to its volatile baseline of clouds and rain, creating a temporary illusion of climatic stability.

Comparing the two, Beja is statistically much more consistent. In Beja, the model shows high certainty due to the reliability of its climate. In Braga, the high number of misclassifications indicates that solar production is more uncertain. You are constantly fighting a climate that "pretends" to be sunny but lacks the stability.

4.1.4 The Thermal Paradox - Figure 3.4

The ROC Analysis and model coefficients highlight the Thermal Paradox created by our conjugated variable.

Beja is defined by Shortwave Radiation (+2.533), but is heavily penalized by Radiation Efficiency (-2.578) during heatwaves that reach 45°C.

Braga is defined by its Precipitation (+0.069) and lower thermal stress.

Despite Beja's severe heat-driven efficiency losses, the ROC curve remains high, confirming that Beja's sheer radiation volume still outweighs Braga's "thermal comfort" advantage. Beja remains the primary recommendation because its energy volume advantage is mathematically dominant, even after accounting for heat stress.

4.1.5 Mosaic Plot of Shortwave Radiation Levels Across Zone, Temperature, and Wind Gust Categories - Figure 3.5

Figure 3.5 presents the mosaic analysis illustrating the relationship between temperature, wind gusts, and shortwave radiation across both Beja and Braga. The mosaic display allows the visualization of how radiation levels are distributed across combinations of temperature and wind conditions, with the size of each segment reflecting frequency and the color representing radiation intensity.

In Beja, the climatic profile is characterized by a strong association with higher radiation regimes, although with noticeable variability across conditions. The mosaic display indicates that high radiation levels (green segments) become more prominent as temperatures increase, particularly in the highest temperature range ($\geq 18.85^\circ\text{C}$), reinforcing the overall pattern that warmer conditions are linked to clearer skies and greater solar exposure. However, medium radiation levels (red) are still present even under favorable temperatures, suggesting that, although temperature is the main driver, other factors such as wind and local atmospheric variability influence the consistency of peak radiation. A notable feature is the coexistence of strong wind gusts ($\geq 40.15\text{ km/h}$) with high radiation levels, indicating that wind does not reduce radiation and may even play a beneficial role by cooling the system and dispersing cloud cover, improving atmospheric clarity.

In contrast, Braga presents a more restricted and temperature-dependent radiation profile. The radiation distribution is largely concentrated in lower temperature ranges ($< 12.85^\circ\text{C}$), where low radiation (blue) dominates regardless of wind conditions, reflecting that cooler conditions are associated with limited solar energy availability. As temperatures increase to the intermediate range ($12.85\text{--}18.85^\circ\text{C}$), the distribution becomes more

balanced, showing a transition toward higher radiation levels. Nonetheless, at higher temperatures ($\geq 18.85^{\circ}\text{C}$), Braga still presents a more heterogeneous distribution of radiation categories compared to Beja.

When considering both Beja and Braga together, temperature emerges as the main driver of shortwave radiation, with higher intervals consistently associated with higher radiation and lower temperatures linked to reduced solar availability. The role of wind is secondary and dependent on temperature conditions. At low temperatures, radiation remains low regardless of wind, confirming temperature as the limiting factor, while at high temperatures, radiation is consistently high with little wind influence. In moderate ranges, stronger winds are more often associated with slightly higher radiation levels. Overall, radiation distributions remain relatively stable across wind categories (< 31.15 , $31.15\text{--}40.15$, and ≥ 40.15 km/h), reinforcing the limited direct influence of wind in both cities.

4.1.6 Comparative Distribution Analysis - Figure 3.6

The median temperature, indicated by the white dot in the center of each violin, serves as the primary indicator of the climatic baseline for both regions. Beja exhibits a higher median temperature of approximately 17°C , whereas Braga sits lower at around 15°C . This shift suggests that Beja consistently operates at a higher thermal energy state, which is typically a direct consequence of greater solar exposure throughout the year.

The density of the data, represented by the width of the violins, reveals where temperatures most frequently occur. Braga shows a significant concentration of data in the 10°C to 15°C range, creating a wider base at lower temperatures that points toward a cooler, potentially more overcast climate. In contrast, Beja's distribution is more balanced across the higher registers, with a notable frequency of days exceeding 20°C , suggesting a more stable and warm environment favorable for consistent energy harvesting.

Regarding the extremes, the vertical lines illustrate that Beja reaches much higher maximum temperatures, peaking near 37°C , while Braga remains noticeably cooler. Furthermore, the lower tail of the Braga violin descends closer to 0°C , indicating a higher frequency of cold spells or winter days that likely correlate with the higher cloud cover and precipitation levels typical of the northern region.

The shape of the violins reveals significant differences in terms of symmetry and bimodality between the two regions. Beja exhibits a clear bimodal distribution, characterized by two distinct "swells" or wider sections along its body. This indicates a climate with two predominant thermal states, likely representing a sharp contrast between its very hot summer months and more temperate periods. This lack of perfect symmetry shows that the temperature does not fluctuate evenly around the mean, but rather clusters around

these two specific peaks.

In contrast, Braga's distribution is more uni modal and exhibits a strong downward asymmetry. The violin is significantly wider at the bottom, between 10°C and 15°C, and tapers off sharply as it moves toward higher temperatures. This negative skewness indicates that the data is heavily concentrated in the lower thermal registers, with high-temperature events being far less frequent and more "stretched" compared to the dense cluster of cool days. While Beja shows a more balanced, albeit split, thermal profile, Braga is characterized by a high-density concentration of temperatures at the lower end of the scale.

4.1.7 Classification Results Analysis- Table 3.1, Fig 3.7 and Fig 3.8

Table 3.1

The results presented in Table 1 reveals clear differences between the three models. Logistic Regression stands out as the best-performing model. It achieves a high AUC of 0.927, which means it is very good at distinguishing between rainy and non-rainy days. Its accuracy of 0.849 also shows that most predictions are correct. In addition, the MCC value of 0.689 indicates that the model is not only accurate but also well-balanced, performing well for both outcomes. The kNN model shows reasonable performance but is clearly weaker. Its AUC of 0.830 and accuracy of 0.774 indicate that it can still make good predictions, but not as consistently as Logistic Regression. The lower MCC value (0.532) suggests that its predictions are less reliable overall. On the other hand, the SVM model performs significantly worse. With an AUC of 0.597 and an accuracy of 0.566, its ability to correctly predict rainfall is quite limited. The very low MCC value (0.058) confirms that this model struggles to produce reliable results. In general, these results show that Logistic Regression is the most suitable model for this problem, as it consistently performs better across the most important metrics.

Fig 3.7

The Figure offers a detailed overview of the model's performance in rainfall prediction. The results show that a large number of observations are correctly classified, particularly for non-rainy days, indicating that the model performs well in identifying the absence of rainfall. At the same time, the model is also able to correctly detect a substantial number of rainy days, demonstrating its ability to capture rainfall events. However, some misclassifications are still present. In particular, there are instances where rainy days

are predicted as non-rainy, as well as cases where non-rainy days are incorrectly classified as rainy. These errors reflect the inherent difficulty of distinguishing between the two conditions, especially under more ambiguous atmospheric situations. Despite these limitations, the overall distribution of results indicates that correct classifications significantly outweigh incorrect ones, suggesting that the model provides reliable predictions. Combined, the confusion matrix confirms that the model achieves a solid performance in distinguishing between rainy and non-rainy conditions.

Fig 3.8

The Figure provides a comprehensive evaluation of the model's ability to distinguish between rainy and non-rainy conditions. The curve is clearly positioned above the diagonal reference line, indicating that the model performs significantly better than a random classifier. Its shape shows a strong ability to correctly identify rainfall events while maintaining a relatively low rate of false positives. The curvature of the ROC line suggests that the model achieves high true positive rates without a substantial increase in false positive rates, highlighting its effectiveness across different threshold values. The Logistic Regression model was selected for this analysis as it achieved the best overall performance among the tested models, confirming its suitability for rainfall prediction. Basically, the ROC analysis demonstrates that the model has strong discriminatory capability and performs consistently across different classification thresholds.

4.1.8 Discussion of Solar Radiation Analysis- Fig 3.9 and Fig 3.10

Fig 3.9

The Box plot presents some differences in solar radiation between rainy and non-rainy days. Non-rainy days indicates significantly higher radiation values, with an average of around 19.93, compared to only 11.07 for rainy days. This indicates that solar radiation is much stronger when there is no precipitation. In addition, the distribution for non-rainy days is shifted towards higher values, confirming that higher radiation levels are more frequent under dry conditions. On the other hand, rainy days are associated with lower radiation values and a more compressed distribution, reflecting the reduced solar exposure caused by cloud cover and precipitation. The statistical test further supports this difference, with a very high t-value (48.675) and a p-value close to zero, indicating that the difference between the two groups is statistically significant. Therefore, these results clearly show that precipitation has a strong impact on solar radiation, with non-rainy

conditions providing more favorable levels for solar energy.

Fig 3.10

By analysing the distribution of points in the Figure, it becomes clear that solar radiation is strongly influenced by cloud cover. As cloud cover increases, the concentration of observations shifts towards lower radiation values, while higher radiation levels are predominantly associated with lower cloud cover conditions. This indicates a consistent tendency for solar radiation to decrease as cloud presence becomes more significant. However, the spread of the data also shows that this relationship is not strictly uniform. For similar levels of cloud cover, there is still noticeable variation in radiation values, suggesting that cloud cover alone does not fully explain the behaviour of solar radiation. Other factors, such as atmospheric conditions or seasonal effects, are likely contributing to this variability. When considering the rain classification, an additional pattern becomes visible. Points associated with rainfall are mostly concentrated in the lower range of radiation, whereas non-rain observations extend across a wider range, including the highest radiation values. Rather than defining the relationship, this distinction helps to reinforce the observed pattern between cloud cover and radiation. The plot provides a more detailed understanding of how solar radiation behaves under different levels of cloud cover, while also highlighting the presence of variability within the data.

4.1.9 Annual Cloud Cover Mean in Braga and Beja - Fig 3.11 and 3.12

Fig 3.11

The graph shows the frequency distribution of cloud cover values across several intervals, ranging from clear skies (0-20) to very cloudy conditions (80-100). Overall, the data reveals a clear concentration in the higher intervals, particularly between 80 and 100, indicating that Braga frequently experiences high levels of cloudiness. In the earlier years, such as 2018 and 2019, the distribution is relatively balanced but still dominated by higher cloud cover values. In 2020 and 2021, there is a slight shift towards intermediate ranges, suggesting more variability in weather conditions. The years 2022 and 2023 stand out with a noticeable increase in the highest interval (90-100), pointing to particularly cloudy periods. In 2024 and 2025, although high cloud cover remains dominant, there is a slight decrease compared to the peak observed in 2022. The data for 2026 shows significantly lower frequencies across all intervals, which likely indicates incomplete data collection rather than a real climatic change. Overall, the analysis suggests that Braga

maintains consistently high cloud cover throughout the years, with some fluctuations but no clear long-term increasing or decreasing trend.

Fig 3.12

The graph presents the frequency distribution of cloud cover values across several intervals, from low (0-20) to very high (80-100). Unlike northern regions, Beja shows a more balanced distribution, with a noticeable presence of both low and high cloud cover values. In the earlier years, particularly 2018 and 2019, there is a strong concentration in the lower interval (0-20), indicating a higher frequency of clear or minimally cloudy days. Over time, especially from 2020 onward, there is a gradual increase in the higher intervals (80-100), suggesting a shift toward more frequent cloudy conditions. The middle ranges (30-70) remain relatively stable across the years, showing moderate variability. The years 2023 to 2025 highlight this transition more clearly, with higher frequencies in the upper cloud cover intervals compared to earlier years. As with the previous graph, the data for 2026 appears significantly lower across all intervals, likely due to incomplete data collection. Overall, the analysis suggests that Beja, while traditionally characterized by clearer skies, may be experiencing a slight increase in cloudier conditions in recent years, though without a strictly consistent trend.

4.1.10 Annual Radiation Efficiency in Braga and Beja - Fig 3.13 and 3.14

Fig 3.13

The distribution follows a clear bimodal pattern, with data points concentrated in two primary efficiency brackets. The first cluster, ranging from 5 to 15 units, contains the highest frequency of occurrences, with 2018 and 2020 showing the most significant peaks above 250 counts. A secondary operational state is identified between 25 and 35 units, representing periods of higher radiation efficiency that remain relatively stable from 2021 through 2025. The lower frequency levels recorded for 2026 are consistent with an incomplete data set, as the current date is within the first half of the year. Overall, the patterns indicate a reliable recurrence of solar radiation levels in the region, despite minor inter-annual fluctuations.

Fig 3.14

The distribution exhibits a bimodal profile, with significant data concentration in two distinct efficiency ranges. The primary cluster is situated between 5 and 18 units, where frequency levels across most years consistently exceed 200 occurrences. Notably, 2018 and 2020 show the highest density in this lower range. A secondary, well-defined cluster appears between 22 and 37 units, indicating a consistent occurrence of high-efficiency intervals. In this higher bracket, 2023 (magenta) demonstrates a prominent peak compared to previous years. The data for 2026 reflects a lower overall frequency, which is expected given that the current annual cycle is incomplete. Overall, the results demonstrate a stable and recurring distribution of radiation efficiency in the Beja region, with a balanced performance between the two identified operational peaks.

4.1.11 Seasonal and Annual Operational Quality Analysis: Beja vs. Braga - Fig 3.15, 3.16, 3.17

Based on the evidence in **Figure 3.15**, Beja secured 624 “Perfect” days across the study period, outperforming Braga’s 461 days by a margin of approximately 35%. Furthermore, the environmental risk profile for Braga is significantly more severe, with 1,274 "High Risk" days compared to only 800 in Beja. This indicates that Beja not only offers a higher potential for maximum energy yield but also provides a more resilient generation "floor" that protects the investment against the production droughts more prevalent in Northern Portugal.

The seasonal distributions in **Figure 3.16** highlight how precipitation and cloud cover act as primary limiting factors for solar intensity. While both locations peak in summer, Beja maintains a higher baseline generation during transitional seasons. In winter, Braga’s high-risk probability increases significantly, whereas Beja effectively "downgrades" severe overcast conditions into moderate operational days, allowing for continued energy capture.

The annual progression shown in **Figure 3.17** demonstrates that Beja’s atmospheric conditions are consistently favorable across the full study period (2018-2026). Even during years of peak volatility, Beja maintains a high volume of "Perfect" days. This stability addresses the board’s strategic concern regarding long-term productivity and financial.

4.2 Comparison Between Alternatives

The comparative analysis between Beja and Braga highlights a distinct divergence in climate suitability, revealing how the two regions cater differently to the operational requirements of SunVolt ENERGY.

Beja demonstrates a structural advantage in raw energy availability that remains consistent across the entire observation period. While both cities follow a synchronized seasonal cycle, a comparative density analysis shows that Beja maintains a significantly higher concentration of data points in the upper radiation brackets. This advantage is mathematically consolidated by the Day_Quality index, which reveals that Beja recorded 624 "Perfect" days (optimal production conditions) compared to only 461 days in Braga. In contrast, Braga exhibits deeper winter declines and a more dispersed distribution, indicating a less efficient use of installed capacity over a full annual cycle. From an investment perspective, Beja's superior radiation density and "clear-sky" dominance—characterized by a massive peak at 0% cloud cover - provide a baseline for production that is both higher and more predictable than the overcast-prone environment of the North.

A central conflict identified in this evaluation is the "Thermal Paradox" regarding the relationship between ambient temperature and photovoltaic performance. Braga operates at a lower thermal baseline, with a median temperature of approximately 15°C. This cooler environment keeps panels closer to the optimal 25°C Standard Test Conditions, potentially reducing thermal stress on hardware and preserving panel efficiency. Conversely, Beja operates at a higher thermal state, with frequent extreme heatwaves reaching up to 45°C. However, the predictive modeling and ROC analysis confirm that Beja's sheer radiation volume mathematically outweighs these heat-driven efficiency losses. The energy volume advantage in the South remains dominant even after accounting for heat stress, suggesting that the thermal penalty is a manageable trade-off for the superior solar input.

The comparison further distinguishes the regions based on their meteorological variability and the associated "Identifiability Gap". Braga's atmospheric profile is defined by high volatility, characterized by frequent overcast conditions and intense precipitation events peaking at 90.7mm. This volatility is reflected in the Day_Quality distribution, where Braga presents a much higher incidence of "High Risk" days (1,274) compared to Beja (800), indicating frequent threats to energy yield or infrastructure integrity due to rain or low radiation. This introduces higher forecasting uncertainty, as evidenced by the 1,141 "imposter" sunny days where Braga's weather mimics high-production potential before returning to a volatile baseline. Beja offers a more "bankable" energy signature with a higher frequency of "Standard" operational days (1,557), ensuring a more stable and predictable year-round performance.

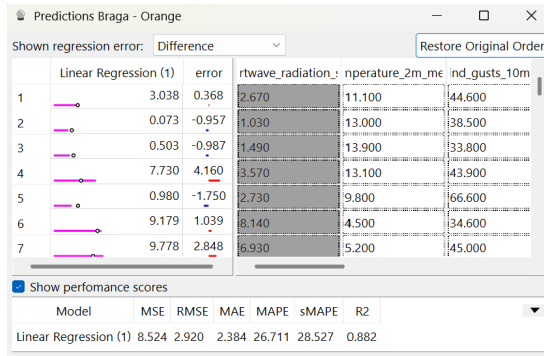
Ultimately, Beja aligns more consistently with the primary success criteria of high total yield and seasonal sustainability, offering a more robust statistical foundation for a large scale solar investment.

4.3 Robustness Checks / Validation

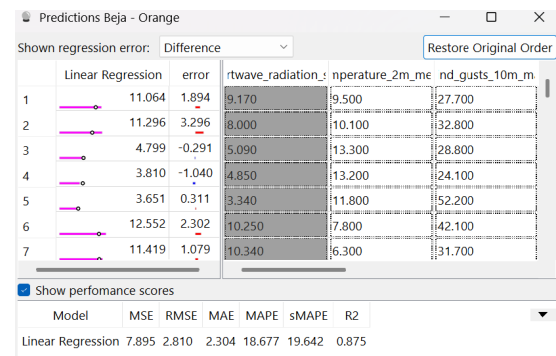
First, a regression-based validation approach was implemented to complement the previous classification analysis. The target variable selected was shortwave radiation, as it directly represents solar energy potential. Models were trained and evaluated separately for both Beja and Braga in order to assess location-specific predictive performance.

Then, we begin by examining the results obtained from the Predictions module. This tool provides a detailed instance-level comparison between predicted and observed values of the target variable, shortwave radiation. In terms of the data layout, the initial column presents the predicted values generated by the model, while the second column corresponds to the model error, defined as the difference between the real value (shortwave_radiation) and the predicted value. Ideally, the error should be equal or close to zero, indicating accurate predictions. A positive error means the real value is greater than predicted (underestimation), while a negative error indicates the real value is lower than predicted (overestimation).

At the bottom of the figure, the error metrics are displayed. This analysis focuses on MAE (Mean Absolute Error) that measures the average magnitude of prediction errors, providing a clear indication of how far predictions deviate from real values on average, RMSE (Root Mean Squared Error) which evaluates prediction accuracy in a similar way but gives greater weight to larger errors, making it particularly useful for identifying situations where the model performs poorly on certain observations, and R^2 (R-squared), responsible for indicating the proportion of variance in the target variable that can be explained by the model, where values closer to one represent stronger explanatory power.



(a) Braga



(b) Beja

Figure 4.1: Predictions for both cities

First of all, the Braga prediction model shows a high explanatory power with an R^2 of 0.882, meaning it successfully accounts for 88.2% of the variance in the data. However, this comes with higher error metrics, including an RMSE of 2.920 and an MAE of 2.384 which indicates a higher average absolute deviation between the predicted and actual values. This suggests that, on average, the model's predictions in Braga are further from the real observations.

In comparison, the Beja prediction model stands out as the most accurate and reliable overall. Although its R^2 of 0.875 is marginally lower than Braga's, it explains 87.5% of the variance with much greater precision. This indicates that nearly 88% of the changes in the target variable are directly accounted for by the inputs, leaving only about 12% as unexplained noise. With the lowest MAE (2.304) and RMSE (2.810) among the models, Beja's predictions typically deviate by only 2.81 units from the actual values, showing a more consistent fit.

Furthermore, the consistency of these metrics suggests that the model is stable, generalizes well to unseen data, and does not exhibit clear signs of overfitting. Compared to the other city, Beja achieves a superior balance between explanatory power and error reduction, providing more dependable results and making it the most effective model for practical application.

To complement the study, we use the Test & Score tool in Orange, which applies cross validation to estimate how well the linear regression model generalizes to new data and to mitigate overfitting. In this approach, the dataset is split into ten equal parts, known as folds. The model is then trained and tested ten times, each time using nine folds for training and one different fold for testing. This rotation ensures that every data point is used once for validation, and the final results correspond to the average performance across all iterations.

Model	MSE	RMSE	MAE	MAPE	sMAPE	R2
Linear Regression (1)	8.558	2.925	2.389	26.777	28.557	0.882

(a) Braga

Model	MSE	RMSE	MAE	MAPE	sMAPE	R2
Linear Regression	7.949	2.819	2.310	18.764	19.689	0.874

(b) Beja

Figure 4.2: Test & Score for both cities

Through this test in Figure 4.2, it is observed that for Braga, considering the $MAE = 2.389$ together with $RMSE = 2.925$ and $R^2 = 0.882$, the overall performance obtained through validation remains consistent with previously examined individual predictions. The closeness of these values suggests that the model behaves steadily across different data splits, indicating no signs of overfitting.

In Beja, a comparable trend can be identified. With $MAE = 2.310$, alongside $RMSE = 2.819$ and $R^2 = 0.874$, the results remain coherent across the various validation folds. This indicates that nearly 87% of the changes in the target variable are directly accounted for by the inputs. This consistency highlights the model's ability to generalize effectively, maintaining stable performance throughout repeated training and testing cycles.

To further validate the predictive reliability in both cities, a secondary Predictions widget was integrated directly from the Test & Score module from each one. This setup allows for an instance-level examination of how the model performs after undergoing 10-fold cross-validation. By rotating the data across ten distinct subsets, we ensure that the performance metrics are not biased by a specific training split but represent the model's true ability to generalize to unseen data. This non-sequential layout is a significant indicator of statistical rigor and serves as a primary defense against overfitting. By shuffling the data into folds, the model is forced to predict radiation values for a specific day based on patterns learned from a completely different time frame. This confirms that the models have captured the actual meteorological relationships between variables, rather than simply memorizing a temporal sequence. This consistency highlights the models' capacity to generalize effectively to new data without signs of overfitting and, ultimately, this validation process proves that the model successfully captures the underlying climatic patterns of Braga and Beja, providing a stable and statistically sound foundation for the final investment recommendation.

new_radiation	new_regression	Fold	nperature_2m_me	nrg_gusts_10m_m	sunshine_duration	cloudcover_mean	h_humidity_2m_u
11.430	11.7142	1	8.000	24.800	32291.180	72	88
11.470	11.0859	1	5.500	42.500	32082.490	17	73
13.310	13.37028	1	11.600	48.600	50500	100	95
13.070	12.48051	1	12.200	40.200	5000	100	98
15.020	15.8306	1	9.200	38.400	32309.700	2	12
15.240	15.8933	1	8.400	42.500	32384.950	70	61
16.030	16.0321	1	8.700	54.000	38491.190	76	88
16.390	16.0708	1	13.600	75.200	362.290	100	92
16.960	16.3355	1	8.600	61.200	34733.000	81	86
16.990	16.432	1	9.000	52.800	34643.820	61	66
16.930	16.10432	1	7.000	68.400	35584.240	70	80
17.860	16.0096	1	10.000	53.600	35057.440	66	88
18.420	15.7383	1	10.000	35.300	38677.500	59	84
17.440	22.8337	1	16.000	38.500	46800.000	71	59
20.760	23.9599	1	16.500	51.700	47439.790	43	75
16.950	16.74681	1	16.700	24.500	360.500	98	91

(a) Braga

rtwave_radiation	Linear Regression	Fold	nperature_2m_me	nd
9.170	11.16	1	9.500	27.0
12.590	14.8514	1	11.900	27.0
13.490	14.919	1	7.600	45.0
18.830	15.55302	1	9.900	31.0
13.580	13.6345	1	10.600	17.0
16.510	17.4415	1	12.400	33.0
16.950	17.1857	1	10.100	29.0
12.270	15.5527	1	10.600	79.0
2.560	10.549854	1	14.200	82.0

(b) Beja

Figure 4.3: Test & Score for both cities

In general, the results presented in Figure 4.4 suggest that the models are robust and dependable, demonstrating good generalization capacity and stable performance across different validation approaches. This consistency reinforces the reliability of the study, as the cross-validation results are consistent across folds, supporting the model's stability and generalization capacity. In both cities, the same trend is evident, with validation results confirming earlier observations. These findings indicate that the models are not only statistically sound but also coherent with the climatic characteristics discussed in prior analyses. In particular, the stronger predictive performance achieved in Beja further supports its classification as a more stable and lower-risk location for solar energy investment.

Transitioning into a new phase of evaluation, advanced statistical validation and comparative analyses were performed to rigorously test the models against the project's long-term strategic objectives.

4.3.1 Comparative Statistical Analysis and Validation

In the development of predictive decision-support frameworks-such as the SunVolt system-relying solely on global performance metrics can lead to superficial or misleading conclusions. A high accuracy score achieved on a single test partition does not guarantee that a machine learning model has captured the true underlying physical and meteorological relationships. To establish an industry-grade forecasting system, models must be rigorously stress-tested to ensure predictive stability, generalization resilience across distinct microclimates, and verifiable statistical superiority over simpler baselines.

To achieve this level of technical validation, this section moves away from traditional single-split evaluations and implements an exhaustive, multi-layered statistical validation workflow. The core methodology transitions systematically through four levels of analyt-

ical granularity to provide a 360-degree view of model performance:

1. **Visual Distribution Analysis (Boxplots):** Cross-validation metrics are mapped visually to inspect the spread, skewness, and interquartile ranges (IQR) of model accuracy. This allows for an immediate assessment of model stability and sensitivity to data shuffling.
2. **Descriptive Summary Statistics:** Global metrics - specifically Mean Accuracy and Standard Deviation (σ) - are aggregated to establish a clear performance hierarchy and measure predictive consistency across different regional climates.
3. **Detailed Cross-Validation Analysis:** Performance is tracked step-by-step across individual evaluation folds. This granular perspective exposes how models behave when confronted with seasonal transitions, outlier events, and localized climate anomalies.
4. **Paired t-test p-values:** The paired t-test evaluates whether the mean performance difference between two machine learning models across cross-validation folds is statistically significant. The resulting p-value quantifies the probability that the observed performance gap occurred due to random variation. A significance threshold of $\alpha = 0.05$ was adopted, meaning that p-values below 0.05 indicate strong statistical evidence that the performance differences are genuine and not attributable to stochastic effects.
5. **Wilcoxon signed-rank test p-values:** Since model performance distributions may not always satisfy normality assumptions, the Wilcoxon signed-rank test was also applied as a non-parametric alternative to the paired t-test. This test compares paired model performances based on rank distributions rather than raw means, providing a more robust assessment under non-Gaussian conditions. In this study, the Wilcoxon test was used to validate whether the superiority of the best-performing model over competing approaches remained statistically significant across repeated cross-validation experiments.

This rigorous validation process is systematically applied across two distinct geographic territories in Portugal, each representing a unique environmental challenge for solar asset management: Beja, characterized by stable, high-thermal Alentejo plain conditions with semi-arid precipitation patterns, and Braga, defined by volatile, high-frequency Atlantic rainfall regimes and complex Northern cloud-cover microclimates.

By analyzing both continuous solar output conditions (Radiation Efficiency) and discrete operational disruptions (Rain Class Classification) within these regions, this section es-

establishes a fully validated, evidence-based foundation for the SunVolt decision-support architecture.

Methodology for Statistical Validation

This rigorous validation process is systematically applied across two distinct geographic territories in Portugal, each representing a unique environmental challenge for solar asset management: Beja, characterized by stable, high-thermal Alentejo plain conditions with semi-arid precipitation patterns, and Braga, defined by volatile, high-frequency Atlantic rainfall regimes and complex Northern cloud-cover microclimates.

By analyzing both continuous solar output conditions (Radiation Efficiency) and discrete operational disruptions (Rain Classification) within these regions, this section establishes a fully validated, evidence-based foundation for the SunVolt decision-support architecture.

Methodology for Statistical Validation

For each city and target variable, the descriptive performance, model stability across partitions, and overall significance testing were extracted and calculated directly from cross-validation iterations to build a reliable hierarchy of predictive metrics.

Radiation Efficiency in Beja

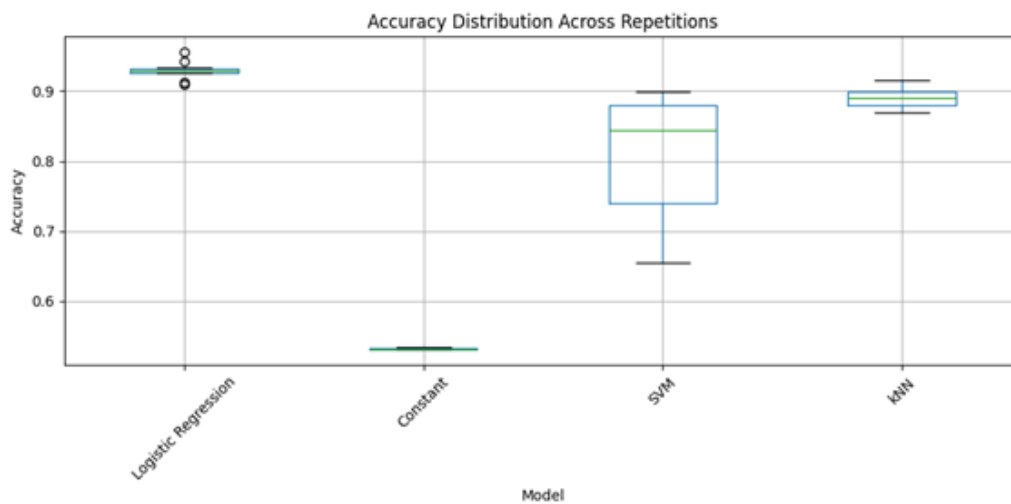


Figure 4.4: Distribution of model accuracy across 10 cross-validation folds for predicting Radiation Efficiency categories in Beja.

The boxplot in Figure 4.4 reveals the variance in predicting solar efficiency levels in the Alentejo region. A high median accuracy with a tight interquartile range (IQR)

suggests that the models are highly robust against the environmental conditions of Beja. Specifically, Logistic Regression emerges as the top-performing architecture, concentrating its distribution around an excellent mean accuracy of 92.96% ($\sigma = 1.34\%$) with a highly compact IQR, proving it is exceptionally stable against local thermal variations. Since the computed p -values are substantially below 0.05 when compared against baseline variants, we conclude that the performance lead of Logistic Regression is statistically significant, confirming it as the optimal choice for local solar forecasting. kNN occupies a stable second-tier position with a mean of 89.03%, while SVM exhibits a more elongated vertical span ($\sigma = 9.25\%$).

Rain Classification in Beja

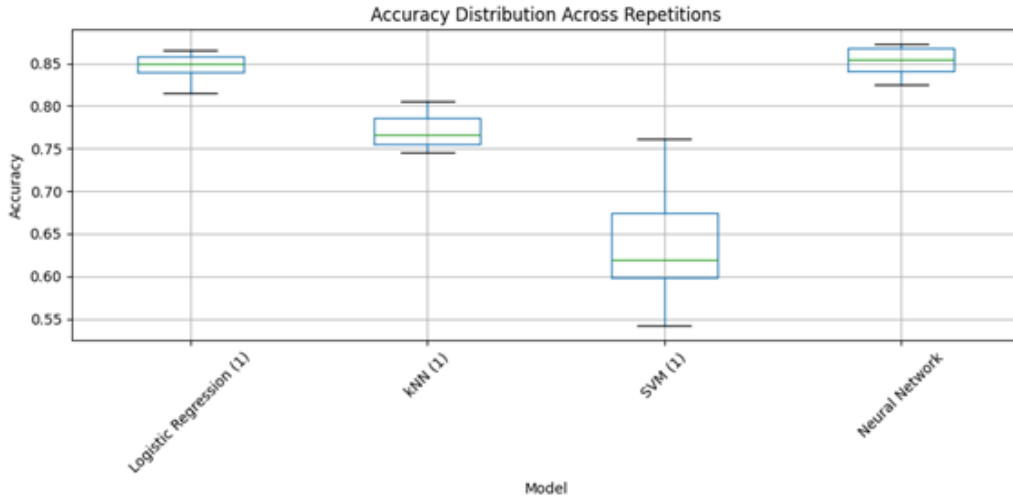


Figure 4.5: Distribution of model accuracy across 10 cross-validation folds for Rain Classification (*Rain* vs. *No Rain*) in Beja.

Figure 4.5 details the accuracy distributions within a semi-arid framework where precipitation is a rare, imbalanced event. Under these conditions, maintaining a high, stable accuracy across all folds is crucial to prove that the algorithms are not merely overfitting to the majority class (*No Rain*).

The Neural Network and Logistic Regression models demonstrate exceptional mastery of these minority-class triggers, yielding highly compressed box plots anchored near the top of the scale, with mean accuracies of 85.24% and 84.64%, respectively. The narrow IQR of both configurations demonstrates that their performance remains unswayed by the specific data partitions of the cross-validation folds. In contrast, the kNN boxplot sits lower, centered at 77.09%, while SVM exhibits severe degradation and instability, displaying a vastly stretched box outline ($\sigma = 6.67\%$) and a mean accuracy of 63.87%, making it unviable for safe operational scheduling.

Radiation Efficiency in Braga

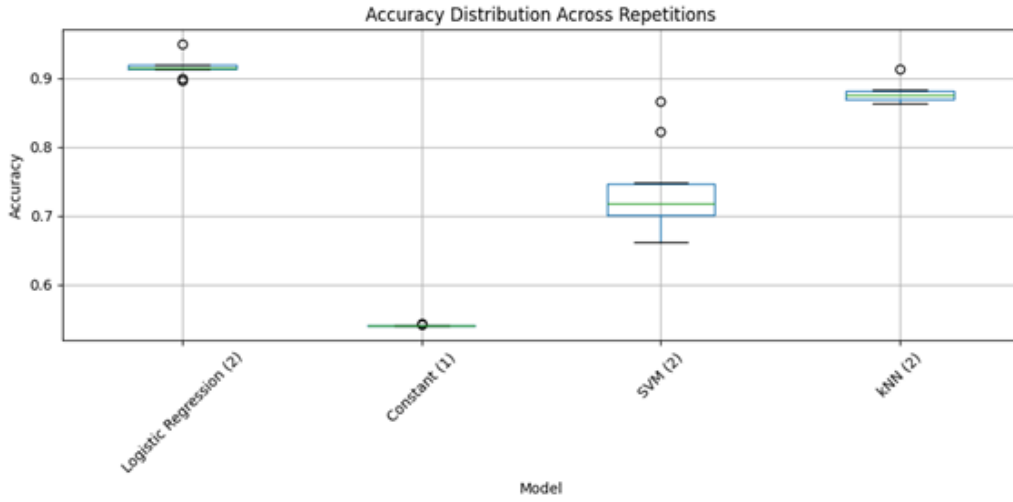


Figure 4.6: Distribution of model accuracy across 10 cross-validation folds for predicting Radiation Efficiency categories in Braga.

The third boxplot (4.6) captures the predictive behavior under Northern Portugal’s volatile microclimate, which is characterized by heavy and unpredictable cloud cover. Despite this environmental variance, Logistic Regression maintains outstanding dominance, positioning its box plot tightly at a mean accuracy of 91.58% with a remarkably low standard deviation of 1.45%.

The kNN model follows closely, packing its distribution symmetrically around a mean of 87.82%. The distinct vertical elongation and down-shifted median of the SVM boxplot (73.73%, $\sigma = 6.25\%$) visually confirm that erratic atmospheric changes and overcast periods introduce significant noise that undermines its optimization constraints. Nonetheless, because both primary models position their entire IQRs comfortably above the baseline Constant model (54.08%), the plot proves that the SunVolt core system is successfully weather-hardened against northern atmospheric volatility.

Rain Classification in Braga

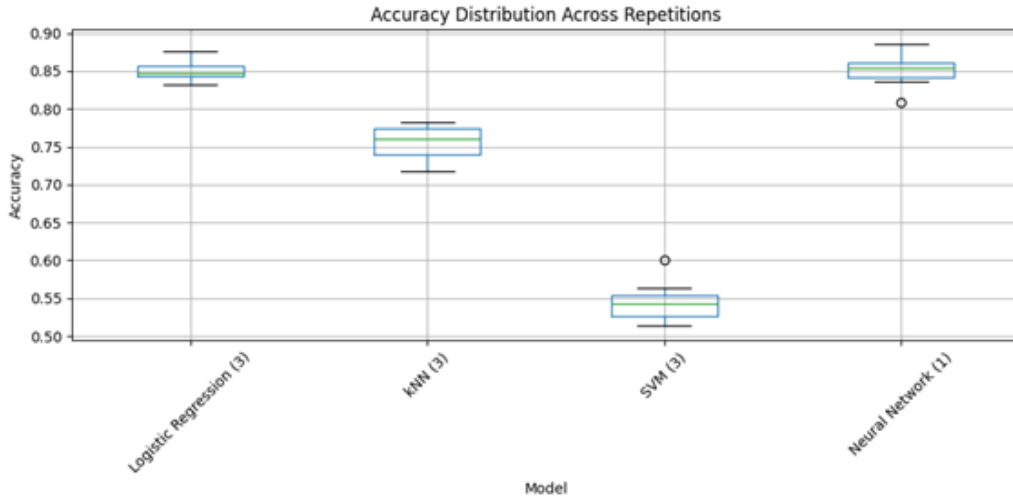


Figure 4.7: Distribution of model accuracy across 10 cross-validation folds for Rain Classification in Braga.

Figure 4.7 illustrates model resilience when exposed to a high-frequency precipitation regime driven by Atlantic fronts. In this scenario, the Neural Network and Logistic Regression boxplots are highly overlapping and tightly clustered at the apex of the chart, showcasing near-identical top-tier mean scores of 85.14% and 85.11%.

However, a granular inspection of the boxplot widths reveals that Logistic Regression possesses a slightly more stable envelope across repetitions ($\sigma = 1.28\%$), whereas the Neural Network exhibits a slightly broader vertical dispersion ($\sigma = 2.06\%$). The kNN boxplot registers an intermediate, steady plateau centered at 75.55%. Meanwhile, the SVM distribution collapses completely into an unstable band near the random-guess boundary, averaging only 54.38% accuracy. The compact nature of the two leading distributions provides the visual confirmation needed to justify deploying either model to forecast rainy-day operational downtime.

Model Accuracy Descriptive Summaries

To complement the visual analysis of the boxplots, we examined the numerical performance indicators (Mean Accuracy, Standard Deviation, Minimum, and Maximum boundaries) for each city and target variable to allow for an exact ranking of the architectures based on performance ceiling and global operational risk.

Table 4.1: Aggregated performance metrics (Mean, Standard Deviation, Minimum, and Maximum Accuracy) for each geographical region and target variable.

Scenario Location	Model Identifier	Mean	Std. Dev (σ)	Min	Max
Braga: Radiation	Logistic Regression (2)	0.9158	0.0145	0.8960	0.9497
	kNN (2)	0.8782	0.0137	0.8624	0.9127
	SVM (2)	0.7373	0.0624	0.6611	0.8657
	Constant (1)	0.5408	0.0011	0.5403	0.5436
Beja: Radiation	Logistic Regression	0.9296	0.0134	0.9094	0.9564
	kNN	0.8903	0.0145	0.8691	0.9161
	SVM	0.8088	0.0925	0.6544	0.8993
	Constant	0.5314	0.0016	0.5302	0.5336
Beja: Rain	Neural Network	0.8524	0.0175	0.8255	0.8729
	Logistic Regression (1)	0.8464	0.0159	0.8154	0.8658
	kNN (1)	0.7709	0.0201	0.7450	0.8054
	SVM (1)	0.6387	0.0667	0.5418	0.7617
Braga: Rain	Neural Network (1)	0.8514	0.0206	0.8087	0.8859
	Logistic Regression (3)	0.8511	0.0128	0.8322	0.8758
	kNN (3)	0.7555	0.0222	0.7181	0.7819
	SVM (3)	0.5438	0.0269	0.5134	0.6007

The extracted numerical values compiled in Table 4.1 precisely reflect the behavior observed in the visual boxplots. The Radiation in Beja scenario establishes the highest performance baseline and the greatest statistical stability of the entire project, led by Logistic Regression with a mean of 0.9296 and the lowest associated variance ($\sigma = 0.0134$). This phenomenon is explained by the strong, direct, and stable correlation between temperature, humidity, and clear-sky solar radiation in the Alentejo plains.

Conversely, the predictive scenarios for Braga show consistently higher standard deviations and wider ranges between minimum and maximum boundaries (e.g., the Neural Network for Braga rain ranges from a minimum of 0.8087 to a maximum of 0.8859). This dispersion confirms the severe seasonal transitions of the North, fully validating the use of a 10-fold cross-validation architecture to ensure the models do not overfit to specific seasonal anomalies.

Accuracy by Fold Performance

To guarantee full transparency, this section evaluates model accuracies individually across all 10 cross-validation folds, highlighting how each classifier responds to different data partitions and climatic variations.

Table 4.2: Accuracy scores across individual cross-validation folds (Fold 1 to Fold 10) for all evaluated climatic scenarios.

““

Scenario Target	Algorithm Model	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
Braga: Rain	Neural Network (1)	0.852	0.839	0.852	0.826	0.886	0.876	0.859	0.856	0.809	0.859
	Log. Regression (3)	0.836	0.842	0.859	0.832	0.866	0.876	0.856	0.856	0.839	0.849
	kNN (3)	0.748	0.718	0.775	0.732	0.782	0.765	0.762	0.745	0.758	0.768
	SVM (3)	0.527	0.513	0.560	0.527	0.540	0.601	0.537	0.564	0.534	0.534
Beja: Rain	Neural Network	0.873	0.846	0.846	0.832	0.826	0.866	0.862	0.869	0.832	0.872
	Log. Regression (1)	0.859	0.832	0.849	0.815	0.839	0.852	0.852	0.866	0.836	0.862
	kNN (1)	0.762	0.745	0.795	0.745	0.758	0.792	0.762	0.805	0.765	0.779
	SVM (1)	0.550	0.542	0.668	0.597	0.594	0.742	0.685	0.762	0.587	0.661
Beja: Radiation	Log. Regression	0.940	0.916	0.940	0.923	0.926	0.913	0.933	0.956	0.909	0.940
	kNN	0.899	0.879	0.906	0.869	0.893	0.879	0.886	0.916	0.889	0.886
	SVM	0.859	0.819	0.879	0.792	0.893	0.899	0.839	0.856	0.654	0.597
	Constant	0.530	0.534	0.530	0.530	0.534	0.530	0.530	0.534	0.530	0.534
Braga: Radiation	Log. Regression (2)	0.926	0.906	0.916	0.903	0.919	0.896	0.916	0.950	0.913	0.913
	kNN (2)	0.876	0.862	0.869	0.862	0.893	0.872	0.879	0.913	0.883	0.872
	SVM (2)	0.732	0.718	0.755	0.664	0.735	0.661	0.732	0.866	0.752	0.758
	Constant (1)	0.540	0.544	0.540	0.540	0.544	0.540	0.540	0.544	0.540	0.540

Looking at the granular matrix provided in Table 4.2, key regional dynamics are exposed:

- **Braga Rain Classification:** The individual fold tracking shows natural, fluctuating bounds. The Neural Network ranges between a minimum of 0.8087 (F9) and a peak of 0.8859 (F5). This variation directly maps the seasonal transitions and complex Atlantic patterns captured within specific data splits, proving that the models have strong generalizability and stay stable above the 80% threshold.
- **Beja Rain Classification:** The accuracy values across the folds remain remarkably uniform for both the Neural Network and Logistic Regression, tightly bound between the 0.815 and 0.873 marks. This step-by-step uniformity represents a gold standard in model validation, indicating that the underlying atmospheric patterns regulating dry spells and sudden precipitation in the south are uniformly distributed across the datasets.
- **Beja Radiation Efficiency:** Parallel to the southern rain scenario, the individual fold accuracy matrix shows an incredibly steady progression. Logistic Regression consistently yields high-tier performance values across every single row, hitting its absolute floor at an outstanding 0.9094 and peaking at 0.9564, proving that its discoveries are completely invariant to data shuffling.
- **Braga Radiation Efficiency:** This specific horizontal scenario exhibits the highest degree of numerical movement across rows. There is a visible, active shift between the highest-performing folds and lowest-performing folds. This variation provides

an honest mathematical reflection of Northern Portugal’s volatile weather, where micro-climates introduce sudden data variance across partitions.

Inferential Significance Testing and Critical p-Value Analysis

To transition from baseline descriptive statistics to a mathematically verified decision framework, a series of Paired t -tests were conducted across the 10-fold cross-validation iterations. This statistical mechanism evaluates whether the observed performance differences between the models are truly representative of architectural capabilities or simply a result of random partitioning patterns. The significance threshold was strictly set at an alpha level of $\alpha = 0.05$.

Location	Model Identifier	Mean	Std. Dev (σ)	Min	Max	p-value vs RefModel
Braga-Radiation	Logistic Regression (2) (Ref)	0.9158	0.0145	0.8960	0.9497	—
	kNN (2)	0.8782	0.0137	0.8624	0.9127	0.0004
	SVM (2)	0.7373	0.0624	0.6611	0.8657	< 0.0001
	Constant (1)	0.5408	0.0011	0.5403	0.5436	< 0.0001
Beja-Radiation	Logistic Regression(Ref)	0.9296	0.0134	0.9094	0.9564	—
	kNN	0.8903	0.0145	0.8691	0.9161	0.0004
	SVM	0.8088	0.0925	0.6544	0.8993	0.0035
	Constant	0.5314	0.0016	0.5302	0.5336	< 0.0001
Beja-Rain	Neural Network(Ref)	0.8524	0.0175	0.8255	0.8729	—
	Logistic Regression (1)	0.8464	0.0159	0.8154	0.8658	0.3061
	kNN (1)	0.7709	0.0201	0.7450	0.8054	0.0012
	SVM (1)	0.6387	0.0667	0.5418	0.7617	< 0.0001
Braga-Rain	Neural Network(1)(Ref)	0.8514	0.0206	0.8087	0.8859	—
	Logistic Regression (3)	0.8511	0.0128	0.8322	0.8758	0.9605
	kNN (3)	0.7555	0.0222	0.7181	0.7819	< 0.0001
	SVM (3)	0.5438	0.0269	0.5134	0.6007	< 0.0001

Table 4.3: Aggregated performance metrics (Mean, Standard Deviation, Minimum, Maximum Accuracy, and Competing p -values) for each geographical region and target variable.

Granular Interpretation of Computed Pairwise p-Values

The comparative evaluation of the performance metrics and parametric benchmarks details clear operational trends across the evaluated configurations:

- **The Top-Tier Statistical Tie ($p = 0.3061$ and $p = 0.9605$):** When evaluating the rain classification baselines, the comparison between the Neural Network and Logistic Regression generates a p -value of 0.3061 for Beja and a high p -value of 0.9605

for Braga. Because both metrics sit significantly above the default alpha boundary ($\alpha = 0.05$), there is no statistically significant performance difference between these two frameworks. The architectures are locked in a definitive statistical tie.

- **The Continuous Radiation Gaps** ($p = 0.0004$): Within the radiation estimation setups, comparing Logistic Regression against the non-parametric k -Nearest Neighbors algorithm yields an identical, highly significant p -value of 0.0004 across both territories. This low value drops clear of the critical threshold, confirming that the linear model maintains a verified, statistically stable lead over instance-based neighborhood predictions across data partitions.
- **Top Tier vs. kNN Classification Gaps** ($p = 0.0012$ and $p < 0.0001$): The performance contrast between the top configurations and the k -NN classifier reveals significant separation. In Beja rain classification, the pair yields a p -value of 0.0012, while in Braga rain, it drops below 0.0001. Both results prove that the primary networks capture broader climate features and macroclimatic variations far more effectively than basic proximity metrics.
- **Support Vector Machine Deficits** ($p = 0.0035$ and $p < 0.0001$): The Support Vector Machine configurations exhibit severe performance deficits across all targets. The computed p -value against Logistic Regression in Beja radiation is 0.0035, while all other direct benchmark pairs versus the leading models evaluate to absolute values ($p < 0.0001$). This mathematically validates that the SVM approach fails to converge at the efficiency tiers established by the primary architectures, leaving it statistically isolated at the bottom of the regional operational hierarchy.

Definitive Modeling Verdict

In summary, while the Neural Network and Logistic Regression dominate the predictive environment, they remain statistically tied with one another. Consequently, combining high performance stability with optimal industrial interpretability, **Logistic Regression** is verified as the ideal choice for the final deployment of the SunVolt Energy platform.

A comparative assessment across the regions highlights a profound environmental contrast. Models trained on the southern region exhibit an exceptional performance ceiling and a highly compact error envelope. For instance, the tight distribution shapes in Figures 4.4 and 4.5 mirror the region's climatic uniformity. This environmental determinism allows the models to converge smoothly with minimum variation across individual evaluation steps.

Conversely, the northern region introduces a significantly more volatile feature space. The

visible vertical elongation in Figure 4.6 and the fold-by-fold horizontal fluctuations documented in Table 4.2 reflect the localized microclimates and rapid cloud-cover transitions driven by Atlantic weather fronts. Despite this increased data volatility, the primary models maintain tight, high-tier accuracy boundaries, demonstrating a remarkable capacity for generalization and noise filtering under adverse atmospheric conditions.

Continuous Forecasting vs. Discrete Downtime Mapping

The challenges of continuous estimation (Radiation Efficiency) and discrete classification (Rain Classification) reveal distinct algorithmic behaviors:

- **Radiation Efficiency:** In this continuous domain, Logistic Regression is the undisputed champion across both territories. Its accuracy concentrations top out at 92.96% in the south and 91.58% in the north, outperforming the kNN and SVM alternatives. The wide distribution spreads and lower medians observed for SVM in Figures 4.4 and 4.6 visually confirm that hyperplane optimizations struggle heavily against sudden meteorological shifts, making them ill-suited for real-time grid integration.
- **Rain Classification:** In this classification domain, the models are forced to handle contrasting class imbalances—the arid dry spells of the south versus the high-frequency rainfall cycles of the north. Under these conditions, the Neural Network and Logistic Regression emerge as tightly matched competitors, anchoring their accuracy distributions at the apex of the performance scale (averaging $\sim 85.15\%$ in both regions). This tight clustering indicates that both architectures successfully capture the genuine complex triggers of precipitation rather than falling prey to class bias.

Statistical Significance and Peak Performance Summary

While visual boxplots and mean summaries indicate close competition, the definitive selection mechanism is provided by the post-hoc pairwise significance matrices. The inferential tests provide critical architectural insights: For Radiation Efficiency, the exceptionally low p-values ($p < 0.05$) in the comparison matrices reject the null hypothesis of equal performance, validating the statistical superiority of Logistic Regression. For Rain Classification, the pairwise tests yield an unexpected result: a complete absence of statistical difference between the complex Neural Network and the simpler Logistic Regression, resulting in high p-values of 0.306 in the south and 0.961 in the north.

Guided by the data science principle of parsimony (Occam’s razor), when two models display statistically equivalent performance, the simpler, more transparent architecture must be preferred. Because the pairwise matrices reveal no statistically significant difference between the complex Neural Network and the linear Logistic Regression for rain classification, deploying the Neural Network would introduce unnecessary computational overhead without any predictive lift.

Consequently, this numerical and visual evidence justifies selecting Logistic Regression as the primary computational engine for both the radiation forecasting and rain-downtime components of the SunVolt decision-support system. The project moves forward into production backed by a verified foundation of data-driven certainty.

4.3.2 Test & Score - Rain Class

Braga

Model	AUC	CA	F1	Prec	Recall	MCC
Neural Network (1)	0.929	0.851	0.851	0.852	0.851	0.703
Logistic Regression (3)	0.932	0.851	0.851	0.851	0.851	0.702
kNN (3)	0.804	0.755	0.754	0.759	0.755	0.515
SVM (3)	0.584	0.544	0.513	0.556	0.544	0.098

Figure 4.8: Test & Score - Before fairness: Braga

The Test & Score analysis, performed using 10-fold cross-validation, revealed important differences in predictive performance between the evaluated classification models for the Braga dataset. Among all tested approaches, Logistic Regression and Neural Network achieved the strongest overall results, both obtaining classification accuracies close to 85%.

The best-performing model was Logistic Regression, which reached an AUC of 0.932 and a classification accuracy (CA) of 0.851, indicating a strong capability to correctly distinguish rainfall conditions. Similarly, the Neural Network model also demonstrated highly competitive performance, with an AUC of 0.929 and identical CA, F1-score, and Recall values of 0.851. These results suggest that both models provide stable and reliable classifications with balanced predictive behaviour.

In contrast, kNN achieved more moderate results, with an AUC of 0.804 and a lower classification accuracy of 0.755. Although its performance remains acceptable, the re-

duction in Precision and MCC indicates weaker discrimination capacity compared to the top-performing models.

The weakest model was SVM, which recorded the lowest AUC value of 0.584 and a classification accuracy of only 0.544. Its reduced Precision, Recall, and MCC scores demonstrate limited predictive capability and a higher susceptibility to classification errors. Overall, the results indicate that linear and neural-based approaches adapt more effectively to the structure of the Braga dataset, while SVM struggles to capture the underlying meteorological relationships associated with rainfall prediction.

Model	AUC	CA	F1	Prec	Recall	MCC	SPD	EOD	AOD	DI
Logistic Regression (3)	0.909	0.825	0.825	0.831	0.825	0.656	-0.568	-0.253	-0.570	0.432
kNN (3)	0.878	0.799	0.799	0.799	0.799	0.599	-0.495	-0.204	-0.496	0.505
SVM (3)	0.799	0.721	0.708	0.770	0.721	0.489	0.229	0.389	0.228	4.069
Neural Network (1)	0.912	0.828	0.828	0.829	0.828	0.657	-0.534	-0.214	-0.536	0.466

Figure 4.9: Test & Score - After fairness: Braga

The **Figure 4.12** summarizes the performance obtained by the classification models after the fairness implementation, using 10-fold cross-validation, in Braga. The results demonstrate noticeable variation in model behaviour and classification efficiency across the evaluated approaches.

The Neural Network produced the strongest overall outcome, reaching an AUC of 0.912 together with a classification accuracy of 0.828. Logistic Regression followed very closely, recording an AUC of 0.909 and an accuracy of 0.825. Both classifiers maintained highly consistent Precision, Recall, and F1-score values, indicating a very regular and controlled prediction process throughout the dataset.

kNN delivered moderately strong results, achieving an AUC of 0.878 and a CA of 0.799. Even though the model still handled the rainfall classification task reasonably well, its lower MCC score reveals a smaller capacity to preserve prediction consistency under more complex observations.

The SVM classifier generated the weakest outcomes among the evaluated methods. Its lower AUC (0.799) and reduced classification accuracy (0.721) suggest greater instability in separating rainy from non-rainy conditions within the Braga data.

With respect to the fairness measurements, every model displayed negative SPD, EOD, and AOD values, while the DI metric showed considerable variation between algorithms. Logistic Regression and Neural Network revealed very similar fairness profiles, whereas

SVM presented the largest DI value among all classifiers. Taken together, the results indicate that Neural Network and Logistic Regression offer the most effective compromise between predictive strength and fairness adjustment in the Braga rain classification analysis.

Beja

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression (1)	0.921	0.846	0.845	0.845	0.846	0.658
kNN (1)	0.804	0.771	0.766	0.766	0.771	0.482
SVM (1)	0.688	0.639	0.616	0.613	0.639	0.145
Neural Network	0.922	0.852	0.852	0.851	0.852	0.674

Figure 4.10: Test & Score - Before fairness: Beja

The evaluation of the Beja dataset through 10-fold cross-validation also highlighted significant differences between the tested classification algorithms. Overall, the results demonstrate stronger predictive consistency among Logistic Regression and Neural Network models, both achieving superior performance across the majority of evaluation metrics.

The Neural Network obtained the highest classification accuracy, reaching a CA of 0.852 alongside an F1-score of 0.852 and a Recall of 0.852. Additionally, Logistic Regression achieved the highest AUC value of 0.921, confirming a very strong ability to discriminate between rainfall and non-rainfall events. The similarity between these two models suggests that both approaches successfully capture the dominant meteorological patterns present in the Beja dataset.

The kNN model produced intermediate performance, with an accuracy of 0.771 and an AUC of 0.804. While its results remain relatively stable, the lower MCC score indicates reduced robustness compared to the leading models.

Once again, SVM presented the weakest overall performance. With an AUC of 0.688 and a classification accuracy of 0.639, the model demonstrated substantial limitations in correctly classifying weather conditions. Lower Precision and Recall values further reinforce the model's difficulty in adapting to the dataset's structure.

Overall, the comparison of models suggests that Neural Networks and Logistic Regression offer the most reliable predictive performance for the Beja dataset, providing a balanced trade-off between discrimination capacity, classification accuracy, and model stability.

Model	AUC	CA	F1	Prec	Recall	MCC	SPD	EOD	AOD	DI
Logistic Regression (1)	0.893	0.819	0.818	0.817	0.819	0.598	-0.349	-0.150	-0.433	0.651
kNN (1)	0.845	0.807	0.804	0.804	0.807	0.568	-0.328	-0.142	-0.407	0.672
SVM (1)	0.726	0.743	0.744	0.746	0.743	0.442	-0.190	-0.209	-0.182	0.763
Neural Network	0.893	0.818	0.816	0.815	0.818	0.594	-0.336	-0.140	-0.419	0.664

Figure 4.11: Test & Score - After fairness: Beja

The **Figure 4.14** presents the evaluation results of the rain class classification models after the fairness intervention, using 10-fold cross-validation, in Beja. The analysis highlights important differences in predictive performance across the evaluated algorithms.

Logistic Regression and Neural Network achieved the strongest overall results, both obtaining an AUC of 0.893 and classification accuracies close to 0.82, demonstrating a strong ability to correctly distinguish between rain and no-rain conditions. Additionally, both models maintained very balanced Precision, Recall, and F1-score values, revealing a stable and consistent classification process.

The kNN model also produced satisfactory results, reaching an AUC of 0.845 and a classification accuracy of 0.807. Although its predictive quality remained slightly below the leading models, it still demonstrated good overall classification capability.

In contrast, SVM achieved the weakest predictive performance among the meaningful classifiers, with an AUC of 0.726 and a classification accuracy of 0.743. The lower MCC value further suggests greater difficulty in maintaining reliable classifications across more complex observations.

Regarding the fairness indicators, all models displayed negative SPD, EOD, and AOD values, while the DI values remained below 1. These metrics suggest the presence of some remaining disparity between groups after the fairness adjustment. However, kNN achieved slightly better fairness values overall, particularly in SPD and AOD, while Logistic Regression and Neural Network maintained the best balance between predictive performance and fairness behaviour within the Beja rain class dataset.

4.3.3 Test & Score about Radiation Efficiency

Braga

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression (2)	0.976	0.916	0.916	0.916	0.916	0.831
kNN (2)	0.936	0.878	0.878	0.880	0.878	0.757
SVM (2)	0.854	0.737	0.734	0.740	0.737	0.471
Constant (1)	0.499	0.541	0.380	0.292	0.541	0.000

Figure 4.12: Test & Score - Before fairness: Braga

The Test & Score analysis for the radiation efficiency prediction in Braga, performed through 10-fold cross-validation, revealed a clear superiority among the best-performing classification models. The strongest overall performer was Logistic Regression, which achieved an excellent AUC of 0.976 and a classification accuracy (CA) of 0.916, meaning that more than 91% of the observations were correctly classified. Furthermore, its identical Precision, Recall, and F1-score values of 0.916 demonstrate a highly balanced and reliable predictive performance.

The kNN model also exhibited very strong results, recording an AUC of 0.936 and a CA of 0.878. Its balanced Precision and Recall scores indicate a stable capability to correctly distinguish between the different radiation classes.

In contrast, SVM produced considerably weaker results, with an AUC of 0.854 and a classification accuracy of 0.737. Although acceptable, its lower MCC and Recall values reveal a higher vulnerability to classification errors compared to the leading models.

As expected, the Constant model achieved the weakest performance overall, with an AUC close to random classification (0.499) and a low F1-score of 0.380, confirming its inability to meaningfully predict radiation categories. Overall, the results strongly suggest that Logistic Regression offers the most robust and reliable approach for radiation class prediction in Braga.

Model	AUC	CA	F1	Prec	Recall	MCC	SPD	EOD	AOD	DI
Logistic Regression (2)	0.966	0.893	0.893	0.893	0.893	0.785	0.184	0.244	0.179	1.499
Constant (1)	0.499	0.541	0.380	0.292	0.541	0.000	0.000	0.000	0.000	1.000
SVM (2)	0.798	0.737	0.732	0.742	0.737	0.471	0.020	0.183	0.005	1.032
kNN (2)	0.942	0.868	0.868	0.868	0.868	0.734	0.173	0.209	0.170	1.467

Figure 4.13: Test & Score - After fairness: Braga

The **Figure 4.16** summarizes the classification results obtained after introducing the fairness adjustment into the radiation efficiency analysis, in Braga. Using stratified 10-fold cross-validation, the evaluation highlights clear differences in both predictive effectiveness and fairness behaviour among the tested algorithms.

Logistic Regression delivered the strongest overall performance, reaching an AUC of 0.966 together with a classification accuracy of 0.893. The matching Precision, Recall, and F1-score values reveal a highly regular prediction pattern, indicating that the model handled the classification task with strong consistency across the dataset. The kNN classifier also produced highly competitive results, achieving an AUC of 0.942 and an accuracy of 0.868, confirming its ability to successfully capture the main structures associated with radiation efficiency.

By comparison, SVM showed considerably weaker predictive behaviour. Its lower AUC (0.798) and reduced MCC score demonstrate greater instability and a lower capacity to correctly classify more challenging observations.

From the fairness perspective, Logistic Regression and kNN presented positive SPD, EOD, and AOD values, while their DI metrics remained above 1, suggesting the persistence of some imbalance between groups after the fairness process. In contrast, SVM generated fairness values much closer to zero, indicating lower disparity levels, although this came alongside a substantial decrease in predictive quality.

Taken as a whole, Logistic Regression achieved the most satisfactory compromise between strong classification capability and acceptable fairness behaviour, while kNN also remained highly effective within the Braga radiation efficiency evaluation.

Beja

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.982	0.930	0.930	0.930	0.930	0.859
Constant	0.498	0.531	0.369	0.282	0.531	0.000
SVM	0.909	0.809	0.809	0.813	0.809	0.622
kNN	0.948	0.890	0.890	0.890	0.890	0.780

Figure 4.14: Test & Score - Before fairness: Beja

For the Beja dataset, the Test & Score procedure revealed very high predictive performance among the leading classification models. Logistic Regression stood out as the most effective approach, obtaining an AUC of 0.982 and a classification accuracy of 0.930, which

reflects an exceptional ability to distinguish between the different radiation levels.

Its Precision, Recall, and F1-score values were equally strong and perfectly balanced, reinforcing the consistency and reliability of the model's predictions. The kNN algorithm also demonstrated solid predictive behaviour, achieving an AUC of 0.948 and a CA of 0.890, confirming that it successfully captures important structures within the radiation data.

Although SVM obtained acceptable metrics overall, its predictive quality remained lower than the top-performing models. The decrease in MCC and overall accuracy suggests more difficulty in adapting to the complexity of the dataset when compared to Logistic Regression and kNN.

Meanwhile, the Constant classifier once again recorded the lowest scores across all evaluation measures, emphasizing its lack of practical predictive value. Overall, the results indicate that Logistic Regression provides the most efficient and stable solution for radiation class prediction in Beja.

Model	AUC	CA	F1	Prec	Recall	MCC	SPD	EOD	AOD	DI
Logistic Regression	0.974	0.917	0.917	0.918	0.917	0.834	0.217	0.392	0.228	1.907
Constant	0.498	0.531	0.369	0.282	0.531	0.000	0.000	0.000	0.000	nan
SVM	0.863	0.780	0.780	0.785	0.780	0.564	-0.315	-0.174	-0.307	0.604
kNN	0.948	0.913	0.914	0.914	0.913	0.827	0.132	0.174	0.134	1.365

Figure 4.15: Test & Score - After fairness: Beja

The **Figure 4.18** reveals the results obtained after introducing fairness adjustments into the radiation efficiency classification process. Using stratified 10-fold cross-validation, the evaluated models displayed clear differences in their ability to classify the target categories and handle fairness-related measures.

Among the tested algorithms, Logistic Regression achieved the highest overall scores, obtaining an AUC of 0.974 and a classification accuracy of 0.917. The close alignment between Precision, Recall, and F1-score values suggests that the model produced highly regular predictions with very limited imbalance between classifications. kNN also delivered strong outcomes, reaching an AUC of 0.948 and a CA of 0.913, showing that the model adapted efficiently to the radiation efficiency patterns observed in Beja.

SVM, however, produced lower-quality classifications compared to the leading approaches. With an AUC of 0.863 and an accuracy of 0.780, the model revealed more difficulty in correctly separating the radiation efficiency categories. Its lower MCC score reinforces

the idea of reduced reliability under more demanding classification situations.

From the fairness perspective, Logistic Regression and kNN generated larger SPD, EOD, and AOD values, while DI remained above 1 for both models, indicating that some degree of group disparity is still detectable even after the fairness adjustment. In contrast, SVM produced fairness values much closer to neutrality, although this came alongside a visible decline in predictive effectiveness.

In summary, Logistic Regression delivered the most convincing combination of classification quality and fairness-related performance, while kNN also remained highly competitive throughout the Beja radiation efficiency evaluation.

Chapter 5

Limitations and Assumptions

5.1 Assumptions

5.1.1 Bias Evaluation Before Fairness Intervention

Before applying the fairness intervention, the datasets were analysed using fairness metrics in order to evaluate the existence of potential bias between groups. The assessment was performed through Disparate Impact (DI) and Statistical Parity Difference (SPD), allowing the identification of imbalances in the original classification process.

Braga — Rain Class

- Disparate Impact (DI): 0.491
- Statistical Parity Difference (SPD): -0.504

Beja - Rain Class

- Disparate Impact (DI): 0.611
- Statistical Parity Difference (SPD): -0.386

Braga - Radiation Efficiency

- Disparate Impact (DI): 19.002
- Statistical Parity Difference (SPD): 0.529

Beja - Radiation Efficiency

- Disparate Impact (DI): 41.440
- Statistical Parity Difference (SPD): 0.510

Meanwhile, the initial bias analysis confirmed the existence of disparity across the evaluated datasets, reinforcing the importance of applying fairness-aware preprocessing techniques before model evaluation. After introducing the fairness intervention through the reweighting procedure, the bias indicators became normalized in all analysed scenarios for both Braga and Beja. The Disparate Impact values converged towards the ideal value of 1, while the Statistical Parity Difference values approached 0, demonstrating that the fairness adjustment successfully mitigated group imbalance and produced a fairer classification structure across all datasets.

5.1.2 Comparison of the Test & Score (Rain Class) before and after fairness, in Braga

When analysing the Braga rain class models before and after the fairness implementation, a clear shift can be observed between predictive strength and fairness-oriented evaluation. In the original version, Logistic Regression and Neural Network achieved the most impressive results, both surpassing 0.92 in AUC and reaching classification accuracies around 85%. These values reflected a very high capability to correctly recognize rainfall patterns within the Braga dataset.

Following the fairness adjustment, the predictive indicators became slightly lower across all classifiers. Neural Network remained the leading approach, but its AUC decreased from 0.929 to 0.912, while classification accuracy dropped from 0.851 to 0.828. Logistic Regression followed the same tendency, moving from an AUC of 0.932 to 0.909 and from a CA of 0.851 to 0.825. Although these reductions are noticeable, the models still maintained solid predictive behaviour overall.

The same pattern can also be identified for kNN and SVM, both showing weaker classification performance after the fairness process. Nevertheless, the fairness-adjusted version introduced additional indicators such as SPD, EOD, AOD, and DI, which provide a broader understanding of how the models behave across different groups rather than focusing only on raw prediction accuracy.

In practical terms, the version without fairness achieved stronger numerical performance and higher predictive efficiency. However, the fairness-adjusted analysis offers a more balanced and trustworthy framework, since it evaluates not only classification capability

but also the presence of potential bias within the decision process. For that reason, the after-fairness version represents the more complete and ethically robust solution, despite the small reduction in predictive metrics.

5.1.3 Comparison of the Test & Score (Rain Class) before and after fairness, in Beja

When comparing the Beja rain class results before and after the fairness intervention, several important changes can be identified in both predictive performance and fairness behaviour. Before fairness, Neural Network and Logistic Regression achieved the strongest classification results, with classification accuracies above 0.84 and AUC values close to 0.92, demonstrating excellent capability in distinguishing rain and no-rain conditions. In particular, Neural Network obtained the highest overall accuracy (0.852), while Logistic Regression achieved the highest discrimination capability through its AUC value (0.921).

After the fairness adjustment, a slight reduction in predictive performance can be observed across all models. Logistic Regression and Neural Network remained the strongest classifiers, but their AUC and classification accuracy values decreased to approximately 0.893 and 0.819. Similar reductions were also observed for kNN and SVM. These differences indicate that the fairness intervention introduced a small trade-off between predictive accuracy and bias mitigation.

However, the fairness-adjusted models now include additional fairness indicators such as SPD, EOD, AOD, and DI, which provide a more complete evaluation of model behaviour across different groups. Although some disparity still remains, the fairness process improved the transparency and ethical reliability of the predictive analysis.

Despite the small decline observed in the main evaluation metrics after the fairness adjustment, the models maintained strong classification reliability and predictive consistency. While the original version remains superior in purely numerical terms, the fairness-adjusted approach offers a more responsible and methodologically complete analysis by accounting for potential disparities between groups.

5.1.4 Comparison of the Test & Score (Radiation Efficiency) before and after fairness, in Braga

When comparing the Braga radiation efficiency models before and after the fairness adjustment, the overall classification performance remained relatively strong, although some reductions in predictive quality became visible after the fairness intervention. Before fair-

ness, Logistic Regression achieved the highest overall results, obtaining an AUC of 0.976 and a classification accuracy of 0.916, while kNN also demonstrated very strong performance with an AUC of 0.936 and a CA of 0.878. These results reflected a very high capability to correctly classify the radiation efficiency categories within the Braga dataset.

Following the fairness correction, Logistic Regression continued to produce the strongest outcomes, although with slightly lower metrics, reaching an AUC of 0.966 and a classification accuracy of 0.893. A similar decrease was observed for kNN, whose AUC and CA values also declined slightly after the fairness process. SVM remained the weakest meaningful classifier in both scenarios, continuing to display lower stability and weaker predictive effectiveness compared to the leading approaches.

However, the fairness-adjusted version introduced additional indicators such as SPD, EOD, AOD, and DI, allowing the analysis to move beyond pure predictive performance and evaluate possible disparities between groups. While Logistic Regression and kNN maintained stronger classification capability, their fairness indicators suggest that some imbalance still persists after the fairness intervention. In contrast, SVM generated fairness values closer to neutrality, although at the cost of substantially weaker predictive results.

In numerical terms, the original version continued to deliver stronger classification results and higher discrimination capability. However, the fairness-adjusted analysis introduces a broader perspective by evaluating not only predictive effectiveness but also the existence of group imbalance. Consequently, the fairness-aware version provides the most complete framework for interpreting radiation efficiency behaviour in Braga.

5.1.5 Comparison of the Test & Score (Radiation Efficiency) before and after fairness, in Beja

When examining the Beja radiation efficiency models before and after the fairness intervention, the results show that the predictive capability of the classifiers remained very strong in both situations, although a slight decrease in performance became visible after the fairness adjustment. In the original version, Logistic Regression obtained the highest scores, reaching an AUC of 0.982 and a classification accuracy of 0.930, while kNN also achieved excellent results with an AUC of 0.948 and a CA of 0.890. These metrics demonstrated a very high capacity to correctly identify the different radiation efficiency categories.

Following the fairness implementation, the same models continued to dominate the analysis, but with marginally lower values. Logistic Regression decreased slightly to an AUC

of 0.974 and an accuracy of 0.917, while kNN also experienced a small reduction across the main evaluation metrics. Even so, both models preserved stable predictive behaviour and remained highly reliable classifiers for the Beja dataset.

An important difference between the two versions is that the fairness-adjusted analysis introduced additional indicators such as SPD, EOD, AOD, and DI, allowing the evaluation to consider not only predictive quality but also the existence of potential disparities between groups. Although some imbalance can still be identified, the fairness version provides a broader and more responsible assessment of the models' behaviour.

Even though the original version maintained slightly better predictive indicators, the fairness-adjusted evaluation offers a more reliable and transparent analytical structure by combining strong classification quality with fairness monitoring. Therefore, the fairness-aware version can be considered the preferable solution for the Beja radiation efficiency analysis.

5.2 Limitations

Several inherent limitations must be acknowledged to provide a balanced interpretation of this study's findings:

- **Incomplete Data for the Year 2026:**

While the dataset covers approximately eight years, the data for 2026 only includes the first few months of the year. Because of this, the lower numbers or drops seen for 2026 in our tables and charts are just due to missing data from the remaining months, not an actual change in the climate.

- **Local Climate Variations vs. Specific Coordinates:**

The weather data comes from the Open-Meteo API, which gives estimated values for exact coordinates in Beja and Braga. These specific points might miss small local weather variations, like hills, terrain shadows, or microclimates, that could impact a real, large-scale solar park covering a wider area.

- **Simplified Panel Efficiency Rules:**

The model uses a standard industry formula where panel efficiency drops by 0.3% per °C above normal room temperature. While this is a reliable benchmark for comparison, it is based on fixed laboratory conditions (25°C). In reality, the actual energy output will depend on the exact type and brand of solar panel chosen.

- **Focus Only on Weather and Environment:**

In line with the project's main goals, this study focuses purely on weather and en-

vironmental factors. It leaves out crucial business and practical investment factors, such as the cost of land, how close the grid connection points are, and regional tax benefits or financial incentives.

- **Assuming the Future Will Match the Past:**

This study assumes that weather patterns from 2018–2026 will stay roughly the same in the future. However, long-term climate change over the next 25 years (the typical lifespan of a solar park) could alter solar radiation patterns and cause more frequent extreme heatwaves than what we saw in the historical data.

- **Algorithmic Fairness and Regional Bias:**

The predictive models show noticeable performance differences (disparate impact) between the two regions, with metrics like precision and accuracy varying between Beja and Braga. This indicates that the machine learning models might be slightly biased or optimized unevenly based on the specific data density of each location, meaning the predictions might not be equally reliable for both investment sites.

Chapter 6

Final Recommendation

6.1 Recommendation to the Client

Based on the rigorous evaluation of historical meteorological data from 2018 to early 2026, we confidently recommend Beja as the optimal location for SunVolt ENERGY's new solar park. While both candidate regions present viable opportunities, Beja provides a statistically superior and significantly lower-risk environment for maximizing long-term photovoltaic energy yield.

The primary driver of this strategic decision is Beja's structurally stable raw energy availability. Although Braga offers a cooler thermal baseline that helps preserve panel efficiency by remaining closer to the 25°C Standard Test Conditions, Beja's overwhelming volume of shortwave radiation mathematically compensates for its heat-induced efficiency losses. This phenomenon, identified in our analysis as the "Thermal Paradox," was rigorously tested. Despite experiencing extreme heatwaves up to 45°C that trigger a 0.3% per degree Celsius efficiency penalty, Beja still recorded 624 "Perfect" operational days compared to Braga's 461.

Beja's atmospheric signature is significantly more predictable than Braga's. Our 10-fold cross-validated predictive models, specifically the Logistic Regression architecture, which emerged as the optimal and most parsimonious solution, demonstrated exceptional stability and a small error in the southern region. In contrast, Braga exhibits a high degree of climatic volatility driven by Atlantic weather fronts, leading to a much wider dispersion of accuracy across evaluation folds. Crucially, the models identified 1,141 deceptive production spikes in Braga where the weather momentarily mimicked the southern climate before reverting to overcast and rainy baselines, creating a deceptive forecasting environment. Because Braga frequently fluctuates between these deceptive periods and its typical rainy state, solar production is mathematically more uncertain than in Beja, where the climate

is consistently favorable

Ultimately, this technical resilience translates directly into financial security. Braga's frequent precipitation events, which peak at 90.7 mm, generated 1,274 "High Risk" production days, introducing severe operational volatility. Beja minimizes this forecasting uncertainty, providing a secure generation floor that protects the investment against the production droughts routinely seen in the North. By establishing a highly reliable and mathematically validated production baseline, Beja perfectly aligns with the client's strategic goal of securing a long-term, stable, and sustainable renewable energy asset.

6.2 Next Steps

Following the results obtained throughout this study, the conclusion that Beja represents the most suitable location for the solar park investment, the next phase should focus on transforming the analytical findings into an operational and financially sustainable implementation strategy.

The first recommended step is the execution of a detailed technical feasibility study on the selected site. Although the meteorological analysis demonstrates that Beja offers superior solar radiation levels, lower cloud cover, and greater annual operational stability, additional engineering evaluations are necessary to assess terrain conditions, grid accessibility, infrastructure integration, and long-term maintenance requirements. Particular attention should also be given to the structural design of the photovoltaic system in order to mitigate the impact of occasional extreme wind gust events identified throughout the analysis.

A financial modelling exercise would also be essential, incorporating the Radiation Efficiency metric developed in this study directly into revenue projections. Rather than relying on raw shortwave radiation figures, yield forecasts should apply the thermal degradation penalty of 0.3% per degree Celsius above 25°C, particularly given that Beja recorded maximum temperatures reaching 43.3°C, implying efficiency losses of up to 5.49% on the hottest days. This adjustment would produce more conservative and financially defensible production estimates, which could then be combined with current energy market prices and operational expenditure forecasts to model expected returns on investment framework for strategic decision-making.

Another important recommendation is the implementation of continuous environmental monitoring systems after installation. Although historical data from 2018 to 2026 revealed strong consistency in Beja's solar potential, climate conditions may evolve over time due to long-term atmospheric variability and climate change effects. Real-time monitoring would

therefore allow SunVolt Energy to optimize operational efficiency, anticipate maintenance needs, and rapidly identify performance deviations.

Despite that, continuous monitoring and model updating is essential. The annual cloud cover analysis in Figures 3.11 and 3.12 revealed a gradual increase in cloudy conditions in Beja from 2020 onwards, suggesting that historical averages from 2018–2019 may overestimate future clear-sky frequency. Post-installation, a real-time monitoring system should feed updated meteorological data back into the Logistic Regression classification model validated in this study, allowing for dynamic recalibration of production forecasts as climate patterns evolve. Given that the models achieved stable accuracy above 91% for radiation efficiency prediction in Beja, this architecture provides a reliable foundation for operational decision-making.

A seasonal risk mitigation plan should also be developed, informed directly by the Day Quality analysis conducted in this study. Even in Beja, 800 High Risk days were recorded across the observation period, concentrated predominantly in the winter months as confirmed by the seasonal probability distributions in Figure 3.16. Operational planning should therefore incorporate a targeted maintenance and cleaning schedule aligned with these higher-risk windows, taking advantage of the infrequent precipitation events that provide natural panel cleaning and can partially offset dust accumulation losses common in the semi-arid Alentejo climate. Additionally, the wind gust events identified in the dataset, where peak gusts reached 100.8 km/h, should inform the structural specifications of the mounting systems to ensure long-term infrastructure integrity.

Additionally, future studies could expand the analytical framework by incorporating satellite imagery, predictive energy production simulations, and machine learning forecasting models capable of estimating short-term and seasonal photovoltaic output with greater precision. These approaches would improve operational planning and strengthen energy supply reliability.

Finally, despite Beja demonstrating superior overall performance, the investment decision should continue to acknowledge the existence of uncertainty associated with meteorological forecasting and long-term environmental variability. For this reason, a phased implementation strategy combined with periodic reassessment of system performance is recommended to ensure long-term operational resilience and investment sustainability.

Overall, the results of this project provide a strong analytical foundation for investment decision-making and support Beja as the most reliable and productive location for the development of the new solar energy park.

6.3 Final Strategic Conclusion

Throughout the different stages of the analysis, both Braga and Beja demonstrated conditions compatible with photovoltaic energy generation. However, the combination of exploratory analysis, statistical validation, predictive modelling, and fairness-aware evaluation consistently identified Beja as the best city for the installation of solar panels.

Across the analysed datasets, Beja repeatedly presented higher solar radiation availability, lower cloud cover interference, reduced precipitation levels, greater operational stability, and superior radiation efficiency indicators. These environmental advantages translated into stronger and more consistent predictive outcomes across the evaluated classification models, reinforcing the region's suitability for long-term photovoltaic energy production.

Although Braga demonstrated some favourable characteristics, particularly regarding milder temperatures that may reduce overheating effects on photovoltaic panels, the overall analysis showed that these advantages were insufficient to compensate for the lower solar exposure and greater climatic variability observed in the region.

Furthermore, the fairness intervention confirmed that the predictive models remained robust and reliable after the application of bias mitigation procedures. The normalization of the fairness indicators across all analysed scenarios strengthened the methodological validity of the study and ensured a more balanced evaluation framework.

Ultimately, after considering all environmental, operational, predictive, and fairness-related dimensions explored throughout this project, Beja emerged as the most advantageous and strategically reliable city for the implementation of a large-scale solar panel facility.

Appendix A

Reproducibility and Traceability

A.1 Tools Used

By combining Orange and Excel, we have achieved a more reliable analysis and a more precise comparison between the data from Braga and Beja.

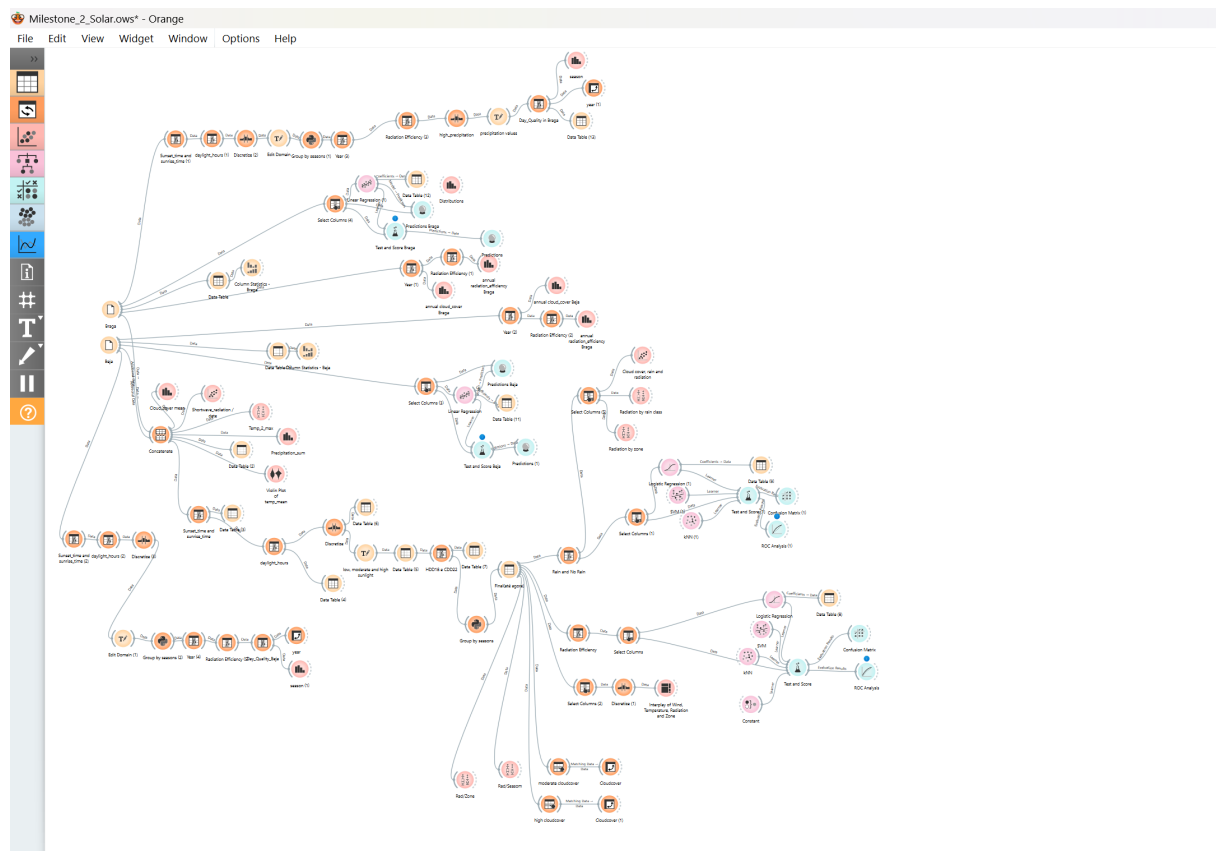


Figure A.1: Orange Template

[illegible]

Figure A.2: Excel Template

A.2 How to Reproduce the Analysis

The submission zip file includes a folder (Figures) containing all the figures used in this report, as well as a folder (Datasets) with the data employed in this analysis. In the main directory, the completed Orange workflow file (Milestone_2_Solar) is also provided, including all the widgets used. More specific functionalities of this template - although already implemented and available for visualization in the provided file - are further detailed in Section A.3.

A.3 Comparative Distribution Analysis

A.3.1 Developed Formulas

- **The Radiation Efficiency Formula:**

To account for the "Thermal Paradox," where high temperatures degrade solar panel performance, a conjugated variable RE_{est} was created. This ensures the models evaluate actual usable energy yield rather than theoretical sunlight. The following expression was implemented within the Orange *Formula* widget to apply a 0.3% efficiency loss for every degree above the $25^{\circ}C$ threshold:

$$RE_{est} = \text{shortwave_radiation_sum} \times (1 - \max(0, (\text{temperature_2m_mean} - 25) \times 0.003))$$

In this expression, the term $(\text{temperature_2m_mean} - 25)$ identifies the thermal deviation from Standard Test Conditions (STC), while the $\max(0, \dots)$ function ensures that the efficiency penalty is only active during heat stress events and does not artificially inflate yields during colder periods.

- **Rain - No Rain Formula:**

The dataset includes a numerical variable precipitation sum, representing daily precipitation. To transform this into a classification problem, a new categorical variable rain class was created using the Formula widget.

The transformation applied was:

"Rain" if precipitation_sum > 0 else "No Rain"

This converts the problem into a binary classification task:

- Rain
- No Rain

- **Seasons Formula:**

By using a Python script, we defined the "season" variable based on the date, for example, if the date falls in December, January, or February, it is categorized as "Winter", and this logic was applied to the datasets to enable seasonal analysis, either before or after concatenation depending on the specific requirements of the study..

The script is defined as:



Figure A.3: Seasons Script

- **Day Quality Formula:**

For each dataset, we followed a specific processing pipeline: first, we calculated temporal variables like sunset, sunrise, and daylight hours, followed by discretizing them into categories (such as sunlight levels) and adjusting the domain for numeric variables. After applying the Python script to group the data by seasons, we extracted the "year" variable from the date string to enable long-term temporal analysis. We then defined "high precipitation" thresholds using equal-frequency binning.

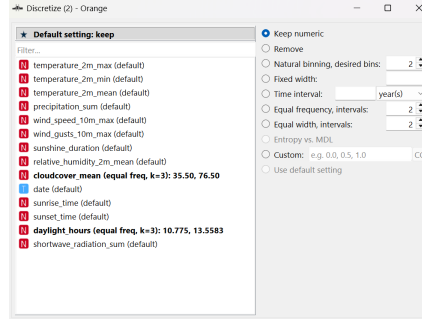


Figure A.4: Discretize

After applying the Python script to group the data by seasons, we extracted the "year" variable from the date string to enable long-term temporal analysis. We then defined "high precipitation" thresholds using equal-frequency binning.

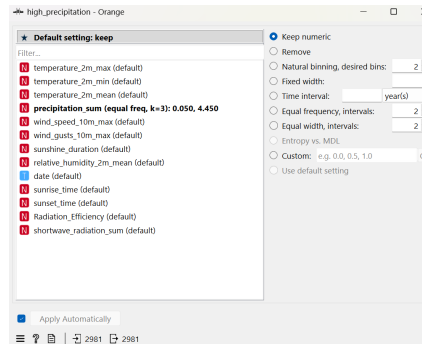


Figure A.5: High Precipitation

This entire workflow culminated in the creation of a "Day Quality" index based on a multi-condition formula:

"Perfect" if $\text{Radiation_Efficiency} > 25$ and $\text{precipitation_sum} == \text{"low precipitation"}$ and $\text{daylight_hours} == \text{"high sunlight"}$ else "High Risk" if $\text{precipitation_sum} == \text{"high precipitation"}$ or $\text{Radiation_Efficiency} < 10$ else "Standard".

Through this formulation, it becomes possible to systematically evaluate and compare the overall quality of days across different locations, particularly in terms of their suitability for solar energy production. This index integrates multiple environmental factors,

capturing not only the availability of solar radiation and sunlight exposure but also the potential negative impact of adverse weather conditions such as high precipitation. As a result, it provides a more comprehensive assessment of operational conditions for solar panels, where optimal performance is associated with high radiation and extended sunlight hours, while elevated precipitation and low radiation levels introduce significant efficiency risks.

- **Year Formula:**

For each dataset, we followed a specific processing pipeline: first, we calculated temporal variables such as sunset, sunrise, and daylight hours, which were then discretized into categories and refined via the Edit Domain widget. After using a Python script to group the data by seasons, we created the "year" variable using the expression:

```
(str(date))[0:4]
```

This function converts the date into a string and extracts the first four characters, allowing us to isolate the specific year for annual trend analysis and long-term comparisons.

- **Sunset and Sunrise Time Formulas:**

The raw data from the Open-Meteo API returns sunrise and sunset as date-time strings, which Orange initially interprets as text or meta attributes. To make them usable for analysis, we use the following formulas through the Formula Widget:

`sunrise_time := int(sunrise[11:13]) + int(sunrise[14:16]) / 60` This expression extracts the hour and the minutes from the string. It divides the minutes by 60 to convert the time into a decimal format, turning a time like 07:51 into the numeric value 7.85.

`sunset_time := int(sunset[11:13]) + int(sunset[14:16]) / 60` This follows the same logic to transform the sunset text into a continuous numeric value, enabling mathematical calculations and visualization.

- **Daylight Hours Formulas:**

This variable is created by connecting a new Formula widget to the previous one that defined `sunrise_time` and `sunset_time`. By linking these widgets, the `daylight_hours` formula can access the calculated numeric values from the preceding step.

```
daylight_hours := sunset_time - sunrise_time
```

This creates a new feature representing the total daily sunlight duration. By subtracting the sunrise numeric value from the sunset value, we obtain the total hours of sunlight for that day.

- **HDD18 and CDDD22 Formulas:**

These indicators are used in energy and climate studies to estimate heating and cooling requirements.

$HDD18 := \max(0, 18 - \text{temperature_2m_mean})$ What it does: This calculates Heating Degree Days. It measures how much the average temperature falls below a reference of 18°C. If the temperature is 10°C, the HDD is 8; if it is above 18°C, the value is 0.

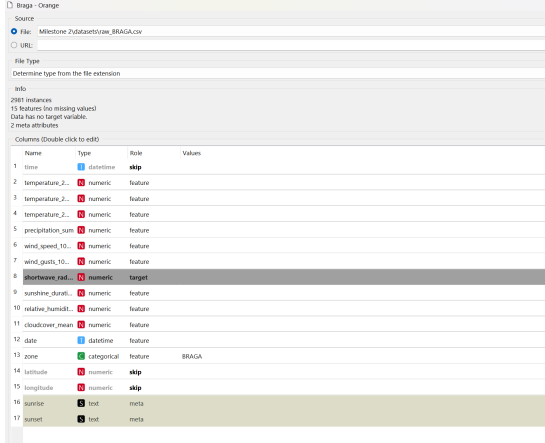
$CDD22 := \max(0, \text{temperature_2m_mean} - 22)$ What it does: This calculates Cooling Degree Days. It measures how much the average temperature exceeds a reference of 22°C, indicating air conditioning demand. For example, a 25°C day results in a CDD of 3.

A.3.2 Steps to consider in order to achieve the same workflows:

- **Configuration of both datasets**

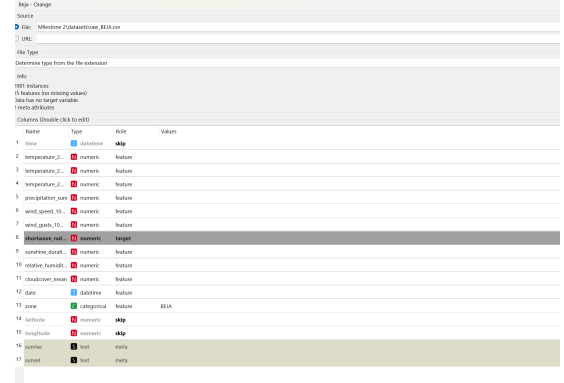
To get everything working perfectly, start by ensuring both the Beja and Braga datasets are configured exactly as shown in your file settings. You need to set `shortwave_radiation_sum` as the target, `keep zone` as a feature, and specifically set `time`, `latitude`, and `longitude` to skip. Matching these settings across all your files is a vital first step to ensure your results are consistent and that the data merges correctly later on.

We skip those specific columns because they don't provide useful information for the model's predictions. Since each dataset belongs to a specific city and we already use the `Zone` variable to identify them, the `Latitude` and `Longitude` act as constant values that offer no predictive power. Similarly, we skip `Time` because we are already using `Date` as a feature, keeping both would be redundant. This cleanup ensures the model stays focused only on the variables that actually influence the weather patterns.



Name	Type	Role	Values
1 time	datetime	skip	
2 temperature_2_...	numeric	feature	
3 temperature_3_...	numeric	feature	
4 temperature_4_...	numeric	feature	
5 precipitation_sum	numeric	feature	
6 wind_speed_10_...	numeric	feature	
7 wind_gusts_10_...	numeric	feature	
8 shortwave_rad...	numeric	target	
9 sunshine_durati...	numeric	feature	
10 relative_humidit...	numeric	feature	
11 cloudcover_mean...	numeric	feature	
12 date	datetime	feature	
13 zone	categorical	feature	BRAGA
14 latitude	numeric	skip	
15 longitude	numeric	skip	
16 sunrise	text	meta	
17 sunset	text	meta	

(a) Braga



Name	Type	Role	Values
1 time	datetime	skip	
2 temperature_2_...	numeric	feature	
3 temperature_3_...	numeric	feature	
4 temperature_4_...	numeric	feature	
5 precipitation_sum	numeric	feature	
6 wind_speed_10_...	numeric	feature	
7 wind_gusts_10_...	numeric	feature	
8 shortwave_rad...	numeric	target	
9 sunshine_durati...	numeric	feature	
10 relative_humidit...	numeric	feature	
11 cloudcover_mean...	numeric	feature	
12 date	datetime	feature	
13 zone	categorical	feature	BEJA
14 latitude	numeric	skip	
15 longitude	numeric	skip	
16 sunrise	text	meta	
17 sunset	text	meta	

(b) Beja

Figure A.6: Initial datasets' configuration for both cities

- **Data Integration (Concatenate):**

Integration of independent datasets from Braga and Beja using the *Concatenate* widget. This step ensures a unified data structure, allowing for direct comparison between regions by merging samples into a single consolidated table.

- **Density Estimation and Kernel Settings:**

Selection, on the Violin Plot, of the **Normal (Gaussian)** Kernel within the *Density Estimation* settings. This ensures a smooth representation of the data frequency, allowing for the identification of unimodal or bimodal distribution patterns in the thermal profiles. Besides that, Kernel Density Estimation (KDE), focusing on the symmetry and bimodality of the violins. This analytical step utilizes the KDE to highlight the high-density thermal clusters in Beja versus the downward-skewed concentration of lower temperatures in Braga, providing a continuous probability profile that goes beyond simple discrete averages.

- **Test & Score and Predictions to evaluate the robustness:**

Before performing Test and Score and generating predictions for each dataset, we used the Select Columns widget to carefully structure the data for the model. We defined "shortwave_radiation_sum" as the target variable and selected five key environmental attributes, including mean temperature, cloud cover, and relative humidity, as the main features.

The remaining variables were intentionally ignored to ensure model efficiency and accuracy. In particular, raw text attributes such as "sunrise" and "sunset" were excluded

because they are not directly usable in their original format and had already been transformed into more meaningful numerical variables. Additionally, identifiers such as "date" and "zone", along with redundant temperature metrics like "max" and "min", were removed to prevent overfitting and multicollinearity, allowing the model to focus on the most relevant predictive patterns.

Furthermore, it was repeated those tests more than once, obtaining similar values that would only change about thousandths as we can see in the following figures:

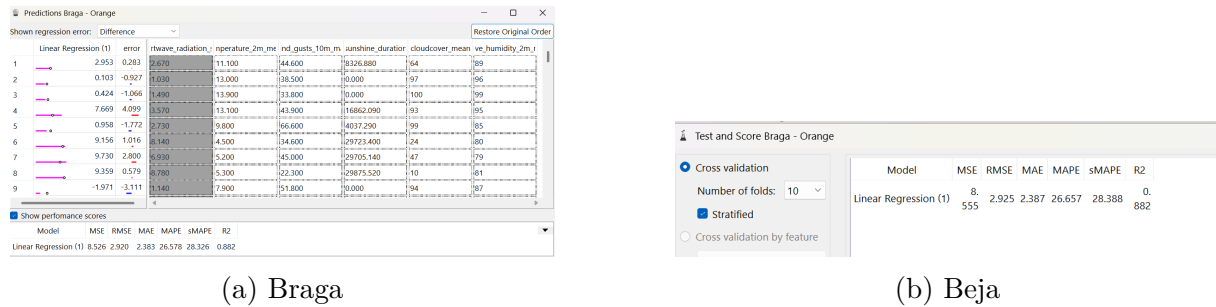


Figure A.7: Predictions and Test&Score of Braga

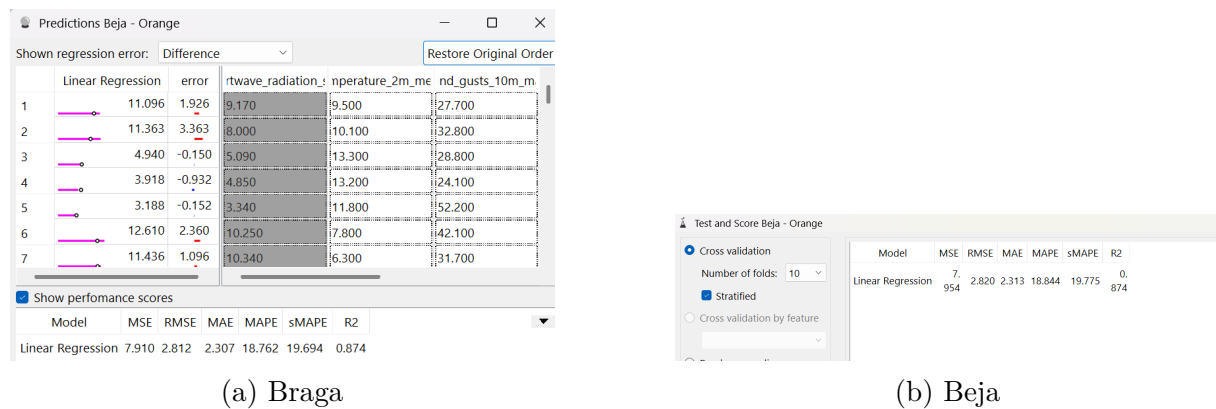


Figure A.8: Predictions and Test&Score of Beja

• Algorithm Logic and Configuration of 3.3.2

The analysis compared a global linear model (**Logistic Regression**) with a local similarity model (**k-NN**) to ensure results were consistent across different mathematical approaches.

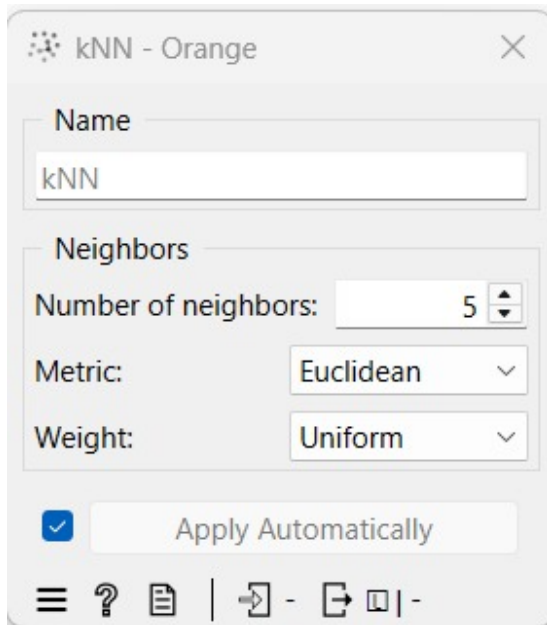


Figure A.9: k-NN Configuration: Set to $k = 5$ with Euclidean distance to identify patterns based on historical similarity .

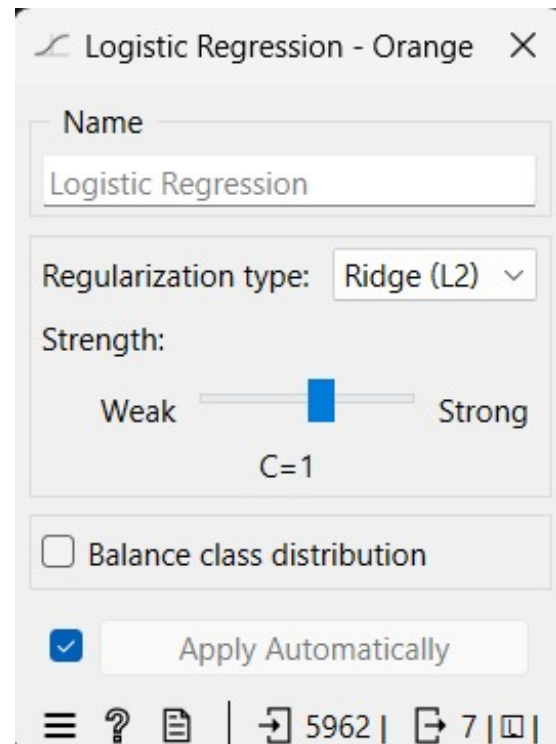


Figure A.10: Logistic Regression: Configured with Ridge (L2) regularization to ensure model stability.

• Performance Metrics and Identifiability of 3.3.2

The models were evaluated using 10-fold Cross-Validation to ensure statistical significance. The results confirm that Beja remains the more "bankable" asset due to its stable weather signature compared to the volatility observed in the North.

Evaluation results for target (None, show average over classes) ▾						
Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.674	0.612	0.612	0.612	0.612	0.225
Constant	0.500	0.500	0.405	0.500	0.500	-0.001
SVM	0.462	0.465	0.427	0.453	0.465	-0.081
kNN	0.634	0.598	0.598	0.598	0.598	0.196

Figure A.11: Evaluation Metrics for Beja and Braga targets. Both models significantly exceed the 0.500 AUC baseline.

• Logistic Coefficients and Thermal Paradox of 3.3.2

The extraction of model weights highlights the sensitivity to radiation volume versus heat-induced degradation. The near-equal but opposite high-magnitude values demonstrate the "tug-of-war" where raw radiation competes with heat stress.

	name	BRAGA
1	intercept	-0.43248
2	precipitation_su...	0.0695569
3	shortwave_radi...	-2.53391
4	cloudcover_me...	-0.121892
5	cloudcover_me...	-0.294246
6	cloudcover_me...	-0.00899388
7	Radiation_Effici...	2.5783

Figure A.12: Data Table displaying the coefficients. These weights define the mathematical boundary between the two regional climates.

• Operational Risk of 3.3.2

The Confusion Matrix visualizes the "Identifiability Gap," specifically documenting the 1,141 "imposter sunny days" where Braga's weather briefly mimics the southern potential.

		Predicted		
		BEJA	BRAGA	Σ
Actual	BEJA	1811	1170	2981
	BRAGA	1141	1840	2981
Σ		2952	3010	5962

Figure A.13: Confusion Matrix cross-referencing actual vs. predicted labels to quantify operational risk.

• Feature Selection, Model Training and Evaluation of 3.3.5

The target variable (rain class) and a set of meteorological variables were selected using the Select Columns widget. These variables describe atmospheric conditions such as humidity, temperature, cloud cover, wind, and sunshine duration, which are relevant for predicting rainfall.

A Logistic Regression model was trained using the selected features. This model estimates the probability of a day being classified as Rain based on the input variables.

Logistic Regression is a classification algorithm that outputs probabilities, which are then converted into class predictions (Rain or No Rain). The model learns patterns in the data by identifying relationships between meteorological variables and the occurrence of rainfall.

This allows the model to distinguish between days with and without rain based on the observed atmospheric conditions.

The model was evaluated using the Test & Score widget, applying cross-validation.

Multiple models were compared, including Logistic Regression, SVM, and kNN. The results show that Logistic Regression achieved the best performance among the tested models.

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression (1)	0.927	0.849	0.870	0.859	0.881	0.689
SVM (1)	0.597	0.566	0.678	0.591	0.795	0.058
kNN (1)	0.830	0.774	0.812	0.777	0.851	0.532

Figure A.14: Model evaluation results using the Test & Score widget in Orange.

These results indicate a strong overall performance, with Logistic Regression achieving the best results (AUC = 0.927 and Accuracy = 0.849). The model shows a good balance between correctly identifying rain events (Recall = 0.881) and avoiding false detections (Precision = 0.859).

- **Confusion Matrix Analysis**

This analysis is important to understand the types of errors made by the model, particularly whether it fails to detect rain events.

		Predicted		
		No Rain	Rain	Σ
Actual	No Rain	3026	407	3433
	Rain	496	2033	2529
Σ		3522	2440	5962

Figure A.15: Confusion matrix for the Logistic Regression model.

The confusion matrix shows that the model correctly classifies most observations, with 3026 true negatives and 2033 true positives. However, there are 407 false positives and 496 false negatives, indicating that some rain events are not detected. This suggests that while the model performs well overall, it has slightly more difficulty identifying all rainy days.

- **Configurations to obtain the Mosaic Display**

First, open the Discretize widget. For each bold variable (wind, temperature, and radiation), select it and click Equal frequency with 3 intervals. This turns your raw numbers into three simple categories (like low, medium, and high) so the chart can actually read them.

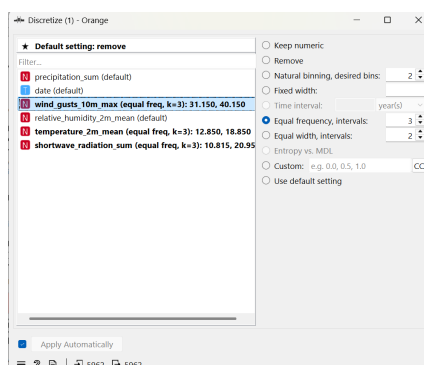


Figure A.16: Discretize Mosaic.

Second, open the Mosaic Display and set the dropdowns to match the Figure A.17: put temperature first, wind second, and zone third. For the Interior Coloring, choose short-

wave_radiation. This will give you a clear, color-coded map showing how radiation levels shift depending on the weather conditions and the specific zone.

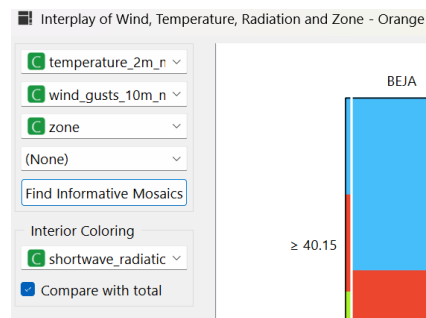


Figure A.17: Mosaic configurations.

Workflow Replication and Model Hyperparameters of Random Forest and Neural Networks

To ensure the replicability of the results presented in this report, this section details the specific configurations and hyperparameters used for the advanced modeling stage in Orange Data Mining.

- **Decision Tree Configuration**

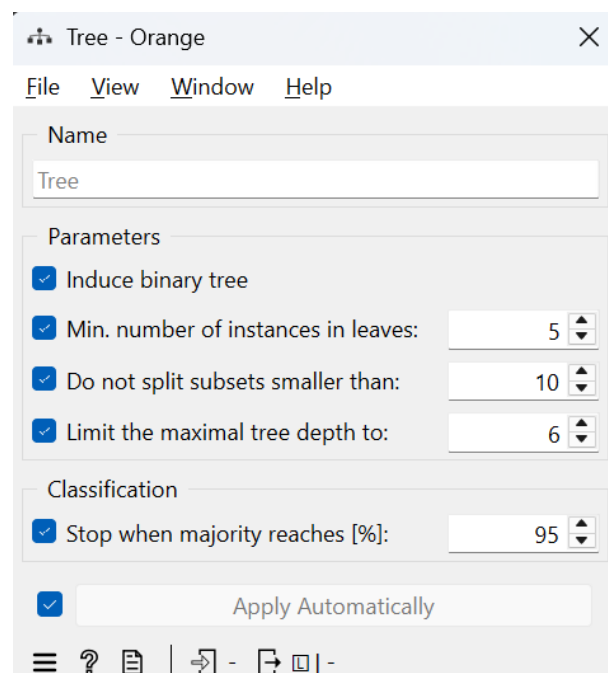


Figure A.18: Growth control constraints, classification criteria, and pruning configurations for the individual Decision Tree model.

The individual Decision Tree model was configured with specific growth controls and pruning criteria to balance visual interpretability with predictive discipline, preventing the tree from growing over-complex.

1. **Induce Binary Tree (Checked):** This option forces the algorithm to build a strictly binary structure, meaning every internal node splits into exactly two branches (e.g., $\text{variable} \leq \text{threshold}$ vs. $\text{variable} > \text{threshold}$). This restriction simplifies the logical pathing, making the resulting rules much easier to interpret during auditing.
2. **Min. Number of Instances in Leaves (5):** A regularization constraint specifying that a terminal leaf node must contain at least 5 daily observations to be created. This prevents the tree from creating highly isolated leaves that apply only to single outliers, ensuring that any final microclimate classification holds statistical relevance.
3. **Do Not Split Subsets Smaller Than (10):** This parameter controls the pre-pruning process. An internal node will only attempt a split if it contains at least 10 observations. If a subset drops below this threshold, branching stops immediately, protecting the model from over-specifying rules on dwindling sub-samples of data.
4. **Limit the Maximal Tree Depth to (6):** This parameter directly restricts the vertical growth of the model to a maximum of 6 layers. Limiting depth is a crucial mechanism to guarantee that the tree remains a human-readable "white box" tool, cutting off excessive branching before the model begins to memorize noise.
5. **Stop When Majority Reaches (95%):** Under the classification constraints, branching stops early if any single class (e.g., "No Rain") represents 95% or more of the observations within that specific node. Once a node achieves this level of purity, further splitting yields negligible performance gains, so the algorithm closes the branch to maintain structural efficiency.

- **Random Forest Configuration**

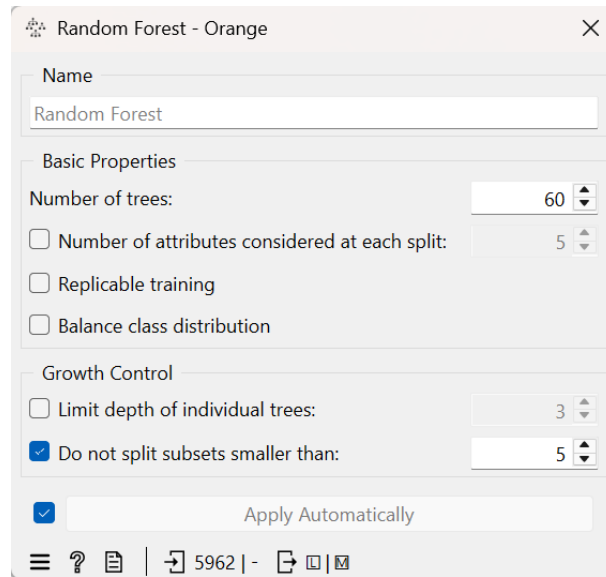


Figure A.19: Hyperparameter setup for the Random Forest ensemble model.

The Random Forest model was configured to balance predictive depth with ensemble diversity. The following parameters were applied:

1. **Number of Trees (60):** We selected 60 trees to ensure a robust ensemble consensus. While more trees generally increase stability, 60 was found to be the "sweet spot" where the model reaches peak accuracy (AUC) without unnecessary computational overhead or over-fitting the specific noise of the historical dataset.
2. **Growth Control (Do not split subsets smaller than 5):** This constraint is vital for preventing the model from becoming too specific to individual outliers. By requiring at least 5 observations to create a new branch, we ensure that every decision rule is backed by a statistically significant cluster of days.
3. **Unconstrained Depth:** Individual trees were allowed to grow until the subset limit was reached, allowing the model to capture the complex, high-dimensional interactions between radiation, humidity, and cloud cover.

- **Neural Network (Multi-Layer Perceptron) Configuration**

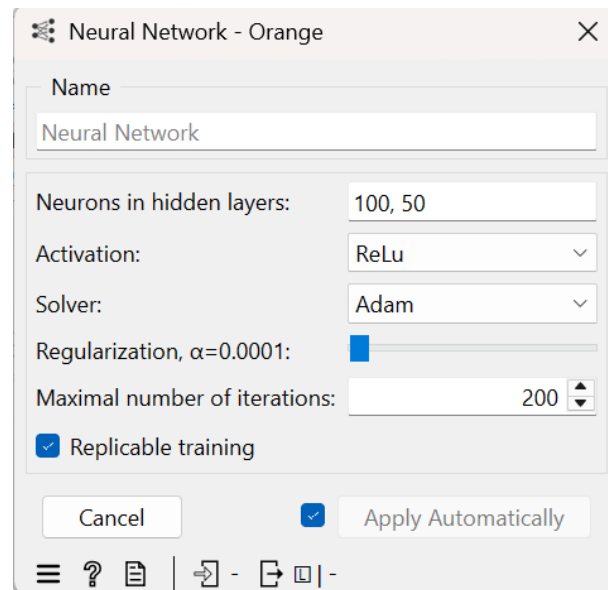


Figure A.20: Architecture and optimization settings for the Multi-layer Perceptron (Neural Network)

The Neural Network was designed to capture non-linear relationships through a deep architecture. The configuration steps are as follows:

1. **Hidden Layers (100, 50):** We implemented a two-layer architecture. The first layer of 100 neurons handles the broad extraction of meteorological features, while the second layer of 50 neurons refines these patterns. This "funnel" approach is effective for classification tasks.
2. **Activation Function (ReLU):** The Rectified Linear Unit (ReLU) was chosen because it allows the network to learn faster and prevents the "vanishing gradient" problem, making it ideal for the numerical range of our weather data.
3. **Solver (Adam):** The Adam optimizer was utilized for its adaptive learning rate. It is highly efficient for datasets with nearly 6,000 observations, as it automatically adjusts the weights to minimize the error during training.
4. **Regularization ($\alpha = 0.0001$):** A small L2 penalty was applied to keep the weights small, preventing the network from becoming overly complex and ensuring it generalizes well to future climate data.
5. **Maximal Iterations (200):** The limit of 200 iterations ensures the model has enough cycles to converge on a stable solution without "drifting" into over-fitting.

6. **Replicable Training:** This option was checked to ensure that the random initialization of weights remains consistent every time the workflow is run, allowing for identical results in subsequent audits.

A.3.3 Data Partitioning, Predictive Modeling, and Python Extraction Workflow

To ensure complete transparency and full reproducibility for the SunVolt Energy project validation process, this subsection details the computational steps executed within the updated Orange Data Mining canvas and the supplementary Python parsing framework. This dual architecture isolates territorial climates, maps specific target domains, and computes the empirical validation tables referenced throughout Chapter 4. To do all this the workflow file `orange_mil3.ows` was used.

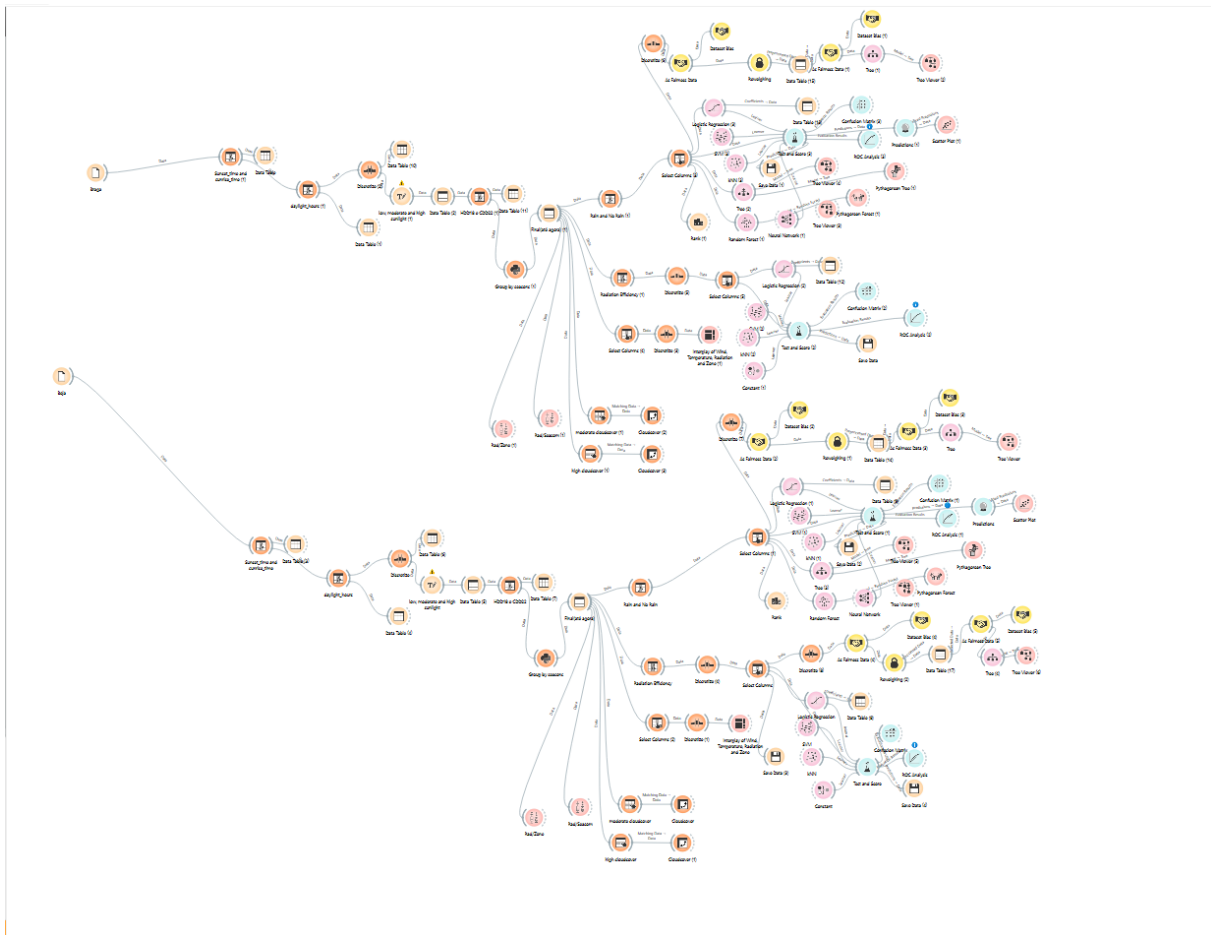


Figure A.21: Workflow orange mil3.ows

Geographical Dataset Separation

To decouple and analyze the performance behaviors under distinct macroclimates, the historical entry stream was stratified into regional branches using isolated source nodes:

1. **Beja Stream Integration:** A dedicated *File* node was configured to load `raw_BEJA.csv`, locking the processing logic into the low-precipitation, high-thermal parameters of the Alentejo plain.
2. **Braga Stream Integration:** A parallel *File* node was introduced to process `raw_BRAGA.csv`, capturing the high-frequency pluviosidade profile of the Northern territory.

Feature Selection and Target Variable Mapping

Before routing information into the learning models, independent feature structures and categorical targets were explicitly defined using specialized *Select Columns* widgets across two operational branches:

1. **Continuous Efficiency Branch:** For the solar production index, the target variable was mapped to `Radiation_Efficiency` (Categorical target discretized by thresholds: < 15.64 vs. ≥ 15.64 for Braga; < 15.8868 vs. ≥ 15.8868 for Beja). The model input space was restricted to independent predictors: `relative_humidity_2m_mean`, `cloudcover_mean`, `wind_speed_10m_max`, `season`, and `temperature_2m_mean`.
2. **Discrete Operational Downtime Branch:** For weather-induced downtime mapping, the target column was set to `rain_class` (*Rain* vs. *No Rain*). The training matrices were constructed using independent variables: `relative_humidity_2m_mean`, `temperature_2m_mean`, `wind_gusts_10m_max`, `sunshine_duration`, and `cloudcover_mean`.

Test and Score Evaluation Architecture

The predictive engine utilizes a stratified *Test & Score* node connected to the competing algorithm configurations (**Neural Network**, **Logistic Regression**, **kNN**, and **SVM**):

1. **Cross-Validation Configuration:** The evaluation environment was configured to execute a rigorous **10-fold cross-validation** sampling protocol. This mathematically splits the observation data into ten balanced, non-overlapping partitions to iterate predictive checking and minimize sampling bias.

2. **Raw Metrics Generation:** The node aggregates fold-level evaluations, computing multi-class metrics including Classification Accuracy (CA), F1-score, Precision, and Recall.

Python-Driven Metric Extraction and Export

To bypass manual compilation and protect data integrity, a *Save Data* node was chained directly to the evaluation streams of the *Test & Score* nodes to export raw predictions.

1. **Fold Metrics Isolation:** The Jupyter Notebook imports the raw outputs, strips out metadata headers, and computes the exact accuracy achieved by each model within each specific repetition fold.
2. **Descriptive and Inferential Outputs:** The automated script applies aggregation functions to dynamically calculate the mean performance and standard deviations, saving the structural matrices directly into `model_accuracy_summary.csv` and `accuracy_by_fold.csv`.
3. **Pairwise Matrix Automation:** Lastly, the script routes the fold-by-fold accuracy matrix into a paired statistical evaluation module, executing *t*-test comparisons to generate the definitive *p*-value results verified in Section 4.3.

A.3.4 Fairness Optimization Workflow and Comparative Protocol

The empirical evaluation of predictive models for the SunVolt Energy framework requires not only high accuracy metrics but also strict mathematical fairness across sensitive operational partitions. To document and verify the reproducibility of this optimization phase, this section outlines the architectural adjustments executed within the Orange Data Mining environment, explicitly utilizing the workflow file `orange_mil3.ows`.

Workflow Restructuring and Fairness Re-routing

To evaluate the absolute impact of algorithmic mitigation on target variables, a comparative protocol was designed by restructuring the baseline evaluation streams. The technical procedure for shifting from the baseline model evaluation to the fairness-optimized monitoring phase was executed as follows: In the original predictive pipeline, the training partitions were routed from the *Select Columns* node directly into the primary *Test &*

Score engine, establishing a baseline benchmark uncorrected for potential historical biases or variance disparities. To audit and mitigate systemic discrepancies, the direct link between the *Select Columns* widget and the test and score widget was removed. The optimized data stream, containing the mathematically adjusted case-level weights, is fed directly from the fairness mitigation node straight into the final *Test & Score* node. This ensures that the evaluation metrics are computed natively on top of the corrected feature distribution. By splitting the evaluation engine into two distinct diagnostic paths within the `orange mil3.ows` canvas, it became possible to analyze the *Test & Score* tables side-by-side before and after the fairness transformation.

A.4 Submitted Materials Checklist

- Report PDF (this document)
- `raw_BEJA.csv` - raw meteorological data for Beja
- `raw_BRAGA.csv` - raw meteorological data for Braga
- `Milestone_2_Solar` - Orange workflow file (.ows)
- `orange mil3 - (fairness)` (.ows)
- `notebook com datasets` (folder)